# 13. Installation and Servicing Air Conditioner using R32

## *13.1 About R32 Refrigerant*

For air conditioning refrigerants such as R410A, the refrigerants were collected back in order to prevent their air dissipation, to curbe the global warming impact, in case they were released into the atmosphere. In the “4th Environmental Basic Plan”, 80% reduction of greenhouse gas emissions by 2050 is required, and due to this requirement, further reduction in the emission of high greenhouse effect gas, such as CFCs, is required. Therefore, the conversion of air conditioning refrigerant into the ones who has smaller greenhouse effect, even if it is dissipated into the atmosphere, became our responsibility.

Nevertheless, in case of air conditioning refrigerant, it would be the best if there is a refrigerant which has smaller impact on global warming, but ensures good energy efficiency and performance, and is safe; however, there is no such refrigerant which satisfies all these conditions. As a result, we have been considering the practical usage, within the safety frame-work, of R32 refrigerant which has short lifetime in the atmosphere, and has smaller effect of global warming, but is slightly flammable.

In 2004, due to the revision of air conditioner safety standards by the International Electro-safety Commission (IEC), the safety standards of air conditioners using slightly flammable refrigerant was issued. In 2010, the regulations of American Society of Heating, Refrigerating and Air-Conditioning Engineers in the United States (ANSI/ASHRAE34) was issued adopting the grades for refrigerants which are difficult to inflame due to their slow burning rates, and as a result have smaller damages in cases of fire. The burning rate of R32 is lower by 10cm / per second, and safety standardization for various usage is now being processed.

## *13.2 Characteristics of R32 Refrigerant*

1. Chemical Characteristics

R32 is one of the refrigerants used in R410A, has almost no toxicity, and chemically stable compound formed by hydrogen, carbon and fluorine.

R32 has short lifetime of 4 to 9 years in case of being released into the atmosphere; therefore, it has smaller greenhouse gas effect but has slight inflammability because of the large proportion of hydrogen.

Chemical Characteristic Table of R32, R410A and R22.

| | R32 | R410A | R22 |
|---|---|---|---|
| Chemical Formula | CH2F2 | CH2F2 / CHF2CF3 | CHCLF2 |
| Composition (mixture ratio wt.%) | Single Composition | R32 / R125A (50 / 50 wt.%) | Single Composition |
| Boiling Point (°C) | -51.7 | -51.5 | -40.8 |
| Pressure (physical) *1 | 3.14 | 3.07 | 1.94 |
| Capacity (physical) *2 | 160 | 141 | 100 |
| COP (physical) *3 | 95 | 91 | 100 |
| Ozone Depletion Potential (ODP) | 0 | 0 | 0.055 |
| Global Warming Potential (GWP) *4 | 675 | 2090 | 1810 |
| Inflammability *5 | Slightly Inflammable (A2L) | Non-inflammable (A1) | Non-inflammable (A1) |
| Toxicity | None | None | None |

*1：Physical property of temperature condition 50°C
*2：Relative value of temperature condition 0/50°C, providing R22=100
*3：Te/Tc/SC/SH=5/50/3/0°C
*4：GWP=Global Warming Potential, each figure is based on “4$^{th}$ IPCC4 Report”
*5：Based on ANSI / ASHRAE std. 34-2010

2. Characteristic of Pressure

As shown in Table 2, R32 does not have much difference in vapor pressure at the same refrigerant temperature comparing to R410A, but comparing to R22, it is higher at 1.6 times more. Thus, the same as in case of R410A, it is necessary to do installation and service using high-pressure tools and components.

Table 2. Saturated vapor pressure comparison table

(Unit: MPa)

| Temperature | Refrigerant | | |
|---|---|---|---|
| | R32 | R410A | R22 |
| -20 | 0.30 | 0.30 | 0.14 |
| 0 | 0.71 | 0.70 | 0.40 |
| 20 | 1.37 | 1.35 | 0.81 |
| 40 | 2.38 | 2.32 | 1.43 |
| 60 | 3.84 | 3.73 | 2.33 |
| 65 | 4.29 | 4.17 | 2.60 |

Reference： Thermal properties table of Japan Society of Refrigerating and Air Conditioning Engineers (60, 65°C)
NIST REFPROP V8.0 (-20 ~ 40°C)

## *13.3 Refrigerant piping installation • Tools used in services*

### 13.3.1 Required Tools

R32 refrigerant air conditioners use the common parts as R410A air conditioners for two-way valves and three-way valves (diameters of service ports); thus, they maintain commonality in the maintenance of the compressive strength, the size of pipe flaring, and the size of flare nuts as R410A. Therefore, for refrigerant pipe installation and services, you can use tools for R410A.

However, mixing of refrigerants is not allowed, so that you have to separate the cylinders for the recovery of refrigerants.

Tools used for installation • relocation • replacement of air conditioning units

| Works | R32 | R410A | R22 |
|---|---|---|---|
| Flaring | Flare tools for R410A (clutch type) | | Flare tools for R22 (clutch type) |
| Connection of pipes | Torque wrench (diameter 1/4 3/8) | | |
| | Torque wrench (diameter 1/2 5/8) *1 | | Toque wrench (diameter 1/2 5/8) |
| Manifold gauge charging hose | R32 & R410A Common (As at November 2013) | | R22 Only |
| Air purging | Vacuum pump + Reducer / expander | | Vacuum pump |
| Gas leakage test | Detection liquid or soup water, HFC detector | | |

*1. Nut diameters of 1/2 5/8, the size of torque wrench common with R410A
For other installation, you can use general tools such as screw drivers (+, -), metal saws, electric drills, long-nose pliers, hole core drills (ø70 or ø65), linen tape, levels, temperature gauges, clamp meters, electric knives, nippers, pipe cutters, reamers or scrapers, spring benders, (diameters 1/4 3/8 1/2 5/8), monkey wrenches, fixing wrenches (17 or 12 mm), feeler gauges, hexagon wrenches (4 mm), testers, megohm testers, etc.

Tools used for services

| Works | R32 | R410A | R22 |
|---|---|---|---|
| Insertion of refrigerant | Digital scale for refrigerant charging, refrigerant cylinders, cylinder adopters and packing *a | | |
| Recovery of refrigerant | Refrigerant recovery devices, refrigerant cylinders, manifold gauges, charging hoses *b | | |

*a. Use cylinder for each refrigerant, cylinder adopter and packing.
*b. Use refrigerant recovery cylinder separately for each refrigerant **(no mixture of refrigerant allowed)**. Please be aware that there are some refrigerant collection devices which do not have self-certification.

### 13.3.2 Tools for R32 (common with R410A)

1. Flare gauges
Use flare gauges when you perform flaring with flare tools (crutch type). Flare gauges are used to set the pipe ends at 0.5 ~ 1.5 mm from clump bars of flare tools.

Flare gauges



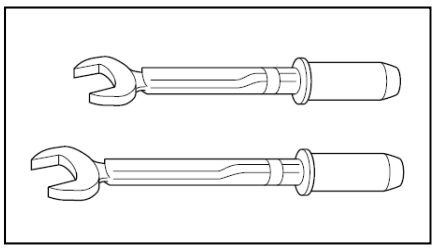

2. Flare tools (clutch type)
Flare tools have larger holes of clump bars in order to set the pipe end at 0 ~ 0.5 mm, and have stronger springs inside to ensure solid flaring torques. These flare tools can be used commonly for R22.

Flare tools (clutch type)



3. Torque wrenches (diameters 1/2, 5/8)
In order to strengthen the compressive strength, the diameters of wrenches change depending on the flare nut sizes.

Torque wrenches



Differences in torque wrenches

| | R32 (common R410A) | R22 |
|---|---|---|
| 1/2 (diameter × torque) | 26 mm × 55 N•m (550 kgf•cm) | 24 mm × 55 N•m (550 kgf•cm) |
| 5/8 (diameter × torque) | 29 mm × 65 N•m (650 kgf•cm) | 27 mm × 65 N•m (650 kgf•cm) |

4. Manifold gauges
R22 gauges cannot be used because of the high pressures.

Each port of manifold has different shapes in order to prevent inserting wrong refrigerant.
*However, the port shape for R410A and R32 is the same; therefore, attention need to be paid not to insert wrong refrigerant.

Differences in high/low pressure gauges

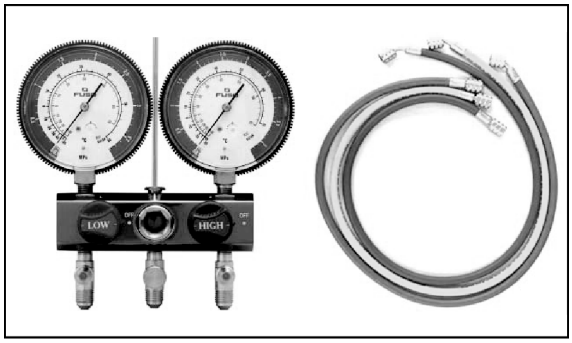
| | R32 (common R410A) | R22 |
|---|---|---|
| High pressure gauges (red) | -0.1 ~ 5.3 MPa<br>-76 cmHg ~ 53 kgf / cm$^2$ | -76 cmHg ~ 35 kgf / cm$^2$ |
| Low pressure gauges (blue) | -0.1 ~ 3.8 MPa<br>-76 cmHg ~ 38 kgf / cm$^2$ | -76 cmHg ~ 17 kgf / cm$^2$ |

Difference in manifold port sizes

| | R32 (common R410A) | R22 |
|---|---|---|
| Port sizes | 1/2 UNF20 | 7/16 UNF20 |

5. Charging hoses
The pressure resistance of charge hoses is increased. At the same time, the material is changed to HFC resistant, and the size of each manifold adopter is changed, as the port size of manifold gauge itself. Further, some hoses are with anti-gas pressure backflow valves placed near the adopters. (hoses with the valves recommended)

Manifold gauges / Charging hoses

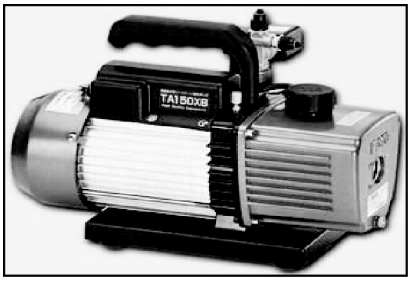



Differences in charging hoses

| | | R32 (common R410A) | R22 |
|---|---|---|---|
| Pressure Resistance | Normal operation pressure | 5.1 MPa (52 kgf / cm$^2$) | 3.4 MPa (35 kgf / cm$^2$) |
| | Burst pressure | 27.4 MPa (280 kgf / cm$^2$) | 17.2 MPa (175 kgf / cm$^2$) |
| Material | | HNBR rubber Internal nylon coating | NBR rubber |

6. Vacuum pump and Vacuum pump adopter
When using a vacuum pump, it is necessary to set a solenoid valve in order to prevent backflow of vacuum pump oil into the charge hoses, and use a vacuum pump with oil backflow prevention function, or use the vacuum pump with vacuum pump adopter.
If vacuum pump oil（mineral oil-based）mixes with R410A (R32), it may cause damage to the machine.

Vacuum pump




Vacuum pump adopter



7. HFC refrigerant_Electric gas leakage tester
R32 refrigerant is often used for other mixed refrigerant (R410A, R404A, R407C etc.). Therefore, the usage of existing HFC detectors is possible, but in order to detect more accurately, we recommend to use detectors specially set and adjusted for R32 detection.

HFC refrigerant_Electric gas leakage tester



8. Digital scale for refrigerant charging
R32 and R410A have high pressure level and their evaporation speed is high.
Thus, if you recover the refrigerant by cylinder charging method, the refrigerant evaporates within the weighing scale glass, which makes reading the scale difficult, rather than liquidating the refrigerant into the cylinder. (Charging cylinders for R22 have different pressure resistance, scale, connection port size; therefore, they are not usable) At the same time, the digital scale for refrigerant charging is strengthened by receiving the weight of the refrigerant cylinders with four pillars at the corners. The connection ports of charging hoses have two separate ports for R22 (7/16 UNF20) and R32/R410A (1/2 UNF20) therefore, they can be used for the insertion of the existing refrigerants.

Digital scale for refrigerant charging




9. Refrigerant cylinders
Refrigerant cylinders for R410A are painted in pink, and the ones for R32 are painted in other colors that might subject to change according to the international standards. R32 is a single refrigerant, so that both liquid and gas insertion are possible. Additional charging is also possible.
(R410A is a mixed refrigerant, so only liquid insertion is possible)

Refrigerant cylinders




10. Connection ports of refrigerant cylinders and packing
Charging ports which fit to the charging hose connection port size (1/2 UNF20) is needed. At the same time, the packing has to be of HFC resistant materials.

Connection ports and packing



## 11. Tools used for refrigerant piping installations and services

| | Tools for R410A | Common with R32 | Possibility of usage for R22 |
|---|---|---|---|
| 1. | Pipe cutters, reamers or scrapers | ○ | ○ |
| 2. | Flare tools (clutch type) | ○ | ○ |
| 3. | Torque wrench (1/4, 3/8) | ○ | ○ |
| 4. | Torque wrench (1/2, 5/8) | ○ | × |
| 5. | Manifold gauges・charging hoses | ○ | × |
| 6. | Vacuum pumps, vacuum pump adopters | ○ Connection 5/16 | ○ Connection 1/4 |
| 7. | Electric gas leakage testers for HFC *1 | ○ | △ |
| 8. | Digital scale for refrigerant charging | ○ | ○ |
| 9. | HCF recovery devices (connection port 5/16) *2 | ○ Connection 5/16 | ○ Connection 1/4 |
| 10. | Refrigerant cylinders (pressure resistant: FC3) | Same specs × | × |
| 11. | Refrigerant cylinders (pink) | Other (colors that might subject to change according to the international standards). | × |
| 12. | Refrigerant cylinder connection ports and packing | ○ | × |
| 13. | Allen wrench (4 mm) Electric knives | ○ | ○ |

*1 Those testers only for HCFC22 (R22), but not for HCF32 (R32) and HCF410A (R410A) cannot be for common use.

*2 Recovery devices which are self-certified for each HCF type can be used.

[Knowledge for the common usage of tools for R410A & R32]

- R410A and R32 machines use different compressor oils.
- If unregulated compressor oil gets mixed into, it may cause damage to the machine function.
- Careful pump down will ensure the recovery of compressor oil, and it will minimize the remaining amount of the oil in the manifold gauge and charging hose.
- If you only perform the recovery of refrigerant and not be able to perform pump down, you have to dispose the compressor oil in the charging hose.

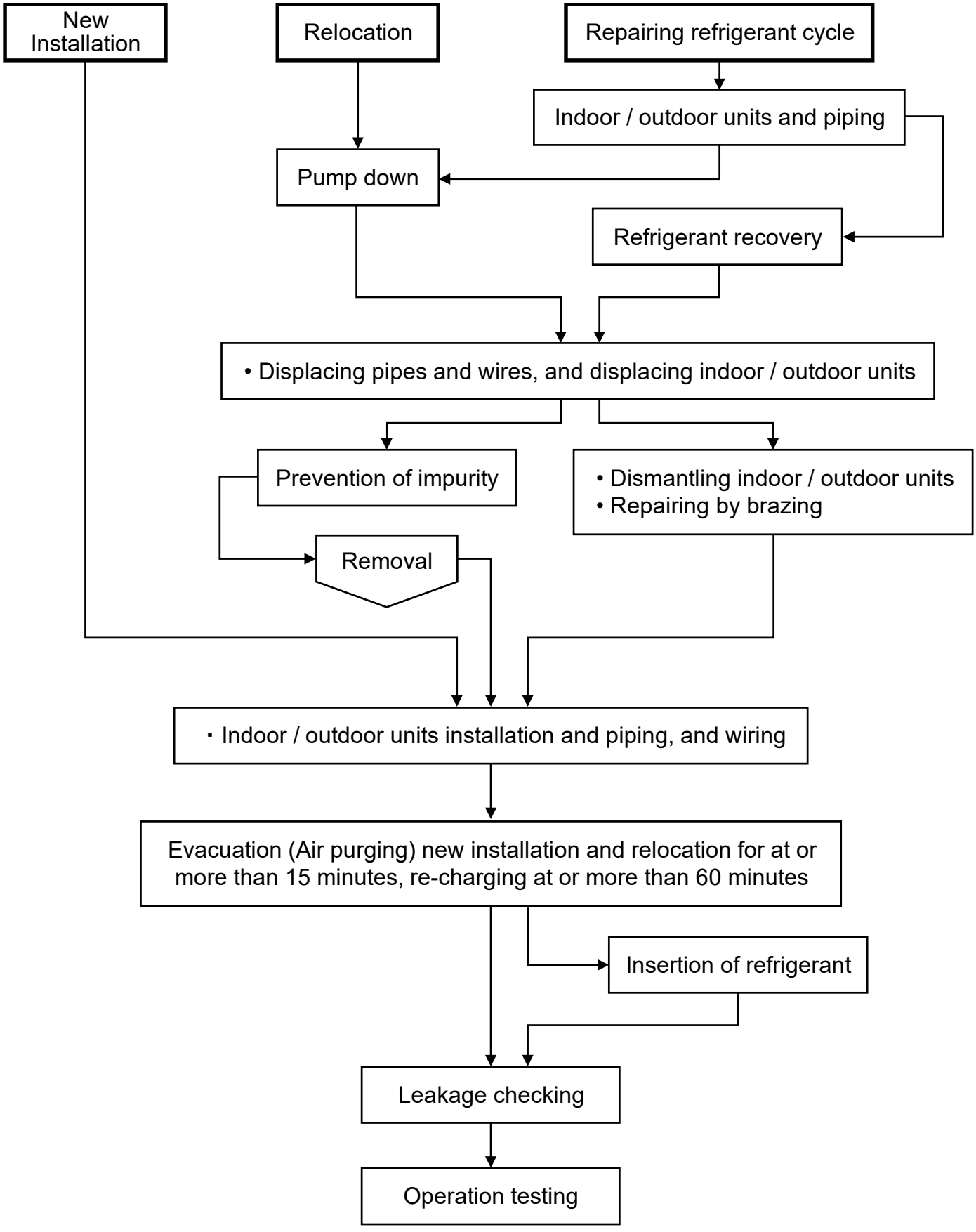
[Precaution of repairing refrigerant cycle]

- In the brazing, open 2-way and 3-way valves, and make sure the refrigerant is completely recovered back and not remaining the system.
- When repairing outside, make sure no refrigerant is in the air, ensure good air flow, and perform the brazing.

[Inserting wrong refrigerant]

- It may cause “not cooling” and “not heating” customer claims because each component (expansion valve, compressor, PCB) of the refrigeration cycle is specially adjusted for R32.
- At the same time, it is not subject to product warranty, if wrong refrigerant was inserted into the system.

## *13.4 New installation, Relocation, Repairing of Refrigerant Cycle System The Procedures*



## *13.5 Piping installation of R32*

### 13.5.1 Pipe materials used and flaring

Copper pipes are used for refrigerant piping. Pipes which comply with JIS Regulations need to be used. Room air conditioners which use R410A and R32 have higher pressure; thus, using pipes which comply with the Regulations is important.
The pipe thickness is regulated by revised JIS B 8607 “Flaring and brazing fittings for refrigerant” and the pipe thickness for R410A, R32 is shown in the table.

Pipe thickness

| O and OL materials | | Thickness (mm) | | |
|---|---|---|---|---|
| Diameter | Diameter (mm) | R410A | R32 | R22 |
| 1/4 | 6.35 | 0.80 | | |
| 3/8 | 9.52 | 0.80 | | |
| 1/2 | 12.70 | 0.80 | | |
| 5/8 | 15.88 | 1.00 | | |

**Caution**

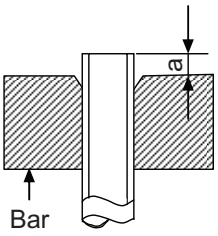
- For connection piping, use copper phosphate seamless pipes (1220T) as regulated in “JIS H 3300” and the pipe thickness is 0.8 mm.
- In the market, there are some pipes of 0.7 mm thickness, but do not use these pipes (0.8 mm thickness has to be strictly followed).
- It is recommended to use pipes whose adhesion amount of oil is at or less than 40 mg / 10 m.
  At the same time, do not use pipes with dent, de-shape, and color change (especially inside).

### 13.5.2 Processing and connection of pipes

For refrigerant pipe installation, be aware of moisture and dirt do not get into the pipes, and make sure of no refrigerant leakage.
◎ The procedure of flaring and precautions

a) Cutting of pipes：use pipe cutter and cut the pipe slowly not to de-shape the pipe.
b) Removal of burrs on the edge of pipe (reamer or scraper)
   If the condition of pipe edge after the deburring is no good or if burrs attaches on the flaring, it may cause refrigerant leakage. Turn the pipe end down and perform deburring carefully.
c) Insert the flare nut (use the nut which is a part of the CZ parts)
d) Flaring
   Ensure the cleanliness of clump bar and pipe, and perform flaring carefully.
   Use the existing flare tools or flare tools for R410A. Be aware that the sizes and dimensions of flaring is different in each flaring tool. If you use the existing flaring tools, use flaring gauge to measure the length of the flaring part.

| Piping size (mm) | Dimensions “a” (mm) R22 | | | Dimensions “a” (mm) R410A/R32 | | |
|---|---|---|---|---|---|---|
| | Flare tools | | Flare tools for R410A | Flare tools | | Flare tools for R410A |
| | Clutch type | Wing nut type | Clutch type | Clutch type | Wing nut type | Clutch type |
| 6.35 (1/4") | 0.5 ~ 1.0 | 1.0 ~ 1.5 | 0 ~ 0.5 | 1.0 ~ 1.5 | 1.5 ~ 2.0 | 0 ~ 0.5 |
| 9.52 (3/8") | 0.5 ~ 1.0 | 1.0 ~ 1.5 | 0 ~ 0.5 | 1.0 ~ 1.5 | 1.5 ~ 2.0 | 0 ~ 0.5 |
| 12.70 (1/2") | 0.5 ~ 1.0 | 1.5 ~ 2.0 | 0 ~ 0.5 | 1.0 ~ 1.5 | 2.0 ~ 2.5 | 0 ~ 0.5 |
| 15.88 (5/8") | 0.5 ~ 1.0 | 1.5 ~ 2.0 | 0 ~ 0.5 | 1.0 ~ 1.5 | 2.0 ~ 2.5 | 0 ~ 0.5 |



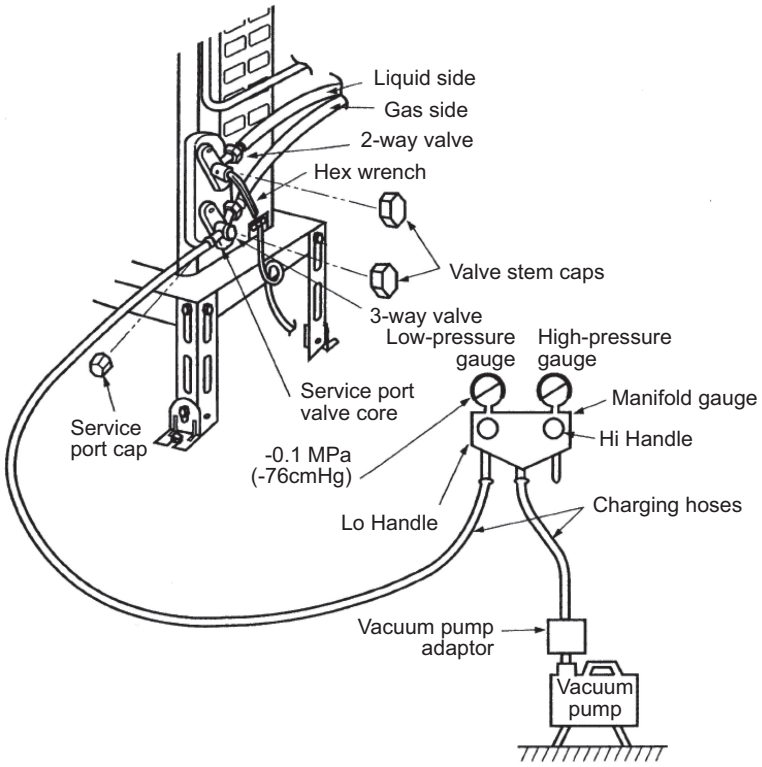

| Nut outer diameter (mm) | |
|---|---|
| R22 | R410A |
| | R32 |
| 17 | 17 |
| 22 | 22 |
| 24 | 26 |
| 27 | 29 |

## *13.6 Installation, Relocation, and Service*

### 13.6.1 Air purge and gas leak test for new installation (using new refrigerant pipes) using vacuum pump

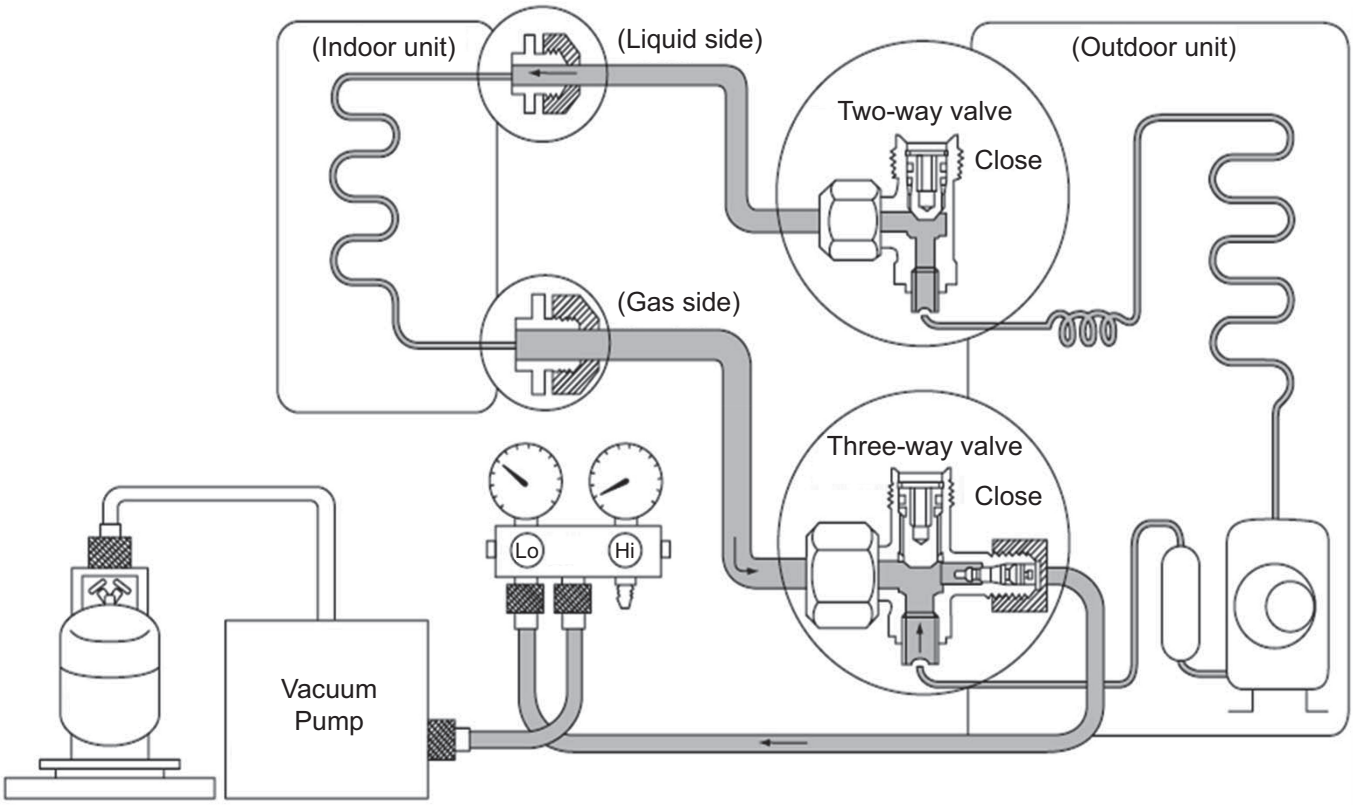
(From the point of view of global environment protection, do not release CFCs into the atmosphere during installation work)

1. Connect the charging hose of manifold gauge to the service port of 3-way valve (pushing insect pin).
2. Fully open the handle Lo of manifold gauge and operate vacuum pump.
   (If the needle of the low-pressure gauge reaches the vacuum immediately, check 1 procedure again)
3. Perform vacuuming 15 minutes or more, and make sure low pressure gauge reaches to −0.1 MPa (-76cmHg).
   When the vacuuming completes, fully open the handle Lo of manifold gauge and stop the operation of vacuum pump, and leave it for 1 ~ 2 minutes. Then, remove the connection side of the charging hose of vacuum pump adopter after checking the needle of manifold gauge does not turn back.
4. Open the stem of 2-way valve to 90° in anti-clock wise, and close the 2-way valve after 10 seconds, and perform gas leakage test.
5. Remove the charge hose from the service port of 3-way valve, and open the stems of 2-way and 3-way valves (open the valves to anti-clock wise carefully, do not use full strength to open)
6. Tighten the service port cap with torque wrench 18 N•m (1.8 kgf•m)
   Tighten the caps of 2-way and 3-way valves with torque wrench 18 N•m (1.8kg f•m)
7. After the tightening of each cap, check gas leakage around the cap.



### 13.6.2 Process of refrigerant recovery

1. Connect the center charging hose of manifold gauge to the in-let side of recovery device.
2. Connect the valves of the discharge side of recovery device and liquid side of refrigerant cylinder with red hose (charging hose).
3. Connect the yellow float switch cable of the recovery device to the refrigerant cylinder.
4. Open the low pressure side valve of manifold gauge.
5. Slightly loosen the charging hose of in-let connecting side of recovery device and perform air purge.
6. Open the liquid valve of refrigerant cylinder and slightly loosen the charging hose in discharging side of recovery device, and perform air purge (the recovery cylinder needs slight inside pressure).
7. Insert electric plug of recovery device into electrical outlet (the fan operation starts).
8. Turn the valve 1 and 2 of recovery device to pressure equalization point.
9. After a few seconds, turn back the valve 1 and 2 to the original position.
10. Turn the switch of the recovery device to “ON”. (the compressor operation starts)
11. When the low pressure of manifold gauge is close to “0”, close the low pressure side valve, turn “OFF” the recovery device switch.
12. Remove the center charging hose of manifold gauge from the recovery device.



### 13.6.3 Relocation

1. Removing the air conditioning unit
   a) Recovery of outdoor unit refrigerant by pumping down
   Press “forced cooling button” (as a general rule, since 1998 the name of cooling testing button is changed, and this name is unified within the air conditioning industry), and then you are able to start cooling operation in which the room temperature is low, and you can recover the refrigerant from the outdoor unit.

      1. Check the valve stems of two-way and three-way valves are open by being turned to anti-clockwise (Remove the caps, and confirm the bars are fully open. Use hexagon wrench <4 mm> to open and close the valves).
      2. Press the “Emergency Operation” button of the indoor units for five seconds and release [Forced cooling operation] (for old models, press “forced cooling” button). Then, operate the air conditioning unit for about 10 minutes.
      3. Turn the stem of the two-way valve to the clock-wise and close the valve.
      4. After about 2 ~ 3 minutes, turn the stem of the three-way valve quickly to the clock-wise, and stop the operation.

**Caution:** In the pump down operation, stop the compressor before removing the refrigerant pipes.

If you do not stop the compressor operation, and if the valve is open and remove the refrigerant pipes, the air may be sucked into the system and causes extreme high temperature in the refrigerant cycle. This may result in rupture or injury, etc.

      5. Attach and tighten the caps of two-way and three-way valves with torque wrench.
      6. Remove the connecting pipes (liquid side and gas side).

   b) Removal of indoor and outdoor units

      1. Remove the connecting pipes and wires between the indoor and outdoor units.
      2. Attach capping flare nuts on the edges of the pipes, connecting the indoor and outdoor units, in order to prevent dust and moisture get into the pipes.
      3. Remove the indoor and outdoor units.

2. Unit installation

Use new refrigerant pipes for the installation, and perform air purging using vacuum pump and gas leakage testing stated in 14.5.1.

### 13.6.4 Replacement of air conditioning units and evacuation (when re-using the existing pipes)

When replacing the air conditioning units, you might use the existing pipes, but it is recommended to perform flaring again. In case of unit replacement, even if the unit is new refrigerant air conditioner, if the refrigerant oil is different, it may cause problem. Further, when re-using the existing refrigerant pipes, it is recommended to evacuate the pipes as much as possible, due to the reason that much refrigerant oil may be attached on the surface of the pipes. If the pipes are used without evacuation, the remaining refrigerant oil may cause under-performance and abnormal refrigerant cycle caused by non-compatibility of those oils.

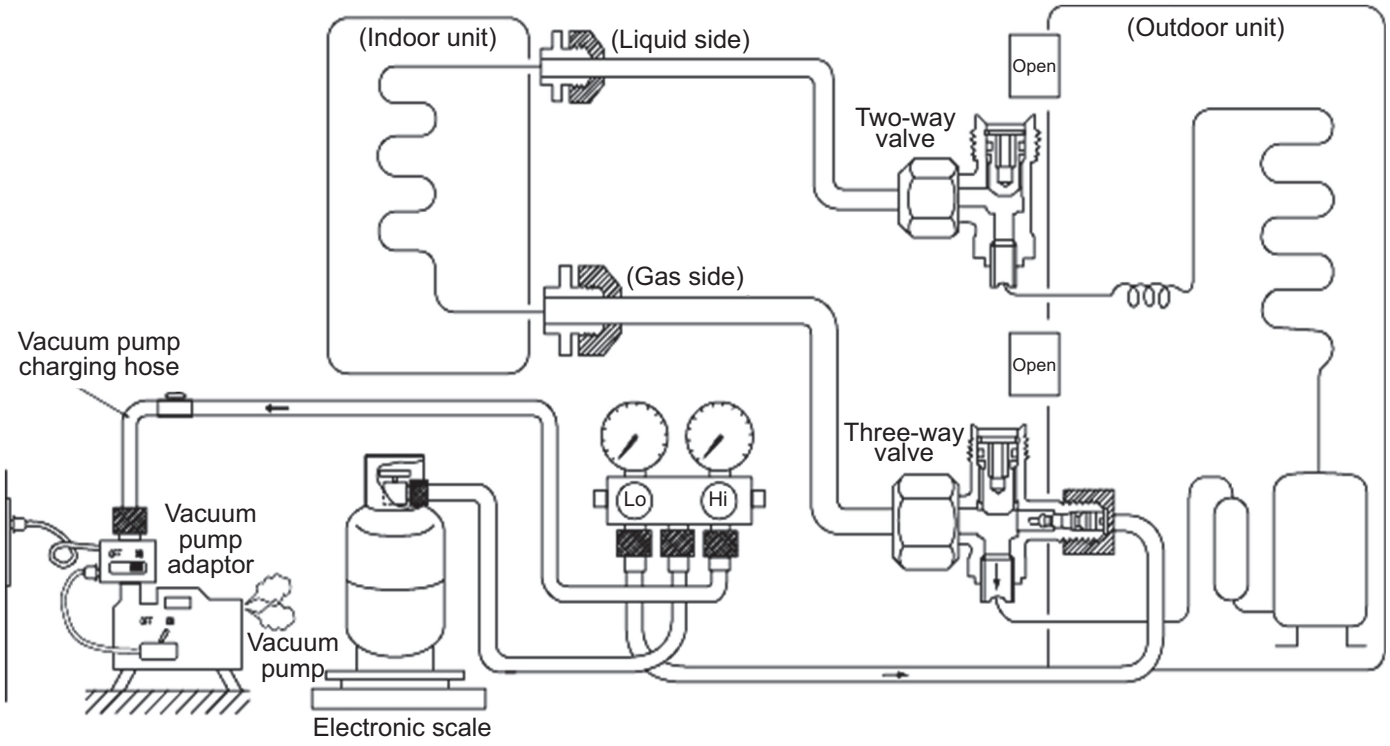
### 13.6.5 Inter-changeability of refrigerant

Do not operate air conditioning units inserting wrong (or mixed) refrigerant (R22, R410A, R32). It may cause mal-function of the units, and at the same time, may cause serious incident such as rupture of the refrigerant cycle.

### 13.6.6 Re-insertion of refrigerant in service

When re-insertion is needed, follow the procedures to ensure the insertion of new refrigerant at correct amount.

1. Attach charging hose (blue) to the service port of the outdoor unit.
2. Attach charging hose (red) to the vacuum pump. Fully open the 2-way and 3-way valves.
3. Place the refrigerant cylinder on the digital scale for refrigerant charging and connect the charge hose (yellow) to the connection port of the vacuum pump and the digital scale. Leave the cylinder valve fully open.
4. Fully open the handles Lo and Hi of the manifold gauge, and switch on the vacuum pump, and then perform evacuation for at or more than one hour.
5. Confirm the compound gauge of -0.1 MPa (-76cmHg) and fully open the handles of Lo and Hi, and switch off the vacuum pump. Leave it for about 1 ~ 2 minutes and confirm the needle of the compound gauge does not turn back.
   Refer to the picture below to follow the procedures below.
6. Remove the charging hose (red) of the manifold gauge from the vacuum pump adopter.
7. After adjusting the digital scale to zero, open the cylinder valve and the valve Lo of the manifold gauge, and insert the refrigerant.
8. If it is not possible to insert the refrigerant at regulated amount at once, operate the cooling mode and gradually insert the refrigerant (recommended amount approx. 150 g / 1 time)
   *Do not insert much refrigerant at once.
9. Close the open/close valve and insert the refrigerant in the charging hose to the outdoor unit.
   *Perform this procedure during operating cooling operation. Close the stem of the two-way valve, and when the pressure of the manifold gauge becomes zero (0), quickly remove the charging hose (blue). Immediately open the 2-way valve, and stop the cooling operation.
10. Final checking • • • Confirm the 2-way and 3-way valves are fully open.
    Attach the caps of the service port and control valve, and then check the gas leakage around the caps.



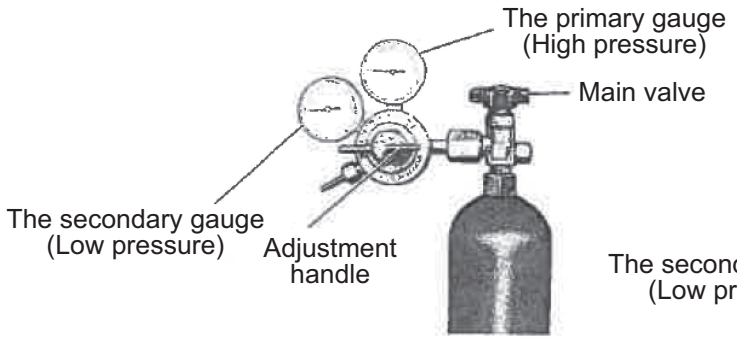
## *13.7 Repairing of refrigerant cycle / Brazing point*

### 13.7.1 Preparation for repairing of refrigerant cycle / brazing

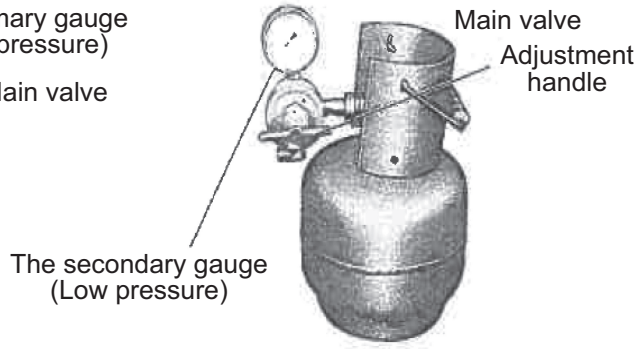
Brazing which is a technique needed for repairing refrigerant cycle requires advanced technique and experience, and this brazing procedure can only be performed by the workers who completed “Gas Welding Skill Training” regulated by the Occupational Safety and Health Act, and went through the training programs of refrigerant operations. Dismantling and re-connecting (assembling) refrigerant system requires working space, and the space has to ensure good air flow and fire prevention (water bucket and fire extinguisher). Moreover, the worker has to ensure the wearing of goggles, grabs, safety shoes, and long sleeve shirts, and be aware of work safety and attempt to prevent secondary defect (quality assurance of products). For brazing the indoor / outdoor unit structural components (heat exchangers, compressors, expansion valves, four-way valve blocks), after the recovery of all refrigerant, confirm that no refrigerant remains in the system, and fully open the 2-way and 3-way valves. When the brazing is conducted outside, check and make sure no refrigerant is contained in the air (be careful with vaporized refrigerant). Furthermore, protect the compressor terminal with metal plates, and heat but use wet clothes to cool down (releasing the heat) the expansion valves, and four way valves (prevent destruction of parts). In brazing, it is important to pour the brazing material without melting the base metal based on capillary action principle. In case of holes and oxidizing caused by overheating, do not perform re-brazing or alteration but replace the parts.

### 13.7.2 Adjustment of vacuum pump pressure

1. Cylinder with adjustment handle
    1. Check and confirm the adjustment handle of the 1st pressure adjuster is loosen (anticlockwise). If cylinder valve is opened when the 1st gauge pressure adjust handle is closed, the 2nd gauge might get broken.



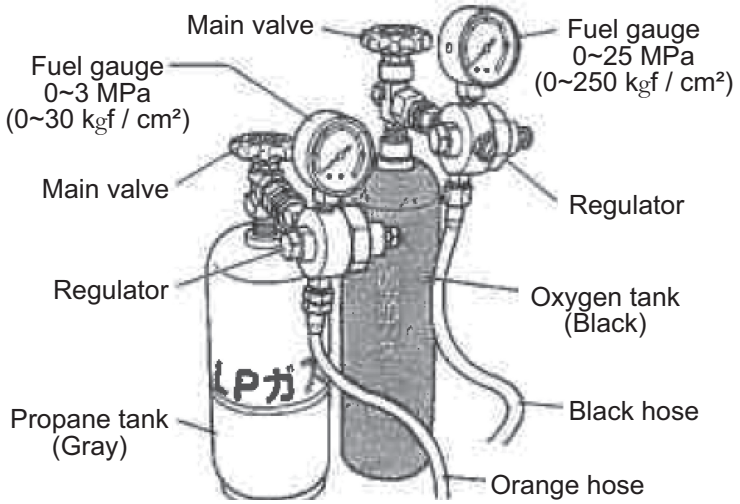
Oxygen regulator



Propane regulator

    2. Open the cylinder valve, and check the remaining amount with the first t side pressure gauge.
    3. Check the pressure of 2nd gauge and turn the adjustment handle to clock-wise direction to adjust the pressure.
        ◎ Oxygen 2nd side gauge pressure · · · · · · · · · · · 0.5 MPa (5.0 kgf / $cm^2$)
        ◎ Propane 2nd side gauge pressure · · · · · · · 0.05 MPa (0.5 kgf / $cm^2$)

2. Cylinder without adjustment valve
   2nd side gauge pressure is adjusted by the adjuster.
   Check the both side valves of the torch and open the cylinder valve to check the remaining refrigerant in the cylinder.
   Caution: Do not attach oil component on the connection port of the adjuster.
   Especially, use an oxygen cylinder adjuster which is no oil substance type. Do not dismantle or repair the adjuster and pressure gauge.



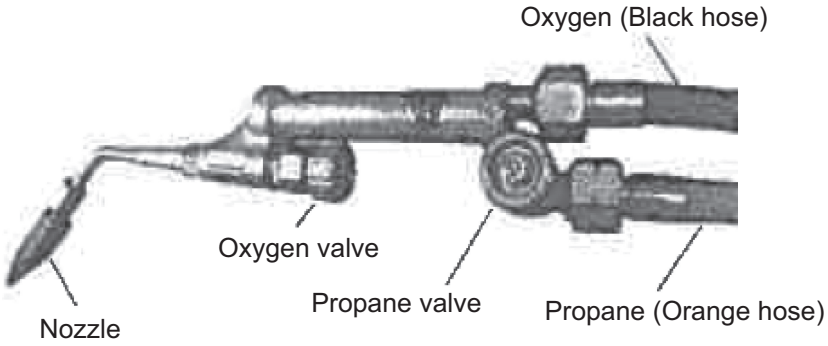

### 13.7.3 Checking of gas provision

Checking there is no fire around the torch, and then confirm the provision of gas.

1. Slightly open the “propane valve” of the torch, and make sure the gas comes out from the torch crater and then close the “propane valve”.
2. Slightly open the “oxygen valve” of the torch and make sure the gas comes out from the torch crater and then close the “oxygen valve”.
   Check there is no gas leakage around the hose connection.

### 13.7.4 Adjustment of flame

1. Slightly open the “propane valve” of the torch and lit with spark lighter.
   This moment, the flame is only by propane and the color is red.
2. Gradually open the “oxygen valve” of the torch to mix oxygen, and adjust the amount of propane and oxygen with the valve to make the flame suitable for brazing work.
   If the white core flame splits into two, the torch crater might be clogged. In this case, remove the crater from the torch and check.



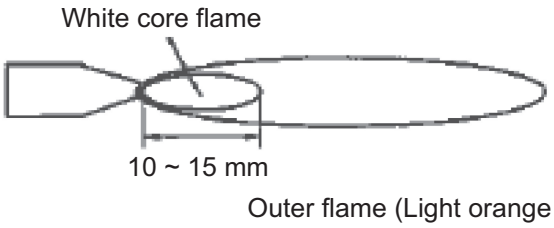
## 13.7.5 Types of flame

Types of flame change based on the proportion of propane and oxygen.

**[Neutral Flame]**

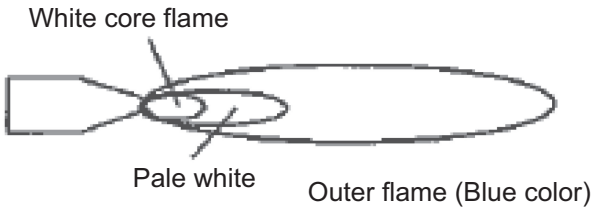
Perform brazing with this flame

(This is a flame when oxygen and propane are mixed at proper proportion, and has lesser effect on the brazed metals)



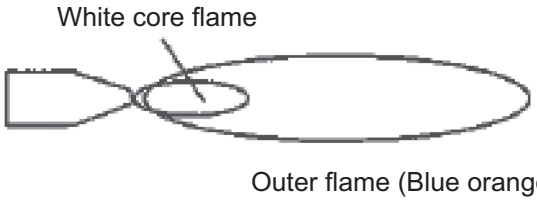

**[Carbide Flame]**

When propane is excessive, the flame has white color flame in between the white core flame and outer flame.
(This is due to the lack of oxygen and the proportion of unburned propane is excessive.
The black carbon created during the brazing work may contaminate the surface of the brazed metal).




**[Oxidizing Flame]**

Oxygen is more compared to the neutral flame. Although the flame size is small, this has the highest flame heat. However, due to the excessive oxygen contained in the flame, the brazing point gets oxidized. (This flame may cause holes, due to the high heat. The pipe may get melt）




## 13.7.6 Closing the flame

**[In case of short break]**

1. Close the “propane valve” of the torch.
2. Close the “oxygen valve” of the torch.

**[In case of finishing work]**

1. As above, close the flame following the procedure of “In case of short break”.
2. Completely close the valves of oxygen and propane cylinders.
3. Release the remaining gas inside the hose by opening the “oxygen valve” and “propane valve” of the torch.

Confirm the 1st and 2nd side gauge pressures of “oxygen” and “propane” cylinder pressure adopter are “zero”.

## 13.7.7 Selection of brazing material

Use BAg brazing material (silver solder) to increase the welding performance.

| Category | JIS Standard Number | Composition of ingredients (%) | | | | | | Temperature (°C) | | | Tensile strength (Reference) | | Characteristics and applications |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Ag | Cu | Zu | Cd | Ni | P | Solidus | Liquidus | Brazing temp | Kgf•cm$^2$ | Base material | |
| BAg | BAg • 1A | 49.0 ~ 51.0 | 14.5 ~ 16.5 | 14.5 ~ 18.5 | 17.0 ~ 19.0 | — | — | approx. 625 | approx. 635 | 635 ~ 760 | 45.5 | S20C | Liquidity is good at low temperature, it is preferable to a small junction of the gap in the universal form. |
| | BAg • 1 | 44.0 ~ 56.0 | 14.0 ~ 16.0 | 14.0 ~ 18.0 | 23.0 ~ 25.0 | — | — | approx. 605 | approx. 620 | 620 ~ 760 | 45.5 | S20C | It has similar performance to the BAg • 1A, and suitable for every base material except the light weight metal. |
| | BAg • 2 | 34.0 ~ 36.0 | 25.0 ~ 27.0 | 19.0 ~ 23.0 | 17.0 ~ 19.0 | — | — | approx. 605 | approx. 700 | 700 ~ 845 | 45.5 | S20C | It is a brazing filler metal in universal form, suitable for a slightly larger gap junction. |
| | BAg • 3 | 48.0 ~ 51.0 | 14.5 ~ 16.5 | 13.5 ~ 17.5 | 15.0 ~ 17.0 | 2.5 ~ 3.5 | — | approx. 630 | approx. 690 | 690 ~ 815 | 35 ~ 70 | SS ~ SUS | It has good corrosion resistance in stainless steel-based brazing, suitable for brazing tungsten carbide, aluminum bronze and copper. |
| BCuP | BCuP-2 | — | remain | — | — | — | 6.8 ~ 7.5 | approx. 710 | approx. 785 | 690 ~ 815 | 21 ~ 24.5 | Cu | Good liquidity, suitable for brazing copper tube. |
| | BCuP-3 | 4.8 ~ 5.2 | remain | — | — | — | 5.8 ~ 6.7 | approx. 645 | approx. 815 | 720 ~ 815 | 21 ~ 24.5 | Cu | Suitable for brazing when the joint spacing is not constant |
| | BCuP-5 | 14.5 ~ 15.5 | remain | — | — | — | 4.8 ~ 5.3 | approx. 645 | approx. 800 | 705 ~ 815 | 21 ~ 24.5 | Cu | When brazing of copper and copper, it is used without a flux, but not possible for brazing basic materials |

Caution
BCuP (phosphorus copper wax) is easy to react with sulfur, and makes a brittle compound water soluble, and causes gas leakage. In hot spring areas, use other brazing materials or paint the surface for protection.

## 13.7.8 Need of flux

**Use flux to protect the base materials.**

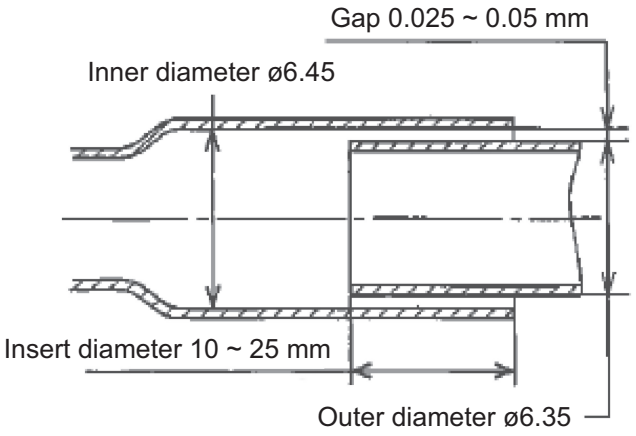
1. Remove impurity and oxide film on the metal base, and improve the flow of the brazing material.
2. Prevent oxidation of the metal surface in brazing.
3. Reduce the surface tension of the brazing material.

## 13.7.9 Need of nitrogen gas

In order to prevent oxidation in the pipe, perform the brazing operation in nitrogen gas flow.
Flow rate 0.05 m$^3$ / h, or pressure reducing valve at 0.02 MPa (0.2kgf / cm$^2$) below.

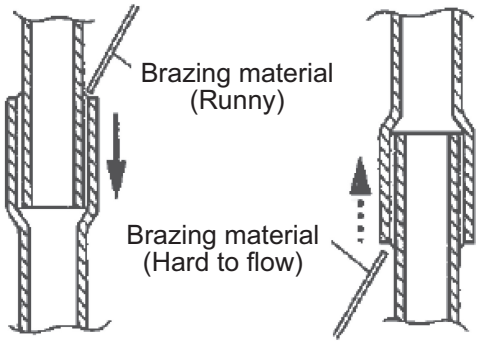
## 13.7.10 Checking of brazing (insert) points

1. No impurity on the brazing point
   If dirt or oil is attached on the brazing point, the brazing filler metal does not reach to junction, and it may cause poor welding.

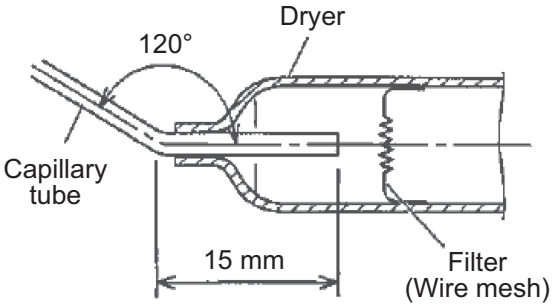



2. Adequate gap space in the brazing point
   The advantage of capillary current situation is used in brazing. If the gap space is too large, this phenomenon may not occur and it may cause poor welding because brazing filler metal does not flow to join the front part.

3. Appropriate size for insertion
   The guideline for pipe insertion dimensions is to three times the diameter of the base material, but you need to decide the insertion size in consideration of the clogging of the brazing material. Generally, for thin pipes, you need to increase the insert size, and for thick pipe vice versa.

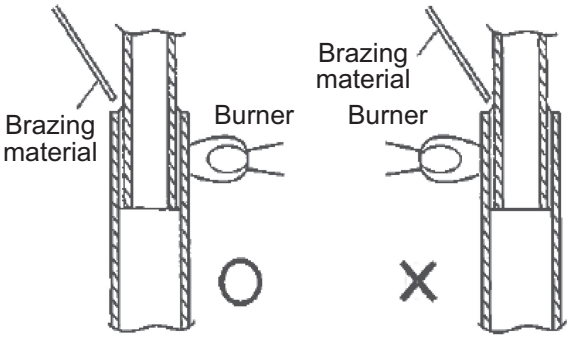



4. Brazing material to flow from top to bottom
   Brazing filler metal will easily flow to the connecting portion by capillary action.
   Further, by bending the brazing portion of [dryer side] of the capillary tube at 15 mm from the tube top to the angle of about 120°, you can prevent the damage of dryer inside and the clogging of brazing material caused by the excessive insertion of capillary tube.




## 13.7.11 Brazing and heating

1. Place the flame to a pipe which has more heat capacity in order to let the brazing material melt by the pipe heat. Heat the pipe up to the melting temperature of the brazing material, but when it is overheating, assess the temperature by pipe color in order not to melt the pipe.




The pre-heating is to heat the base material until the melting temperature, and requires certain training to distinguish the color of the heated base material in order not to melt the material.
The color and temperature of copper tube

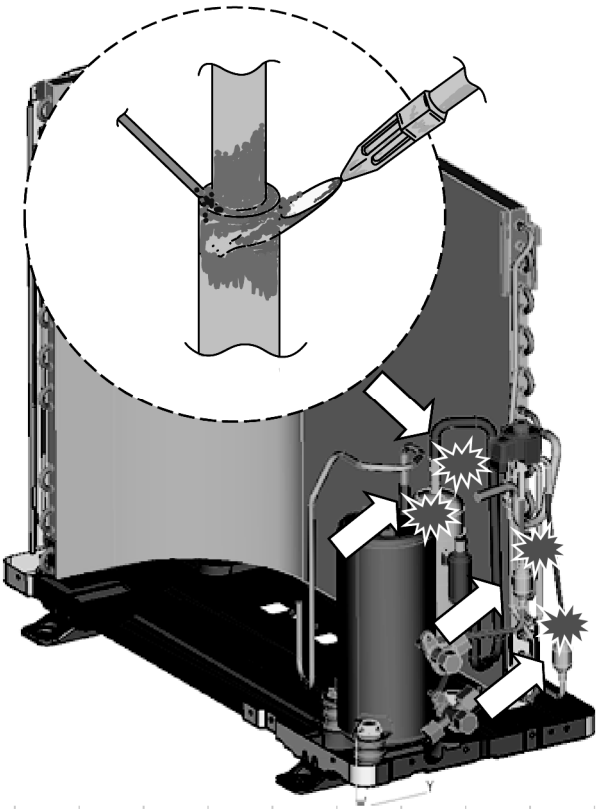
- Becoming red color • • • • • • • 480°C
- Dull red • • • • • • • 650°C
- Cherish red • • • • • • • 760°C
- Brightening cherish red • • • • • • • 870°C

(Reference)
Melting temperature of copper • • • • • • • Approx. 1083°C
Maximum temperature obtained in propane and oxygen • • • • • • • Approx. 1083°C
The important point is to heat the bonding part uniformly within a short period of time until reaching to the brazing temperature in the following manner.

2. Apply the flame on to the side with better heat transmission. If the pipe thickness is consistent, by heating like 30% iron and 70% copper, the copper pipe inside reaches to brazing temperature.
   Iron pipes have low heat transmission and only the part the flame is applied get high temperature, and this causes oxidization of the pipe. The flow of the brazing filler is affected negatively.

3. Apply the flame on to the side of larger heat capacity.
   When brazing a thin tubes such as capillary tube and dryer, etc., caution has to be taken to apply the flame to the dryer side (thick pipe side), in order to prevent burn out by the heat.

4. When brazing the compressor connection pipes (suction and discharge), remove the sound insulation plate and the fan, and place the compressor stand vertically (to prevent the leakage of compressor refrigerating machine oil), and apply the flame from the compressor body side.



## 13.7.12 Terminologies of brazing

Pin holes → Small holes are generated on the surface of the brazing metal.
Wet temperature → Liquidus temperature at which the brazing material starts flowing out by heating, generally it is the liquidus-line temperature.
Blow holes → Hollows made by gas in the brazing material of brazing portion (gas reservoirs).
Pits → As a result of blow holes, small dents generated on the outside surface of welding.
Voids → The blazing material does not reach completely to the brazing part. It cannot be identified from outside.

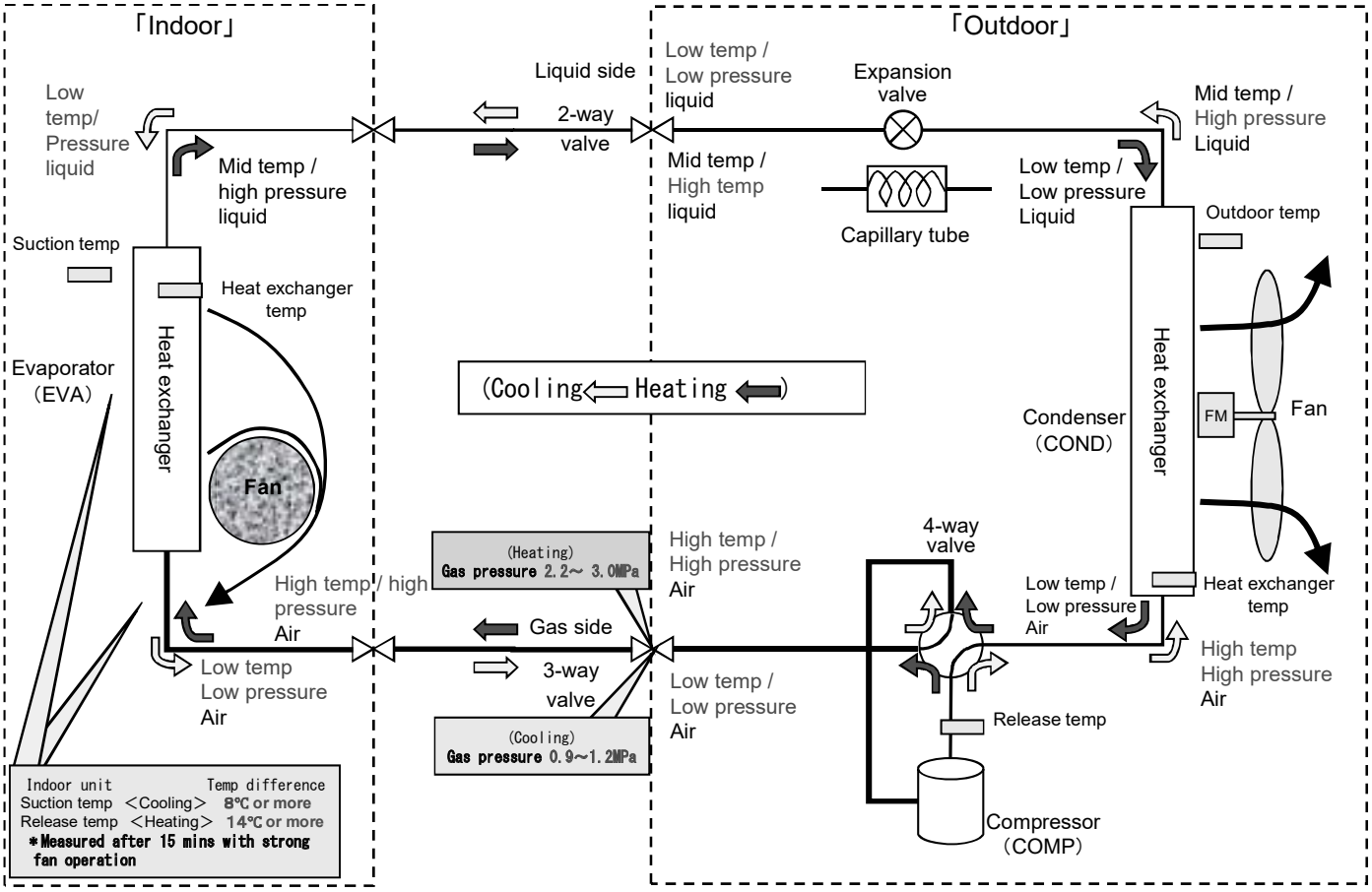
## *13.8 <Reference> Analysis method for no error code, no cooling / no warming*

### 13.8.1 Preparation for appropriate diagnosis

In order to obtain appropriate operation characteristics, minimum 15 minutes or more operation time [testing operation (rated operation)] is required.

1. Method of rated operation (rated operation)
   For the models which have two buttons of “emergency operation and forced cooling operation”, press forced cooling button once. For the models which have only emergency operation button, press the button once for 5 seconds and when hear “beep” sound, release the button. Then, cooling operation starts.
2. Checking the mal-functions of indoor / outdoor units
   1) Any obstacles against heat release and air suction? (short circuit)
      (Forget to remove the outdoor unit cover or fallen leaves blocking the outdoor unit)
   2) Are the indoor unit air filters clean? (obstructing heat suction)
   3) Is the setting temperature on the remote controller correct? (is the setting temperature set at lower/higher than the room temperature?)

### 13.8.2 Understanding and verification of refrigerant cycle



1. Measuring temperature
   1) Indoor unit suction temperature, release temperature, temperature difference, → Measure by thermometer
   2) 2-way valve pipe temperature in cooling mode is low temperature (benchmark：5 ~ 10°C), in heating mode is medium temperature (benchmark：25 ~ 35°C).
   3) 3-way valve pipe temperature in cooling mode is low temperature (benchmark：7 ~ 15°C) in heating mode is high temperature (benchmark：38 ~ 50°C).
2. Measuring electric current
   - Measuring electric current in operation → check by clump meter (refer to table of technical characteristic guideline)
3. Meauring pressure
   - Measuring gas pressure → check the pressure by manifold gauge (refer to table of technical characteristic guideline)
4. Any sound from the expansion valve?
   (when starting the operation and the outdoor unit is turned on, the expansion valve is re-set, check if there is any edged sound or clack sound)

### 13.8.3 Guidance for diagnosis of refrigerant cycle

| Comparison with normal operation | Cooling mode | |
|---|---|---|
| | High | Low |
| Refrigerant pressure | Excess insertion of refrigerant | Clogged capillary, expansion valve malfunction |
| | Heat releasing obstruction | Clog by moisture |
| | Dirty condenser, attachment of impurity | Lack of refrigerant gas |
| | Compressor malfunction | |
| Operation electric current | Excess insertion of refrigerant | Lack of refrigerant gas |
| | Heat releasing obstruction | Compressor malfunction |
| | Dirty condenser, impurity | Mixture of air |
| | | (Insufficient evacuation) |
| 2-way valve temperature | Excess insertion of refrigerant | Clogged capillary, expansion valve malfunction |
| | Compressor malfunction | Lack of refrigerant gas |
| 3-way valve temperature | Lack of refrigerant gas • Compressor malfunction | Excess insertion of refrigerant |
| | Clogged capillary, expansion valve malfunction | |

| Suction temperature & Release air temperature | Temperature difference at or less than 8°C in cooling operation • • • Causes | | |
|---|---|---|---|
| | • Heat releasing obstruction | • Dirty condenser | • Attachment of impurity |
| | • Lack of refrigerant gas | • Excess insertion of refrigerant | |
| | • Mixture of air | • Mixture of moisture | |
| | • Clogged capillary | • Expansion valve malfunction | • Compressor malfunction |

◎ Above all are based on the condition that the installation work is properly performed (no issues in indoor / outdoor pipe connections, etc.)

# 14. Operation and Control

## *14.1 Basic Function*

Inverter control, which equipped with a microcomputer in determining the most suitable operating mode as time passes, automatically adjusts output power for maximum comfort always. In order to achieve the suitable operating mode, the microcomputer maintains the set temperature by measuring the temperature of the environment and performing temperature shifting. The compressor at outdoor unit is operating following the frequency instructed by the microcomputer at indoor unit that judging the condition according to internal water setting temperature and water outlet temperature.

### 14.1.1 Internal Water Setting Temperature

Once the operation starts, control panel setting temperature will be taken as base value for temperature shifting processes. These shifting processes are depending on the Air-to-Water Heat pump settings and the operation environment. The final shifted value will be used as internal water setting temperature and it is updated continuously whenever the electrical power is supplied to the unit.

### 14.1.2 Heating Operation

#### *14.1.2.1 Thermostat Control*

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > 2°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) <-3°C.

#### *14.1.2.2 Thermostat Control (Outdoor Ambient Temperature)*

Stops provide heating to room side during high outdoor ambient condition.
Control content:

- Heating operation and water pump will turn OFF when outdoor ambient temperature > outdoor thermo off temperature + 3°C.
  (Outdoor thermo off set temperature is set by control panel. Thermo off set temperature is between 5°C ~ 35°C)
- Heating operation will resume when Outdoor ambient temperature < Outdoor thermo OFF set temperature + 1°C.

#### *14.1.2.3 Heat Mode Operation*

Operation of heat pump provide heating capacity to room side by hot water through heating panel, floor heating or fan coil unit.

1 3 ways valve control:
   - 3 ways valve switch and fix to room side.
2 Heat pump operates follow normal heating operation.
3 Back up heater operate follow normal operation.
4 2 ways valve control:
   - 2 ways valve opens.

### 14.1.3 Cooling Operation

#### *14.1.3.1 Thermostat control*

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > -1.5°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) >3°C.

#### *14.1.3.2 Cool Mode Operation*

1 3 ways valve control:
   - 3 ways valve switch and fix to room side.
2 Heat pump operates follow normal cooling operation.
3 Room heater DOES NOT operate during cool mode.
4 2 ways valve control:
   - 2 ways valve is closed.

## 14.1.4 Target Water Temperature Setting

### *14.1.4.1 Target Water Temperature Control of Standard System (Optional PCB not connected)*

There are 2 types of temperature control selection which are Compensation and Direct.

- Temperature control type selection by installer:
  1 Compensation : WLo, WHi, ODLo, ODHi can be set at installer menu.
  2 Direct : Direct Water Temperature Set
- Remote control setting by user:
  1 Compensation : Shift value ±5°C from the compensation curve
  2 Direct : Direct water temperature set change

*This setting is only able to set when the room sensor select as Water Temperature.
*Instead of water temperature, user will set target room temperature when room sensor select as Room Thermistor OR Internal Room Thermostat.

- Target water temperature is calculated as below condition.
  - Target water temperature = A (Base temperature) + B (shift temperature)

| A (Base Temperature) | Compensation | Direct |
|---|---|---|
| | Value from the curve + User shift value set | Direct value from user setting |

  - B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

| B (Shift Temp.) | B shift value depend on the room sensor selection at remocon as table below |
|---|---|
| **Sensor selection** | |
| Water temperature | B = 0 |
| External Room thermostat | B = 0 |
| Internal Room Thermostat & Room Thermistor | **Cool Mode:**<br>B = 0 ; when Zone OFF or Zone Room Thermo OFF<br>B = 1* (room set temp (R/C) - actual room temp)<br>Max/Min Regulation of B: (Max = 5 ; Min = -5) |
| | **Heat Mode:**<br>B = 0 ; when Zone OFF or Zone Room Thermo OFF<br>B = follow Heating PI control contents |

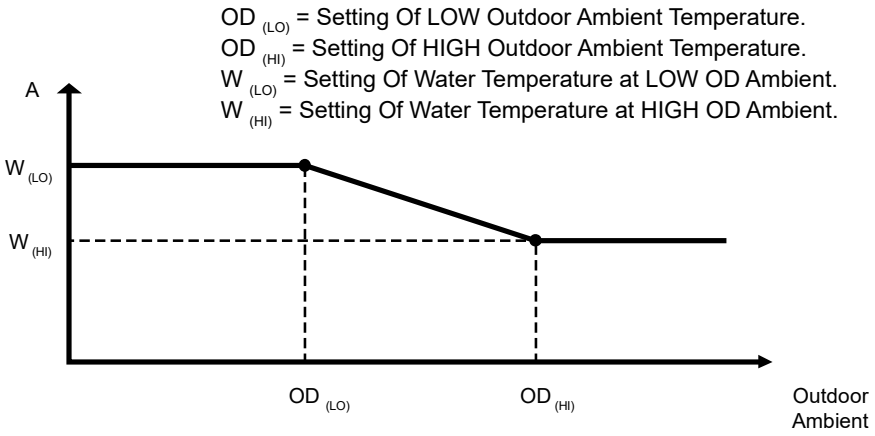
- Maximum/minimum regulation of Target Water Temperature.

| | Heating | Cooling |
|---|---|---|
| Max | 55°C (Below Ambient -20°C) *1<br>60°C (Ambient -15°C ~ 0°C or Above Ambient 25°C) *1<br>65°C (Ambient 5°C ~ 20°C) *1*2 | 20°C |
| Min | 20°C | 5°C |

Update as below:-
*1 Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.
*1 Between outdoor ambient 5°C and 0°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*1 Between outdoor ambient 20°C and 25°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*2 Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

Compensation Type: (Operation under Heat Mode and Cool Mode)

- The set temperature defines the parameters for the ambient (Outdoor temperature) dependent operation of the unit. The water temperature is determined automatically depending on the outdoor temperature. Default setting is the colder outdoor temperature will result in warmer water and vice versa. The user has the possibility to shift up and shift down the target water by remote control setting.

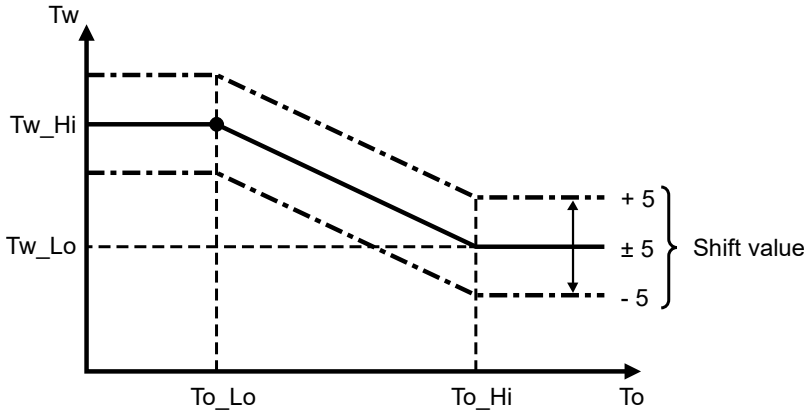



- Outdoor ambient is updated every 30 minutes when operation ON.
- Setting water outlet temperature always follow $W_{(LO)}$ or $W_{(HI)}$ whenever is higher if outdoor ambient sensor or indoor comminication error happen.

However, when powerful mode is requested by remote control during heating mode, the higher value of WLo or WHi will be used for A calculation.

* There are 2 compensation curves (for heating and cooling). During heating mode, the heating curve is used and during cooling mode, the cooling curve is use.

- Compensation curve set shift value:



### 14.1.5 Target Water Temperature at Extension System (Optional PCB is connected)

Target water temperature is calculated as below.

- Heat Mode:
  - When buffer tank selection is “YES:”
    Target water temperature = Target buffer tank temperature + [2°C]
  - When buffer tank selection is “NO”
    - If both zone 1 and zone 2 is active
      Target Water Temperature = Higher zone target water temperature of Zone 1 and Zone 2.
    - If only one zone is active
      Target Water Temperature = Zone target water temperature of active zone.
- Cool mode:
  - When buffer tank selection is “YES”
    - If both zone 1 and zone 2 is active
      Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2 + [-3°C]
    - If only one zone is active
      Target Water Temperature = Zone Target Water Temperature of active zone + [-3°C]
  - When buffer tank selection is “NO”
    - If both zone 1 and zone 2 active
    - Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2
    - o If only one zone is active
    - Target Water Temperature = Zone Target Water Temperature of active zone

### 14.1.6 Target Zone Water Temperature Control

Purpose:- To control zone mixing and zone pump according to the zone sensor temperature

#### *14.1.6.1 Target Zone 1 water temperature setting control*

- Start condition
  - Heating zone 1 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 1 is ON by remote control or Timer or Auto Mode.
- Cancel condition
  - Heating zone 1 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 1 is OFF by remote control or Timer or Auto mode.
- Target Zone 1 water temperature is calculated as below condition.
  - Target Zone 1 water temperature = A (Base temperature) + B (shift temperature)

| A (Base Temperature) | Compensation | Direct |
|---|---|---|
| | Value from the curve + User shift value set | Direct value from user setting |

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.

- B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

| B (Shift Temp.) | B shift value depend on the room sensor selection at remocon as table below |
|---|---|
| **Sensor selection** | |
| Water temperature | B = 0 |
| External Room thermostat | B = 0 |
| Internal Room Thermostat & Room Thermistor | **Cool Mode:**<br>B = 0 ; when Zone OFF or Zone Room Therma OFF<br>B = 1* (room set temp (R/C) - actual room temp)<br>Max/Min Regulation of B: (Max = 5 ; Min = -5) |
| | **Heat Mode:**<br>B = 0 ; when Zone OFF or Zone Room Thermo OFF<br>B = follow Heating Room Temperature PI control logic |
| Pool Function Selected | B = Delta value setting from remocon |

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool "B" value)

** Pool function also can be select at Zone 1 when optional PCB is connected and Zone 1 system is select.

- Maximum/minimum regulation of Target Water Temperature.

| | **Heating** | **Cooling** |
|---|---|---|
| Max | 55°C (Below Ambient -15°C) *<br>60°C (Below Ambient -10°C) * | 20°C |
| Min | 20°C | 5°C |

* Between outdoor ambient -10°C and -15°C, max target decreases gradually from 60°C to 55°C.

- Target Zone 1 Water Temperature (Heat mode) during SG ready control
  - If buffer tank selection is "NO" then following shift is carried out.
    - While digital input is "10" or "11" then,
      Final Target Zone 1 water temperature
      = Target Zone 1 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is "YES"
    - No shift of Target Zone 1 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to "Buffer tank temperature control"

- Target Zone 1 Water Temperature (Cool mode) during SG ready control
  - Final Target Zone 1 water temperature = Target Zone 1 water temperature + SG Cool Setting

### *14.1.6.2 Target Zone 2 water temperature setting control*

- Start condition
  - Heating zone 2 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 2 is ON by remote control or Timer or Auto Mode.

- Cancel condition
  - Heating zone 2 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 2 is OFF by remote control or Timer or Auto mode.

- Target Zone 2 water temperature is calculated as below condition.
  - Target Zone 2 water temperature = A (Base temperature) + B (shift temperature)

| **A (Base Temperature)** | **Compensation** | **Direct** |
|---|---|---|
| | Value from the curve + User shift value set | Direct value from user setting |

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.

- B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

| B (Shift Temp.) | B shift value depend on the room sensor selection at remocon as table below |
|---|---|
| **Sensor selection** | |
| Water temperature | B = 0 |
| External Room thermostat | B = 0 |
| Internal Room Thermostat & Room Thermistor | **Cool Mode:**<br>B = 0 ; when Zone OFF or Zone Room Thermo OFF<br>B = 1* (room set temp (R/C) - actual room temp)<br>Max/Min Regulation of B: (Max = 5 ; Min = -5) |
| | **Heat Mode:**<br>B = 0 ; when Zone OFF or Zone Room Thermo OFF<br>B = follow Heating Room Temperature PI control logic |
| Pool Function Selected | B = Delta value setting from remocon |

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool "B" value)

** Pool function also can be select at Zone 2 when optional PCB is connected and Zone 2 system is select.

- Maximum/minimum regulation of Target Water Temperature.

| | **Heating** | **Cooling** |
|---|---|---|
| Max | 55°C (Below Ambient -15°C) *<br>60°C (Below Ambient -10°C) * | 20°C |
| Min | 20°C | 5°C |

* Between outdoor ambient -10°C and -15°C, max target decreases gradually from 60°C to 55°C.

- Target Zone 2 Water Temperature (Heat mode only) during SG ready control
  - If buffer tank selection is "NO" then following shift is carried out.
    - While digital input is "10" or "11" then,
      Final Target Zone 2 water temperature
      = Target Zone 1 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is "YES"
    - No shift of Target Zone 2 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to "Buffer tank temperature control"

### *14.1.6.3 Zone Temperature Control Contents*

- During Standard System (Optional PCB not connected)
  - Only 1 zone temperature control is available
  - This zone room temperature is control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temperature)
  - Target Zone Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor. Target Water Temperature will set same as Target Zone Water Temperature
  - Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
  - Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- During Extension System (Optional PCB connected)

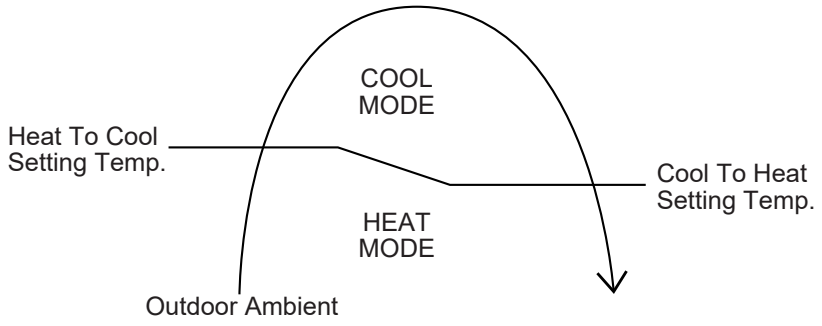
- Buffer Tank connection select "NO" &
  - One zone system is select
    - This zone room temperature control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temp.)
    - Target Zone Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor.
    - Target Water Temperature will set same as Target Zone Water Temperature
    - Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

  - 2 zone system select
    - Each zone room temperature is control by each sensor which select from either one of the 4 room sensor
    - Target Zone 1 & 2 Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor.
      * Zone Mixing Valve & Zone pump will operate to achieve Target Zone Water Temperature which refer to zone sensor.
      * Zone Sensor will detect if zone sensor is open or short.
    - Target Water Temperature will set same as the active & higher zone water temperature setting. (When cooling mode, lower zone water temp setting)
    - Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat).

- Buffer Tank Connection select "YES" &
  - 1 zone system or 2 zone system select
    - Each zone control by each sensor which select from either one of the 4 room sensor
    - Target Zone 1 & 2 Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor. Each zone have their own Target Zone Water Temperature.
      * Zone Mixing Valve & Zone pump will operate to achieve each Target Zone Water Temperature which refer to zone sensor
      * Zone Sensor will detect if zone sensor is open or short.
    - Target Buffer Tank Temperature will be set as active & higher zone water temperature setting + Buffer Delta T. (Heating) Target Buffer Tank Temperature will be set as active & lower zone water temperature setting (Cooling).
    - Target Water Temperature will set as Target Buffer Tank Temperature + [2°C] (Heating) Target Water Temperature will set as Lower or Active Target Zone Water Temp. + [-3°C] (Cooling)
    - Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM thermo OFF
      - Heat mode: ROOM thermo OFF (Buffer Tank Temperature > Target Buffer Tank + [0°C]
      - Cool mode: ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat)

### 14.1.7 Auto Mode Operation

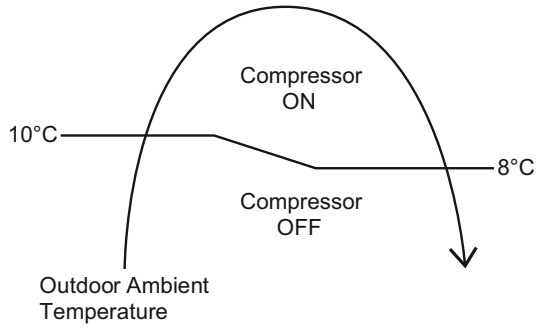


- Control details:
  - To enable the unit to operate either heat or cool mode automatically, heat to cool set temperature and cool to heat set temperature can be set by control panel.
  - Automatic operation is judged based on control panel setting temperature and outdoor ambient temperature.

  * Minimum setting of heat to cool set temperature is 1°C higher than cool to heat set temperature.

- Judgement control:
  - If outdoor ambient temperature < Heat to Cool Set Temperature, unit will operate in Heat Mode or else the unit will operate in Cool Mode.
  - If current operation is Cool mode, outdoor ambient temperature > Cool to Heat Temperature, unit will maintain Cool mode operation or else the unit will operate Heat mode.
  - If current operation is Heat mode, outdoor ambient temperature > Heat to Cool Temperature, unit will maintain Heat mode operation or else the unit will operate Cool mode.
  - Every 60 minutes the outdoor ambient temperature is judged.
  - When Auto + Tank mode is selected, operation mode switching is judged by both outdoor ambient temperature and indoor air temperature.

### 14.1.8 Auto Cooling Mode Operation Limit




- Auto Mode Cooling Only operation will start once the outdoor ambient temperature reaches 10°C and compressor will continue to run until the outdoor ambient temperature drops to 8°C.
- Due to this limitation,
  If Heat to Cool temperature is set lower than 10°C, the compressor will not operates until the outdoor ambient temperature reaches 10°C or higher.

### 14.1.9 Tank Mode Operation

- 3 ways valve direction
  - 3 ways valve switch to tank side during Tank Thermo ON condition. Switch 3 ways valve to room side when tank achieve Tank Thermo OFF temperature.

- Tank Thermo ON/OFF Characteristic
  - Tank Thermo OFF

  Case 1: <u>Internal Tank Heater is select and Tank Heater ON</u>
  - Tank temperature > Tank Set Temperature continuously for 15 seconds.
  - Water outlet >75°C

  Case 2: <u>Tank Heater OFF OR External Heater is select</u>
  - When heat pump OFF due to water thermos & Tank temperature > Tank water set temperature for continuously 20 seconds. **OR**
  - Tank temperature > Tank set temperature + 1°C for continuously 20 seconds.

- Tank Thermo ON

Case 1: Internal Tank Heater is select and Tank Heater ON
- Tank temperature < Tank set temperature + R/C (Tank re-heat temperature)

Case 2: Tank Heater select OFF OR External Heater is select
- Tank temperature < Tank water set temperature + R/C (Tank re-heat temperature)

* When tank thermo ON, water pump will ON for 3 minutes then only heat pump turn ON.
* Tank water set temperature = tank set temperature or tank limit temperature whichever lower.

| Outdoor ambient temperature | Tank limit temperature |
|---|---|
| OD < -17°C | 48°C |
| -17°C <= OD < 36°C | 52°C |
| 36°C <= OD | 50°C |

- 2 ways valve close
- Heat pump Thermostat Characteristic
  - Heat pump Water Outlet set temperature is set to below table:

| Outdoor ambient temperature | Heat pump water outlet temperature |
|---|---|
| OD < -17°C | 54°C |
| -17°C <= OD < -3°C | 56°C |
| -3°C <= OD | 59°C |

Characteristic of heat pump thermos ON/OFF under tank mode condition:

Water Outlet Thermo Condition
- Heat pump thermos OFF temperature:
  1. Heat pump thermo OFF temperature = Target Water outlet temperature + (2°C)
  2. Water outlet temperature > heat pump thermo OFF temperature for continuously 90 seconds, heat pump OFF but water pump continue ON.

- Heat pump thermo ON temperature:
  1. Heat pump thermo ON temperature = water inlet during thermo OFF time + [-3°C]
  2. When water outlet temperature < heat pump thermo ON temperature, heat pump ON.

Water inlet thermo protection condition
- Heat pump thermo OFF temperature:
  1. Water inlet temperature > [60°C] for continuously 30 seconds, heat pump OFF, water pump continue ON.

- Heat pump thermo ON temperature:
  1. Heat pump thermos ON temp = water inlet temperature < [60°C].

Thermo ON/OFF for Heat Pump in Tank Operation:

When tank temperature achieve heat pump OFF condition, refer below condition:
Condition 1: When Internal Tank Heater is select and Tank Heater ON
- Heat pump will turn OFF, water pump continue ON and room heater will continue ON if tank temperature below tank heater thermo ON condition. 3 ways valve will only switch to room side after tank temperature reach tank heater thermo OFF condition.

Condition 2: When Tank Heater select OFF **OR** when External Tank Heater is select and Tank heater ON
- If tank temperature achieve tank thermo OFF, heat pump turn OFF, water pump turn OFF, room heater OFF and 3 ways valve switch to room side.

When tank temperature achieve heat pump ON condition, water pump ON, heat pump ON and room heater turn OFF.

Heat pump OFF condition at Tank Mode

- Tank temperature > tank water set temperature continuously for 20 seconds after heat pump thermos OFF due to water thermo. (Heat pump turn OFF but water pump continue ON and room heater turn ON to achieve tank set temperature) **OR**
- Tank temperature > tank set temperature + [1°C] for continuously 20 seconds. (Heat pump OFF, water pump OFF, room heater OFF and 3 ways valve switch to room side)

Heat pump ON condition at Tank Mode

- Tank temperature < tank water set temperature + R/C setting (Tank re-heat temp)
  (Water pump turn ON OR continue ON, heat pump ON and 3 ways valve switch to tank side or maintain at tank side)

Internal heater control

- Internal heater only operates to tank side if the Internal Tank Heater is select, Tank heater ON and backup heater is enable.

Internal heater turn ON condition:

- Tank temperature < tank set temperature **AND**
- Heat pump thermos OFF **AND**
- 20 minutes from previous heater off **AND**
- Internal tank heater selects USE from control panel.

Internal heater turn OFF condition:

- Tank temperature > tank set temperature for continuously 15 seconds **OR**
- Heat pump thermo ON **OR**
- Mode change or operation is off by control panel.

External Heater control

- External heater only operate to tank side if tank heater ON & External Heater select.

### 14.1.10 Heat + Tank Mode Operation

1. 3 ways valve control:
   - 3 ways valve switch to room side during room heat-up interval and switch to tank side during tank heat-up interval. Both modes will switch alternately. Tank mode is the initial running mode of Heat + Tank mode.

2. Heat pump operation control:
   - During room heat-up interval
     - Follow normal heating operation.
       Switching to tank side depends to below cases:
       Case 1:
       [Previous switch from tank interval to room interval due to thermo OFF]
     - Switch to tank heat-up interval when Tank temp < Tank thermos ON temp (Room heat-up interval ends)
       Case 2:
       [If heating operation at room side is less than 30 minutes and switch to tank side 3 times consecutively]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
       Case 3:
       [Previous switch from tank interval to room interval due to tank interval timer is complete]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.

- During Tank heat-up interval
  - Tank interval is the first mode running when heat + tank mode is select.
  - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
  - Heat pump operates according to normal tank mode operation.

3 Room heater control:
- During heating heat-up interval
  - Follow normal room heater control operation.

4 Tank heater control:
- During heating heat-up interval
  - If External tank heater is select.
    - External heater ON/OFF according to external tank heater operation control.
  - If internal tank heater is select.
    - Internal tank heater will not function under heating heat-up interval.
- During tank heat-up interval
  - If External tank heater is select.
    - Once heating heat-up interval switched to tank heat-up interval, tank heater is turn OFF and tank heater delay timer will start counting.
    - Tank heater will turn ON after tank heater delay timer is fulfill and tank temperature is lower than tank set temperature.
    - Tank heater delay timer is clear when switching to heating heat-up interval.
  - If internal tank heater is select.
    - Internal tank heater will turn ON after heat punp thermo off to boil tank temperature to tank set temperature.

- 2 ways valve control is open

- Indoor water pump control:
  - Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

### 14.1.11 Cool + Tank Mode Operation

1 3 ways valve control:
- 3 ways valve switch to room side during room cooling interval and switch to tank side during tank heat-up interval. Both mode will switch alternately. Tank mode is the initial mode of cool + tank mode.

2 Heat pump operation control:
- During room heat-up interval
  - Follow normal cooling operation.
    Switching to tank side depends to below cases:
    Case 1:
    [Previous switch from tank interval to room interval due to thermo OFF]
  - Switch to tank heat-up interval when Tank temperature < Tank Thermo ON temperature (Room interval will ends)
    Case 2:
    [If cooling operation at room side is less than 30 minutes and switch to tank side for 3 times consecutively]
  - Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
    Case 3:
    [Previous switch from tank interval to room interval is due to tank interval timer is complete]
  - Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.

    *Tank Thermo ON temperature:

| | |
|---|---|
| Internal Tank Heater select USE | <Tank set temperature + R/C setting (Tank re-heat temperature) |
| Others | <Tank water set temperature + R/C setting (Tank re-heat temperature) |

- During tank heat-up interval
  - Tank interval is the first mode running when the cool + tank mode is select.
  - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
  - Heat pump operates according to normal tank mode operation.

3 Room heater control:
- During room cooling interval
  - Room heater is OFF and not operates.

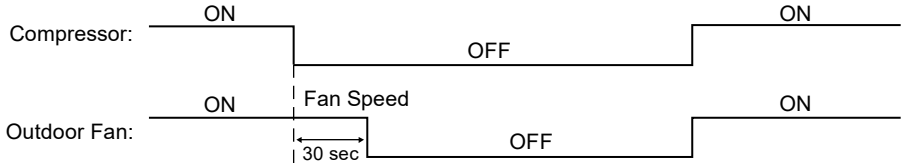
4 Tank heater control:
- During room cooling interval
  - If External tank heater is select.
    - External heater ON/OFF according to external tank heater operation control.
  - If internal tank heater is select and tank heater ON
    - Internal tank heater will not function under room cooling interval.
- During tank heat-up interval
  - If External tank heater is select.
    - Once room cooling interval switch to tank heat-up interval, tank heater will turn OFF and tank heater delay timer will start counting.
    - Tank heater turns ON after tank heater delay timer is fulfill and tank temperature is lower than tank set temperature.
    - Tank heater delay timer is clear when switch to room cooling interval.
  - If internal tank heater is select and tank heater ON.
    - Internal tank heater will turn ON after heat pump thermos off to boil tank temperature to tank set temperature.

5 2 ways vale is close.

6 Indoor water pump control:
- Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

### 14.1.12 Outdoor Fan Motor Operation

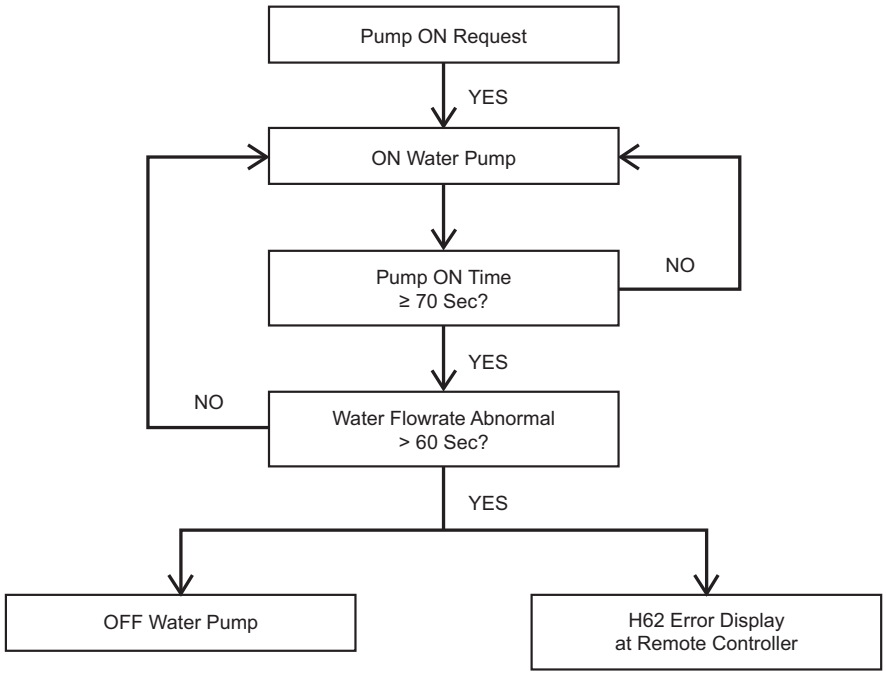
Outdoor fan motor is adjusted according to operation condition. It starts when compressor starts operation and it stops 30 seconds after compressor stops operation.



## *14.2 Water Pump*

### 14.2.1 Water Pump Control

- Once the indoor unit is ON, the water pump will be ON immediately and no error judgment for 70 seconds. However, during this 70 seconds operation, if there is any abnormality cause at outdoor or malfunction, the compressor should be OFF immediately and restart delay after 3 minutes.
- The system will start checking on the water flow level after operation start for 70 seconds. If water flow level is detected low continuously 60 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H62 error occurs).
- When error happens, the power has to be reset to clear the error.
- If there is no error indication, the water pump shall be continuously running.
- The water pump will remain ON when compressor OFF due to thermostat OFF setting is reached.
- Water pump will OFF when room thermo **OR** tank thermo **OR** buffer tank thermo OFF.
- Water pump will delay 15 seconds to turn OFF when request to OFF except during anti-freeze deice activate or air purge mode.



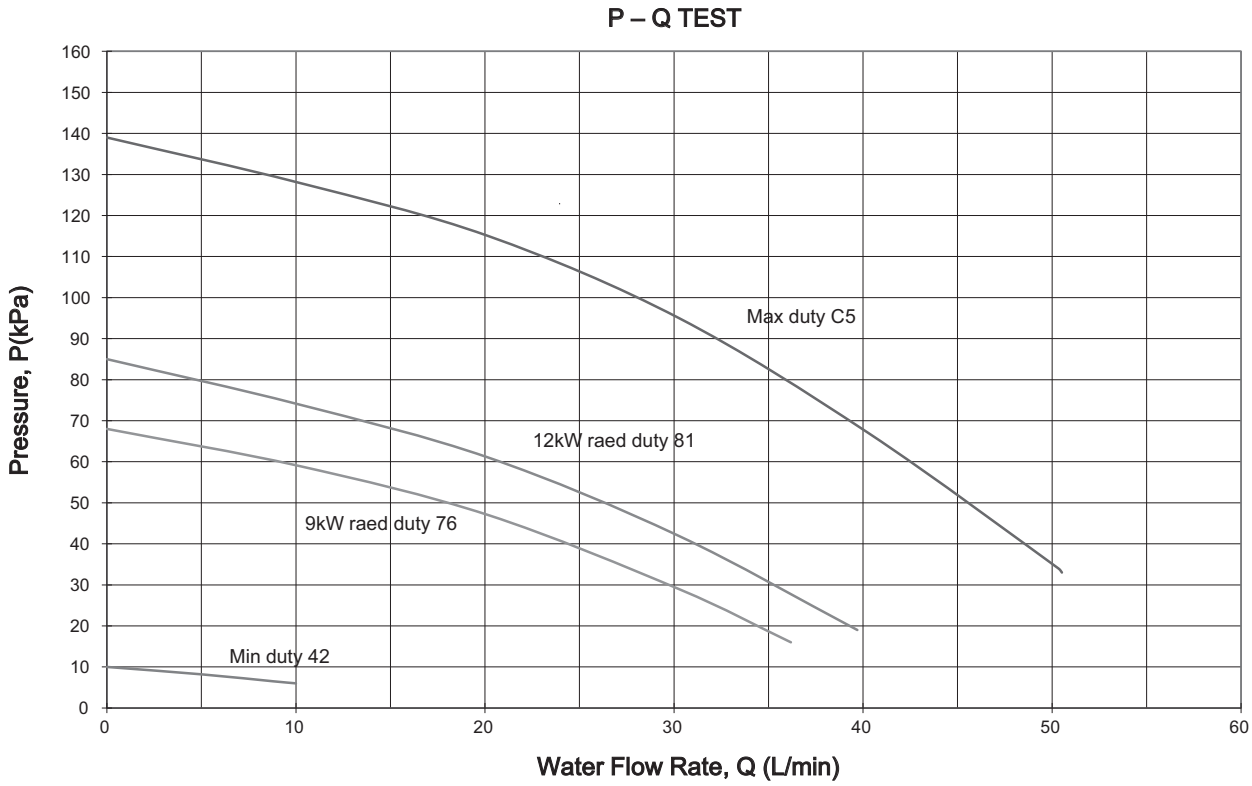

**Maximum pump speed setting on remote control**

- When Pump flowrate setting is ΔT
  Standard pump speed is automatically controlled to get the designed water temperature different between water inlet and outlet (ΔT). The maximum pump speed setting limitation can be adjusted by the installer according to water circuit pressure drop condition.
- When pump flowrate setting is Max. Duty
  Indoor water pump speed will operate at the maximum pump speed setting at room side operation. The maximum pump speed setting can be adjusted by the installer according to water circuit pressure drop condition.

However, the following sequences do not follow maximum pump duty setting by remote control.

- Pump down mode
- Air purge mode
- Normal deice

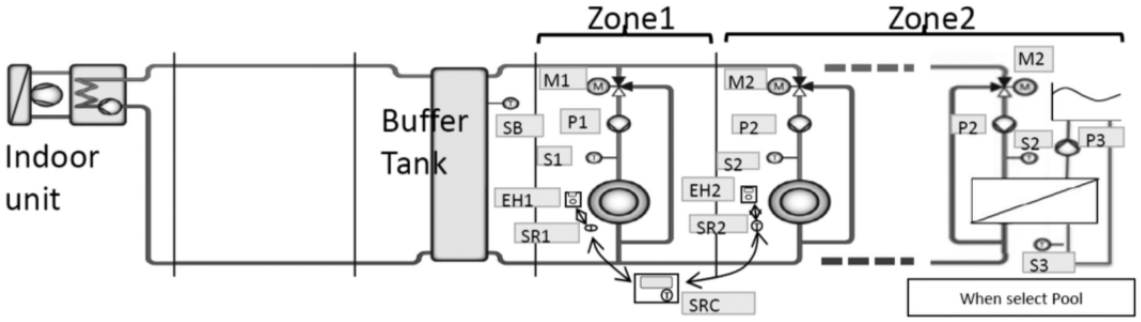
1) P-Q graph for different pump HEX duty




## 14.2.2 Zone Water Pump Control

Purpose:

- Water pump install at each zone to circulate the water inside each zone during buffer tank connection selected “YES” or 2 zone systems.




Content:

- AC type water pump install for this zone water pump control. When optional PCB connected, 230V output will drive this zone pump.
- There are three pump can be connected through Optional PCB. (Zone 1 Pump, Zone 2 Pump, & Pool Pump)

  * Zone 1 pump [P1] use to circulate zone 1 water circuit & Zone 1 mixing valve [M1] adjust to control the Zone 1 target water temperature.

  * Zone 2 pump [P2] use to circulate zone 2 water circuit & Zone 2 mixing valve [M2] adjust to control the Zone 2 target water temperature.

  * When Pool Function select as Zone 2 circuit, [P2] use to circulate water to heat exchanger which use to transfer heat to pool water.

  * Pool pump [P3] circulates the pool water through the heat exchanger to get warm water.

- Zone 1 and Zone 2 water pump start condition:
  - Zone room request ON
    (eg. Zone 1 thermo ON, only zone 1 pump will turn ON)
- Zone 1 and Zone 2 water pump stop condition:
  - Zone room request OFF
- Pool water pump start condition:
  - Pool Zone request ON **AND**
  - Pool function is selected
- Pool water pump stop condition
  - Pool zone: Zone room request OFF **OR**
  - Pool function is cancel

  * Zone 1 & Zone 2 water pump need to turn OFF when antifreeze deice pump stop control activate and turn ON back after the antifreeze deice pump stop control end under setting of "NO" buffer tank connection.

**Zone Pump Prohibit ON control:**
- Start condition:
  Zone 1 water temperature ≥ 75°C continuously for 5 minutes
  *stop zone water pump operates if the zone water fulfilled.
- Cancel condition:
  After 30 minutes from start condition fulfilled.
  *zone water pump operates according to normal condition.

**Zone Pump Control during Anti-Freeze**
Zone pump control during Zone Anti-Freeze Control:
- When Zone Anti-Freeze Flag=1, Zone Pump Turn ON.
- When Zone Anti-Freeze Flag=0, Zone Pump Turn OFF.

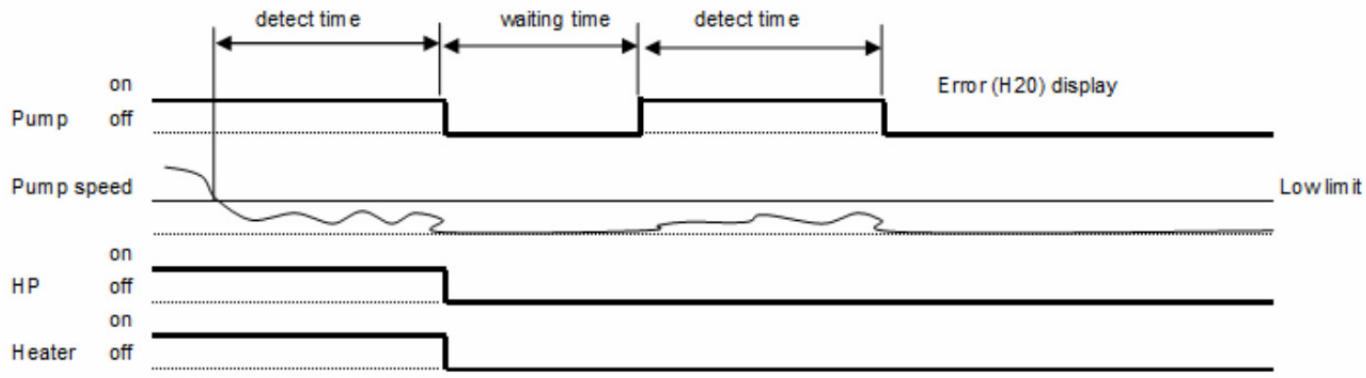
Zone pump control during Indoor Anti-Freeze Control:
- Zone pump only ON/OFF if the Extension PCB connected and Buffer Tank select "NO" condition

  When Indoor Anti-Freeze flag=1, Zone Pump Turn ON
  When Indoor Anti-Freeze flag=0, Zone Pump Turn OFF

* Pool Water Pump will not affected by both Indoor anti-freeze control or zone anti-freeze control.

### 14.2.3 Water Pump Speed Feedback Error

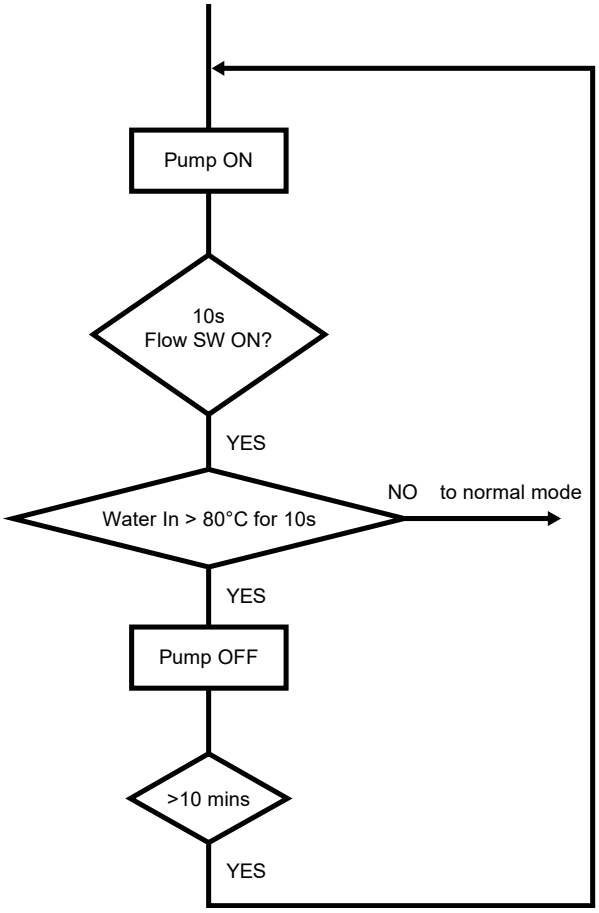
- Basically pump speed feedback is control by micon.
- When pump speed is below low limit or over high limit for a few seconds, micon detect pump error and system is stopped.
- Error detection conditions:
  - Detect abnormal water pump speed for continuous 10 secs.
  - Current pump speed < 700 rpm or
  - Current pump speed > 6000 rpm for 10 seconds.
- Control contents:
  - When error occurs, water pump, heating and heater is stopped for 30 seconds then restart again (Retry control).
  - When micon detect error again, system is stopped and error code [H20] is displayed at control panel.



## 14.3 Indoor Unit Safety

### 14.3.1 Indoor Unit Safety Control

1 When water pump is ON, the system will start checking flow switch status (ON/OFF).
2 If the flow switch ON for 10 seconds, the system will check on the water inlet temperature for 10 seconds.
If the water inlet temperature not exceeds 80°C, the water pump shall be continuously running with normal mode.
If the water inlet temperature exceeds 80°C for continuously 10 seconds, the water pump will be OFF immediately.
3 After water pump OFF for more than 10 minutes, it will be ON back and the indoor unit safety control checking is restarted.




## 14.4 Auto Restart Control

1 When the power supply is cut off during the operation of Air-to-Water Heatpump, the compressor will re-operate after power supply resumes.

## 14.5 Indication Panel

| LED | Operation |
|---|---|
| Color | Green |
| Light ON | Operation ON |
| Light OFF | Operation OFF |

Note:
- If Operation LED is blinking, there is an abnormality operation occurs.

## *14.6 Indoor Back-Up Heater Control*

### 14.6.1 Indoor Electric Heater Control

1 Normal Heating Mode
- Heater On condition:
  a. Heater switch is ON
  b. After Heatpump thermo ON for **[Remote Control Set Delay Time]** mins
  c. After water pump operate **[1]** min
  d. Outdoor air temperature < Outdoor set temperature for heater
  e. When water outlet temperature < Water set temperature + [**Remote Control Heater ON Setting]**
  f. **[20]** minutes since previous Backup heater Off
     * When heatpump cannot operate due to error happens during normal operation, heater will go into force mode automatic
     * Heater need to operate during deice operation

- Heater Stop Condition:
  a. When outdoor set temperature > outdoor set temperature + **[+2°C]** for continuous 15 secs OR
  b. When water out temp> water set temperature + **[Remote Control Heater OFF Setting]** for continuous 15 secs OR
  c. Heater switch is Off OR
  d. Heatpump thermo-off or OFF condition

2 Force Heater Mode
- Heater On condition:
  a. After water pump operate **[2]** mins
  b. When water outlet temperature < water set temperature + **[Remote Control Heater ON Setting]**
  c. **[20]** minutes since previous Backup heater Off

- Heater Stop condition
  a. Force mode off **OR**
  b. When water outlet temperature > water set temperature + **[Remote Control Heater OFF Setting]** for continuous 15 secs

* Do not operate heater at the following situation
1 Water outlet temperature sensor, and water inlet sensor abnormal
2 Flow switch abnormal
3 Circulation pump stop condition

### 14.6.2 Room Heater Operation during Deice

**Purpose:**
- To protect the indoor Heat Exchanger from ice forming and prevent heat exchanger plate breakage.

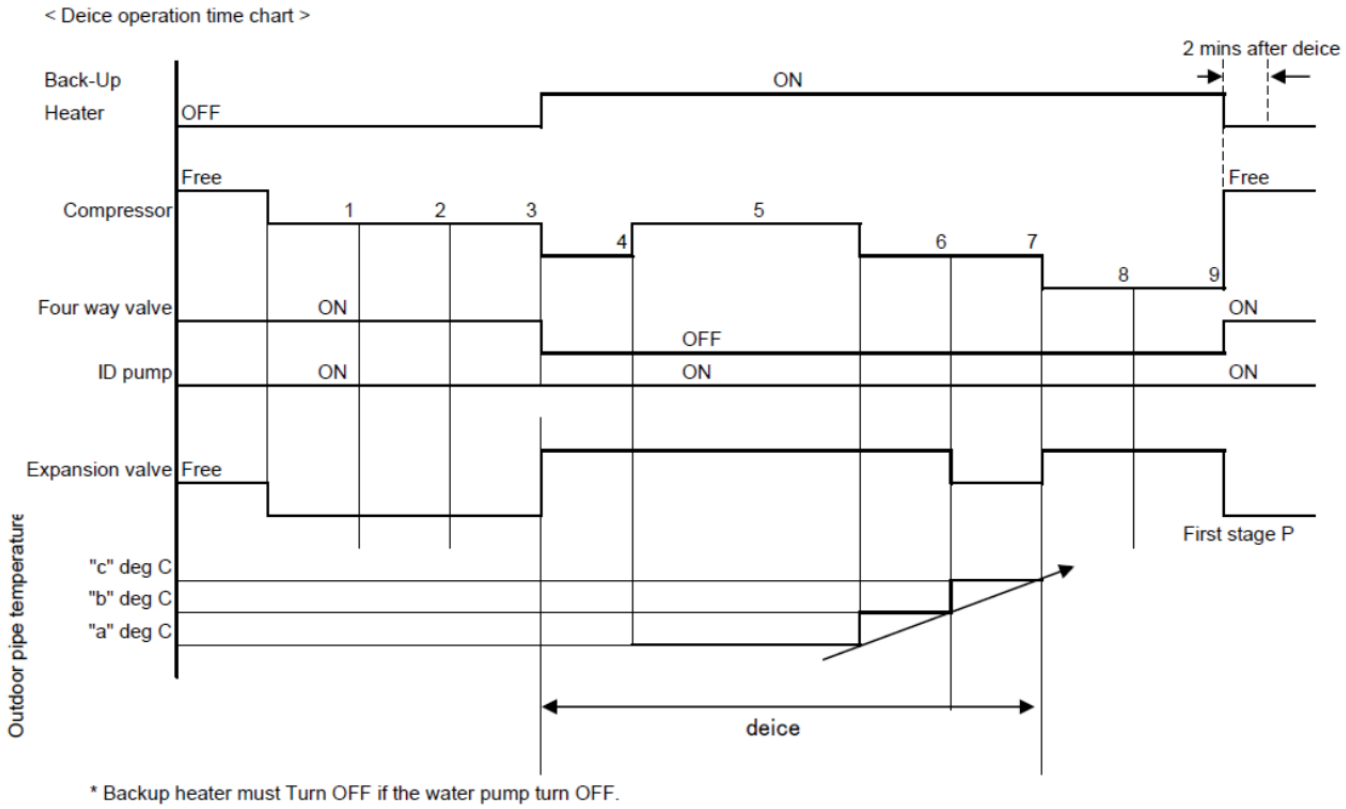
**Control content:**
- This Heater protection control will activate only if the backup heater is enable at custom setup by remote controller. Once fulfil the start condition, room heater will turn ON together (base on max heater capacity selection) and stop together if stop condition is fulfilled.
  * If the heater is request to turn ON OLP feedback will be detected.

**Starting conditions:**
- During normal deice operation 4~9
- Water outlet temperature < 7°C
- Water Flow rate < 20L/min
- Water outlet temperature < 10°C or
  Outdoor air temperature < -30°C or
  Water inlet temperature < 10°C

Heater operates when 1 ~2 fulfilled **OR** When 1 and 3 ~ 4 is fulfilled.
However, this control does not relate to Heater ON/OFF button on remote control.

**Stop condition:**
- When normal deice end or
- Water outlet temperature > 45°C

However, room heater keeps ON if indoor electric heater control activate.




## *14.7 Tank Heater Control*

### 14.7.1 Tank Heater Remote Control Setting

1 Tank heater selection:
External: - Booster Heater use to heat up tank when select external
Internal: - Backup Heater use to heat up tank when select internal
* When select External Tank Heater, Heater Delay ON Timer need to set. (range 20 min ~ 3 hrs)

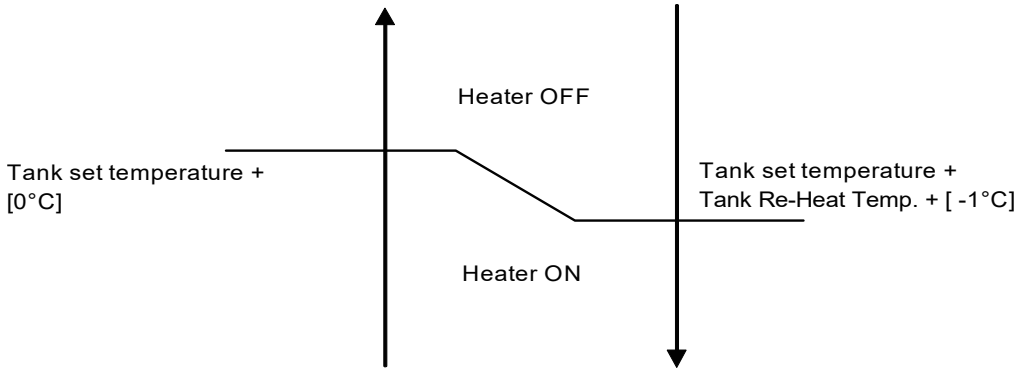
2 Tank Heater ON/OFF selection by user.

The remote control Tank set Temperature range will change according to the External and Internal Tank Heater use.

| Tank Heater Selection | Range |
|---|---|
| External | 40 ~ 75°C |
| Internal | 40 ~ 65°C |

### 14.7.2 External Heater Control at Tank Side

Heating operation condition:




- Tank heater Turn On condition:
  1 External Heater selected for Tank heater by remote controller.
  2 Tank Heater select ON by user.
  3 Tank mode operation ON (Tank mode, Heat + Tank, or Cool + Tank).
  4 After TANK HEATER DELAY TIMER fulfill during heat pump startup time in tank mode, or during switching from heating heat-up interval to tank heat + up interval in heat + tank mode.
  5 Tank temperature < tank set temperature +[Remote controller Set Tank Re-heat Temp] + [-1°C].
  6 20 minutes since previous heater off.

  * TANK HEATER DELAY TIMER is clear when tank heat-up interval end.

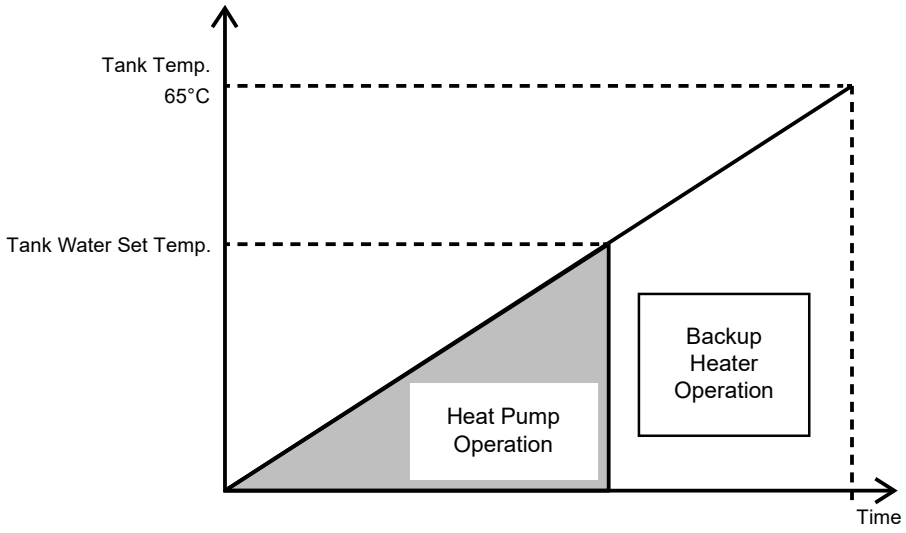
- Tank heater Turn Off condition:
  1 Tank temperature > tank set temperature + [0°C] for continuous 15 seconds.
  2 When BOOSTER HEATER DELAY TIMER start count after switch from heating heat-up interval to tank heat-up interval.
  3 Tank Heater select OFF by user.
  4 Tank Mode Operation OFF.

### 14.7.3 Internal Heater Control at Tank Mode

- Internal heater turn On condition:
  1 Internal Heater select for Tank heater by remote controller
  2 Tank Heater select ON by user.
  3 Backup Heater Enable
  4 Tank Temperature < Tank Set Temperature
  5 Heat Pump Thermo OFF
  6 20 minutes from previous heater off.

- Internal heater turn OFF condition:
  1 Tank Temperature > Tank Set Temperature + [0°C] for continuous 15 seconds. **OR**
  2 Heat Pump Thermo ON **OR**
  3 Mode Change or Operation OFF by remote controller **OR**
  4 Water outlet temperature >75°C.

  * Backup Heater Turn ON/OFF all together according to the selected heater capacity.




## *14.8 Base Pan Heater Control (Optional)*

- To enable the base pan heater function, control panel initial setting has to be manually adjusted by activating Base Pan Heater menu.
- There are 2 optional start condition can be selected, Type A or Type B.
- Control details:
  1 Type A: (Default Auto Mode)
  <u>Start conditions:</u>
  - When outdoor air temperature ≤ 3°C during heating and deice operation is ON.

  <u>Control contents:</u>
  - Base pan heater is ON during deice operation and continues ON for 10 minutes after deice operation ends.

  <u>Cancel condition:</u>
  - When outdoor temperature > 6°C after deice end or
  - When operation is not at heating mode or
  - Base pan heater ON timer count is completed.

  2 Type B: (ON Mode)
  <u>Start conditions:</u>
  - When outdoor air temperature is ≤ 5°C and operates in heating mode, base pan heater is ON.

  <u>Cancel conditions:</u>
  - When outdoor air temperature is > 7°C or
  - When operation is not at heating mode.

## *14.9 Heater Turn ON/OFF priority Control*

Purpose:

- To allow only one heater between room and tank heater to turn ON at the same time due to both heater power supplies is sharing the same ELCB. Prohibit two heater sources to turn ON at the same time to avoid overcurrent.

Start condition:

- Tank heater function select YES by remote controller

Control contents:

- When there is only one heater source (Room Heater or Tank Heater) request to turn ON, operate same as normal room heater and tank heater operation.
- When both backup heater and booster heater request to turn ON at same time, will turn ON only ONE heater source. Under normal condition, tank heater has priority to turn ON except below condition.
  1 Backup Heater Priority to Turn ON when
    - Request Backup heater turn ON at Hex Protection control during deice.
    - Request Backup Heater turn ON at Hex Protection Control during low water temperature.
    - Request Backup Heater turn ON at Anti-frost control.
    - Heat pump error and Force Mode operate.
- When switching from booster heater to backup heater OR backup heater to booster heater, need to delay 5 seconds in between Turn OFF one heater source and Turn ON another heater source.

## *14.10 Force Heater Mode*

Purpose of Force Heater Mode

- As a backup heat source when heat pump error. Force heater Mode control backup heater to heat up the room circuit, and turn ON back up heater or booster heater to boil up tank water base on the tank heater selection (internal or external)

Force Heater Control start condition:

- Force heater request ON by user OR auto turn ON by remote controller during error **AND**
  (Force Heater mode can be operate regardless of mode selection, remocon will send the latest mode selection and force bit to indoor. Indoor will judge to turn ON heater to room side if it is heat mode selected, and turn ON heater to heat up tank water base on tank heater selection)
- During Error Happen (exclude the error list below)

<u>Error List not allow Force Heater Operation</u>

| | | | |
|---|---|---|---|
| H12 | Capacity Mismatch | H76 | Indoor-Remote Controller Communication Error |
| H20 | Abnormal Water Pump | H95 | Abnormal Voltage Connection |
| H62 | Abnormal Water Flow | F37 | Abnormal Water Inlet sensor |
| H70 | Abnormal Back-up Heater OLP | F45 | Abnormal Water Outlet sensor |
| H74 | PCB Communication Error | | |
| [ When tank mode operate with external heater selected & tank heater select ON] | | | |
| H72 | Abnormal tank sensor | H91 | Abnormal tank heater OLP |

Force Heater Control Stop Condition:

- Force Heater request OFF **OR**
- Operation OFF request **OR**
- Mode change request **OR**
- Power reset **OR**
- Error of above list happens during force heater operation.

Control contents:

1) After fulfil start condition, indoor will operate the force heater operation according to below mode condition
   Heat mode Only : Turn ON backup heater to achieve room heat pump target water temperature.
   Heat + Tank mode : Turn ON backup heater to heat up room **OR** Turn ON Heater to Boil up tank water.
   Cool mode Only : Water pump and backup heater will OFF in force heater mode
   Cool + Tank mode : Operate pump and internal Heater OR External heater to Boil up tank water.
   Tank mode Only : Operate pump and internal Heater OR External heater to Boil up tank water.
   * For heat mode condition, backup heater will only turn ON if the backup heater is enable regardless of Room Heater Selection
   * For tank mode condition, If internal heater selected backup heater will turn ON to boil up tank water. If external heater selected, booster heater will turn ON to boil up tank water regardless of tank heater selection

Room Side: (Heat Mode)

1) When force heater mode start condition fulfilled, turn ON water pump and turn ON backup heater follow below control.
2) Operate the 3 ways valve at room side only and turn ON 2 ways valve as heat mode operation.
3) Turn ON the zone pump and mixing valve if system select 2 zone system or Buffer tank connect YES, control according to normal zone pump and mixing valve control.
4) When Force heater mode stop condition fulfilled, turn OFF heater as below condition and turn OFF water pump after pump delay time.

Backup Heater On Condition:

a) When Force Heater Control start condition fulfill **AND**
b) After water pump operate 2 minute **AND**
c) When water outlet temperature < water set temperature + [Remote Control Heater ON Setting] **AND**
d) 20 minutes since previous Backup heater Off **AND**
e) Backup Heater Enable

Backup Heater Stop condition

a) Force mode off **OR**
b) Operation off **OR**
c) When water outlet temperature > water set temperature + [Remote Control Heater OFF Setting] for continuous 15 secs **OR**
   * ON/OFF follow normal heater sequence.

Tank side (Tank mode)

1) When tank mode select and force heater bit received, turn ON backup heater (INTERNAL) or Booster Heater (External) depend on the tank heater selection.
2) If tank heater selection is INTERNAL, follow normal thermo judgement to switch 3 ways valve to tank side and room side.
3) If tank heater selection is EXTERNAL, only turn ON booster heater according to tank thermo.

**Tank Heater selection is INTERNAL:**

Backup Heater ON Condition:

a) After water pump operate 2 mins **AND**
b) When tank temperature < Tank set temperature - [Remocon Set Tank Re-heat Temp] **AND**
c) 20 minutes since previous backup heater OFF **AND**
d) Backup heater enable

Backup Heater OFF Condition:

a) Force mode OFF **OR**
b) When tank temperature > Tank set temperature for continuous 15 secs **OR**
c) Tank Operation OFF

**Tank Heater selection is EXTERNAL:**

Backup Heater ON Condition:

a) Force Heater mode ON **AND**
b) Tank temperature < tank set temperature + [Remocon Set Tank Re-heat Temp] - 1°C, **AND**
c) 20 minutes since previous heater off.

Backup Heater OFF Condition:

a) Tank temperature > tank set temperature for continuous 15 sec
b) Force mode OFF
c) Tank Mode Operation OFF
(During tank interval or tank mode condition, water pump and 3 ways valve will OFF)

## *14.11 Powerful Operation*

Powerful mode is use to increase the capacity of heat pump to achieve higher target temperature. Powerful mode is applicable when heat mode is operating.

**Remote control setting:**

On quick menu of remote control, there is 4 options of powerful mode can be select.

– OFF : Cancel powerful mode
– 30 minutes : Set powerful for 30 minutes
– 60 minutes : Set powerful for 60 minutes
– 90 minutes : Set powerful for 90 minutes

**Control contents:**

During the time set by remote control, powerful will activate according to 2 shift up controls. However, this function is applicable only for heating. Remote control will transmit the signal to indoor unit once this function is select then transmit OFF signal to indoor when the timer is complete. Indoor will transmit signal to outdoor for frequency control.

Indoor setting temperature shift

- If system is standard system (Optional PCB is not connected)
  - Target water temperature will shift up to Wlo or Whi whichever higher.
- If system is extension system (Optional PCB is connected)
  - Target water Zone 1 and Zone 2 temperature will shift up to Wlo or Whi whichever higher.

  * If “Direct Type” temperature control is select, this powerful shift up setting is not effective.

- **Start condition**
  - Powerful function is select by remote control.

- **End Condition**
  - OFF/ON button is pressed.
  - Powerful function is OFF by remote control.

## *14.12 Quiet Operation*

Quiet mode is use to reduce the noise of outdoor unit by reducing the frequency or fan speed.

**Quiet level**

There are 3 level (Level 1, Level 2, Level 3) to set by quick menu function on remote control.

**Control content**

Once the quiet function is select, the remote control will transmit the signal to indoor and outdoor unit.

**Start condition**

- Quiet mode is set on remote control.
- Quiet mode is request ON by weekly timer.

**Stop condition**

If any of below condition is achieve.

- OFF/ON button is pressed.
- Quiet mode is OFF by remote control.
- Quiet mode is request OFF by weekly timer.

## 14.13 Sterilization Mode

- Purpose:
  - To sterilize water tank by setting the required boiling temperature.

- Remote control setting
  - Days for sterilization function to start can be select.
  - Time of selected day to start sterilization function.
  - Boiling temperature (External heater is 55°C ~ 75°C, Internal heater is 55°C ~ 65°C)
  - Maximum operation time is 5 minutes to 1 hour.

- Start condition
  - Tank connection set to “YES” by remote control
  - Sterilization function selects “YES”.
  - Sterilization signal received from remote controller by timer.
  - Tank mode request ON.

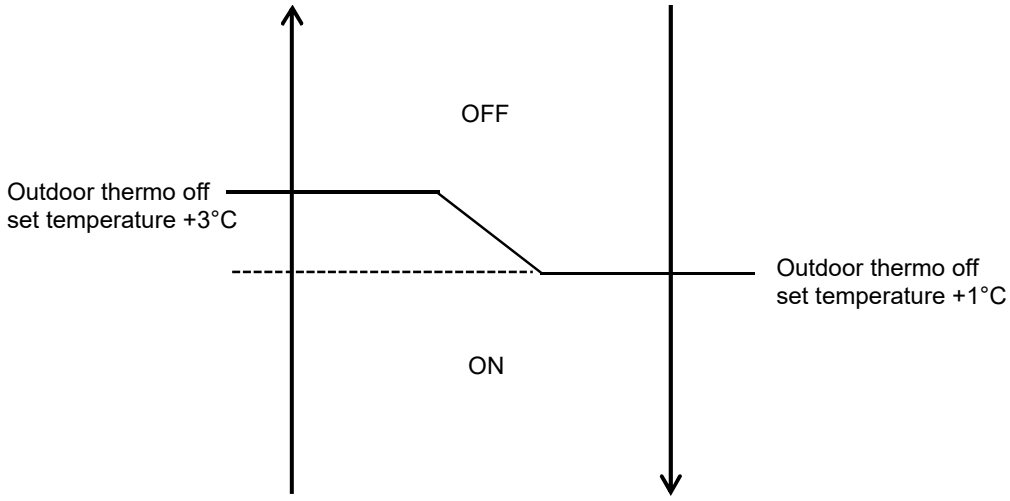
- Stop condition
  - When boiling timer is completed. Boiling timer (Remote control set maximum operation time) start counting once tank achieve boiling set temperature **OR**
  - After 8 hours of operation since sterilization start.
  - Tank mode request OFF.

- Control content:
  - During sterilization function activation time, target tank set temperature will internally change to boiling set temperature.
  - During sterilization activates, heat pump and heater (external or internal) will operate as normal tank mode to achieve the boiling set temperature.
  - Sterilization operation will end when stop condition is fulfill.
  - After sterilization is complete, tank set temperature will resume to normal operation.

* Tank temperature may not achieve boiling set temperature if tank heater is select OFF **OR** external compressor switch.

## 14.14 Outdoor Ambient Thermo OFF Control

- Purpose
  - Stop provides heating to room side during high outdoor ambient condition.




- Control content
  - Heating outdoor ambient thermos OFF control only applicable when heat pump operate in heat mode. (This control will not activate when running in tank side)
  - Heat pump and water pump will turn OFF when outdoor ambient is higher than outdoor thermo OFF set temperature.
  - Heat pump thermos ON when outdoor ambient < outdoor thermos OFF set temperature + 1°C.

## 14.15 Alternative Outdoor Ambient Sensor Control

Purpose of the Alternative Outdoor Ambient Sensor:

- It is some possibility that the air to water heat pump unit will install at a location where the original ambient sensor is expose to direct sunlight. Therefore, another optional ambient sensor can be connect to indoor PCB and locate at new and better reading location to improve the heat pump performance.

Control Detail:

- Remocon can select either the extra outdoor ambient sensor is connected or not. (YES/NO)
- The alternative outdoor ambient sensor will connect to indoor unit main PCB terminal.

    - when alternative sensor select NO
        - Original Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
        - Data communication direction : OUTDOOR send outdoor temperature reading to INDOOR.
        - Error judge : OUTDOOR will judge the original outdoor sensor error (F36 display if error detect). No judge error on alternative outdoor sensor

    - when alternative sensor select YES
        - Alternative Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
        - Data communication direction : INDOOR send outdoor temperature reading to OUTDOOR.
        - Error judge : INDOOR will judge the Extra outdoor sensor error only after operation ON request received from remocon.
          (F36 display if error detect). No judge error on original outdoor sensor.

## 14.16 Force DHW mode

Purpose:
When user want to use hot water now, user can press this force DWH mode under the quick menu to operate tank only mode to boil up the tank temperature.

Remocon setting:
Force DHW function can be activate under quick menu.

Control Content:

- When press the Force DHW function during operation OFF condition:
    - When receive this Force DHW bit from remocon, indoor will run tank only mode regardless of the mode selection.
    - After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to operation OFF with previous mode selection.

- When press the Force DHW function during operation ON condition:
    - When receive this Force DHW bit from remocon, indoor will memories the running mode and run tank only mode regardless of the mode selection.
    - After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to previous memories running mode.

      * when operation OFF or mode change request from remocon during force DHW mode operation, End force DHW mode and follow the new request operation.

      * Once receive force DHW mode from remocon, indoor direct start tank mode and consider tank thermo ON. Thermo OFF only when achieve tank thermo OFF depend on the Tank System Setting.

## 14.17 SMART DHW mode

Panasonic All In One model provide the option to choose STANDARD DHW Mode or SMART DHW Mode for Tank Heat Up according to requirement. SMART DHW mode comparatively consume lower tank heat up power but longer re-heat time than STANDARD DHW Mode.

SMART DHW control

- During SMART DHW start time 20:00 (Default Setting) to SMART DHW stop time 05:00 (Default setting)
  Heat pump re-heat the tank water only when tank temperature drop below 20°C (Default setting)
- Time between 05:00 to 20:00
  Heat pump reheat the tank water when tank temperature as below condition

Condition 1: Tank Heater ON
Reheat when tank temperature below tank set temperature + R/C (Tank re-heat Temperature) - 3°C

Condition 2: Tank Heater OFF
Reheat when tank temperature below Tank set temperature or tank limit temperature (Whichever lower) + R/C (Tank re-heat Temperature) -3°C

* SMART DHW start time, stop time and SMART ON Temperature can change in CUSTOM menu.

STANDARD DHW Mode

- Heat pump always reheat the tank water when tank temperature as below condition

Condition 1: Tank Heater ON
Reheat when tank temperature below tank set temperature + R/C (Tank re-heat temperature)

Condition 2: Tank Heater OFF
Reheat when tank temperature below Tank set temperature or tank limit temperature (Whichever lower) + R/C (Tank re-heat temperature)

| Outdoor ambient temperature | Tank limit temperature |
|---|---|
| OD < -17°C | 48°C |
| -17°C <= OD < 36°C | 52°C |
| 36°C <= OD | 50°C |

## 14.18 DHW Capacity Setting

DHW Capacity is heat pump heating capacity output control during tank boiling operation. There are two DHW capacity setting (VARIABLE & STANDARD) which can be set in remote control.

VARIABLE DHW Capacity:

- Heat pump operate with efficient (Low) Capacity to boil tank temperature during re-heat process. And heat pump regulated to operate with high capacity to boil tank temperature when tank temperature drop below 25°C.

STANDARD DHW Capacity:

- Heat pump operate according to outdoor rated heating capacity during tank boiling process.

## *14.19 Anti Freeze Control*

- Anti freeze protection control menu can be set YES or NO by control panel.
- In heatpump system, there are 3 types of anti freeze control:
  1. Expansion tank anti-freeze control
     - Expansion tank anti freeze heater ON condition:
       - Outdoor ambient temp. < 3°C
     - Expansion tank anti freeze heater OFF condition:
       - Outdoor ambient temp. > 4°C

  2. Water pump circulation anti freeze control
     - Water pump turns ON when **ALL** below conditions are fulfilled:
       - Heatpump OFF (stand by) OR error occurs.
       - Water flowing flag is ON.
       - Water flow switch is not abnormal.
       - Outdoor ambient temp. < 3°C OR outdoor ambient temp. sensor is abnormal.
       - Water inlet/outlet temp. < 6°C.
       - After 5 minutes from previous water pump OFF.
     - Water pump turn OFF when **ANY** of below conditions is fulfilled:
       - Outdoor ambient temperature ≥ 4°C.
       - During -5°C < outdoor ambient temp. < 4°C
         - After water pump ON for 4 minutes, and water inlet temp. ≥ 8°C.
         - Else, shift to back up heater anti freeze control.
       - During outdoor ambient temp. < -5°C
         - After water pump ON for 4 minutes, and water inlet/outlet ≥ 20°C.
         - Else, shift to back up heater anti freeze control.
     - However, if flow switch is abnormal (H62), then water pump circulation anti freeze control will not activate.

  3. Back up heater anti freeze control:
     - Back up heater turn ON when **ALL** below conditions is fulfilled:
       - Water inlet/outlet temp. < 6°C.
       - Water pump circulation anti freeze control activated and water pump ON for 4 minutes.
     - Back up heater turns OFF when ANY of below conditions is fulfilled:
       - Water inlet/outlet temp. > 28°C.
       - Water pump circulation anti freeze control deactivated/water pump OFF.
     - However, if back up heater is abnormal (H70) then back up heater anti freeze control will not activate.

### 14.19.1 Zone Anti-Freeze Control

- If buffer tank selection is “NO” and Anti- Freeze function select “NO” from remote control, this control cannot activate.

Start condition:

- After [5] min from previous Zone pump off. **AND**
- Outdoor air temp < [3] °C **OR** Outdoor sensor is abnormal. **AND**
- Zone water temperature < [6]°C **OR** Zone Sensor Short or Open

Cancel condition:

- After water Zone pump ON [4] min **AND**
- Outdoor air temp ≥ [4]°C **OR**
- During -5 °C ≤ Outdoor air temp < [4] °C
  Zone water temperature sensor > [8] °C **OR**
- During Outdoor air temp < [-5] °C
  Zone water temperature sensor > [20] °C
  *However, Zone water temperature sensor is Open or Short, Condition 3 and 4 is ignored.

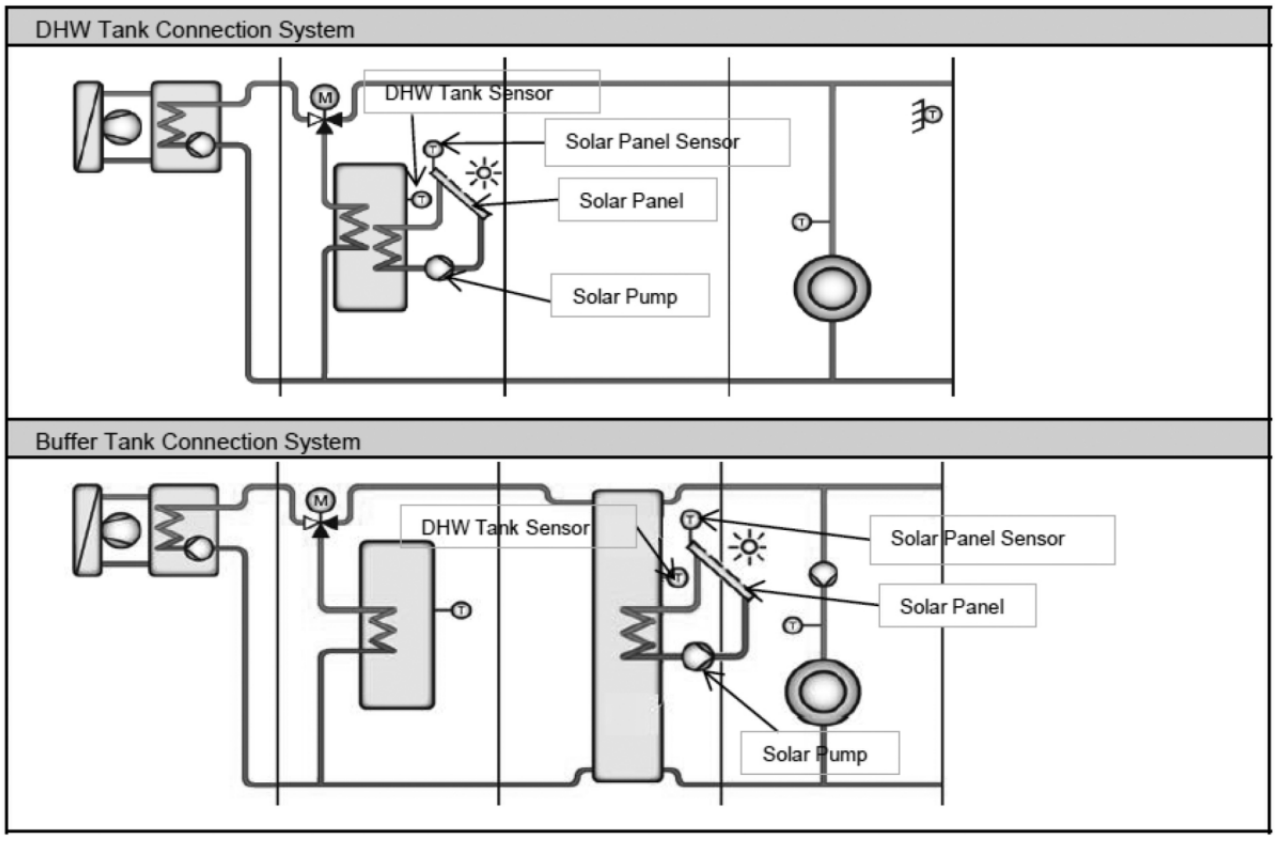
## *14.20 Solar Operation (Optional)*

### 14.20.1 Solar Operation:

Solar function:

- This function allow user to control the solar pump to operate depend on the solar sensor reading compare to the tank installed. Solar pump will circulate the hot water energy store at solar panel to heat up the DHW Tank or Buffer Tank.

Solar Connection Diagram:




- Solar function can only enable when the Optional PCB is connected.
- Few part as below need to install to control the solar operation:
  - Solar Panel
  - Solar Pump
  - Solar Panel Sensor
  - Tank Sensor (Buffer tank sensor OR DHW Tank sensor depend on the connection direction)
    * During Solar Connection to the system, installer need to alert on the high water temperature may flow to the zone circuit or DHW piping circuit. Therefore pipe which withstand higher water temperature need to be installed.

- Solar remote control setting
  1 Solar Setting can only be set when the optional PCB connection is select "YES"
  2 By remote controller, Setting as below list can be set for solar function operation (Installer Menu)
    - Solar Function ("YES" or "NO)
    - Tank Connection Direction ("DHW TANK" or "BUFFER TANK")
    - Delta T turns ON: Difference temperature setting between solar panel sensor and Tank to turn on solar pump. (Range :5 ~ 15°C)
    - Delta T turns OFF: Difference temperature setting between solar panel thermistor and Tank to turn off solar pump. (Range :2 ~ 10°C)
    - Outdoor temp for Anti-Freeze : Outdoor temp to start Anti-Freeze control for solar circuit. (Range : -20 ~ 10°C)
    - Tank Temperature HI Limit Set (Range : 70 ~ 90°C)

### 14.20.2 Solar Operation Control

- Solar function can only be activate if the solar function selection “YES” from remote control. To achieve hot water from solar panel, indoor need to control the solar pump and circulate hot water from solar panel.

  Under normal case:
  - Solar pump start condition:
    - Solar panel temperature > Delta T turn on setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **AND**
    - Tank temperature (DHW or Buffer) < Solar HI Limit Temp (R/C) **AND**
    - Operation ON with heat mode (apply to solar connect to "Buffer Tank" case)

    * Condition c) ignore if the solar system is connect to DHW tank (control active under operation OFF time for Tank connection case)
  - Solar pump stop condition:
    - Solar panel temperature < Delta T turn OFF setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **OR**
    - Tank hot water temp >= Solar HI Limit Temp (R/C) + [2]°C

  Under solar Anti-freeze protection control:
  - Solar pump start condition:
    - Outdoor temp < Outdoor temp setting for Anti-Freeze (R/C)
  - Solar pump stop condition:
    - Outdoor temp > Outdoor temp setting for Anti-Freeze + [2]°C

**However, During Cool mode this function cannot activate if Tank selection is “Buffer Tank".
**Solar pump can operate even if Heat pump is under error stop.

- Solar operation during error:
  - During Tank sensor (DHW or Buffer depend on selection) abnormal, Solar operation will not able to function.
  - During Solar Panel sensor detect OPEN (not include SHORT), Solar operation will not able to function too.

## *14.21 Boiler Bivalent Control*

- Boiler is an additional or alternative heat source to heat up the room when necessary.
- Purpose of this control is to turn ON and turn OFF the Boiler output signal when boiler heating capacity needed in the system.
- Boiler is possible to connect to DHW Tank and Buffer Tank depends on the installer.
- Boiler operation parameter need to be set on Boiler itself, indoor do not control the boiler operation direction and operation.
- There are Alternative mode, Parallel mode, & Advance Parallel mode available to select by installer to fit to the total system.
- There is two option of control pattern can be set by remote controller : AUTO OR SG ready Mode

**AUTO Control Mode:**
- There are Alternative mode, Parallel mode, & Advance Parallel mode available to select by installer to fit to the total system.

**Bivalent control selection by remote controller**

Remote control setting value:
1 Outdoor Ambient Set = (Range: -15°C ~ 15°C)

- **Alternative Mode**
  - Only one heat source operates at one time, either heat pump or boiler depends on condition.
  - External pump will turn ON when the external pump selection is ON when boiler is ON even heat pump is OFF.

  Control detail:
  During Operation ON at Heat mode or Tank mode or Heat + Tank Mode
  - Boiler signal turn ON and heat pump and water pump turn OFF when:
    - Outdoor ambient < Outdoor Ambient Set **AND**
    - Boiler prohibit flag = 0

** However indoor water pump can operate when Anti-freeze control condition fulfilled.

- Boiler signal turn OFF and heat pump and water pump turn ON when:
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Boiler prohibit flag = 1

- **Parallel Mode**
  - Parallel mode allows heat pump and boiler ON at the same time. Boiler operates as an additional heating capacity when low heat pump capacity at low ambient condition.

Control detail:
During operation ON at Heat mode or Tank mode or Heat + Tank mode
- Boiler signal turns ON when:
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when:
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Boiler prohibit flag = 1

- **Advance Parallel Mode**
  - Advance parallel mode allow heat pump to operate and turn ON boiler only when ambient and temperature condition is fulfilled.

Remote control setting value:
1 Outdoor Ambient Set = (Range : -15°C ~ 15°C)
2 Selection of boiler connection direction. (Heat only, DHW only, Heat & DHW)
3 Setting data under Heat Direction
  - Start Temperature | START_TEMP |
  - Start Delay Timer | START_TIMER |
  - Stop Temperature | STOP_TEMP |
  - Stop Delay Timer | STOP_TIMER |

4 Setting data under DHW Direction
  - Delay Timer | DELAY_TIMER |

Control detail:
During operation ON at Heat Mode
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Buffer tank temperature < Target Buffer Tank Temperature + [START_TEMP] for [START_TIMER] **AND**
  - Heat pump operate at room side **AND**
  - Connection of Boiler to Heating Select "YES" From installer menu **AND**
  - Buffer Tank connection select "YES" **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [-2°C] **OR**
  - Buffer Tank temperature > Target Buffer Tank temperature + [STOP_TEMP] for [STOP_TIMER] **OR**
  - Heat pump not at room side. **OR]**
  - Boiler prohibit flag = 1

During operation ON at Tank Mode
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Heat pump operate at tank side for continuous | DELAY_TIMER | **AND**
  - Connection of Boiler to DWH Tank select "YES" from installer menu. **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Heat pump not operates at tank side. **OR**
  - Boiler prohibit flag = 1

**SG ready Control Mode:**
- Using same SG ready from Sub Board input to control boiler ON/OFF output.

* When this SG ready is select for bivalent control, default SG ready function will change to control bivalent output.
- Remote controller can set the External Pump ON/OFF like bivalent alternative mode.

**Control Content**

Indoor will follow the SG ready bit input to control ON/OFF heat pump and boiler

00 : Heat pump OFF, Boiler OFF 10 : Heat pump OFF, Boiler ON

01 : Heat pump ON, Boiler OFF 11 : Heat pump ON, Boiler ON

* External pump will turn ON when the external pump selection is ON when boiler is ON even heat pump is OFF.

**Boiler prohibit flag control**

Purpose:

- For product safety. Boiler signal is OFF when water temperature is too high.

Start condition:

- Water outlet ≥ 85°C continues for 5 minutes.
- Water inlet ≥ 85°C continues for 5 minutes.
- Zone1 water temp ≥ 85°C continues for 5 minutes.
- Zone2 Water temp ≥ 85°C continues for 5 minutes.

Contents:

- After start condition fulfilled, set boiler prohibit flag = 1

Cancel condition:

- After 30 minutes from start condition fulfilled.

Contents:

- Set boiler prohibit flag = 0

## *14.22 External Room Thermostat Control (Optional)*

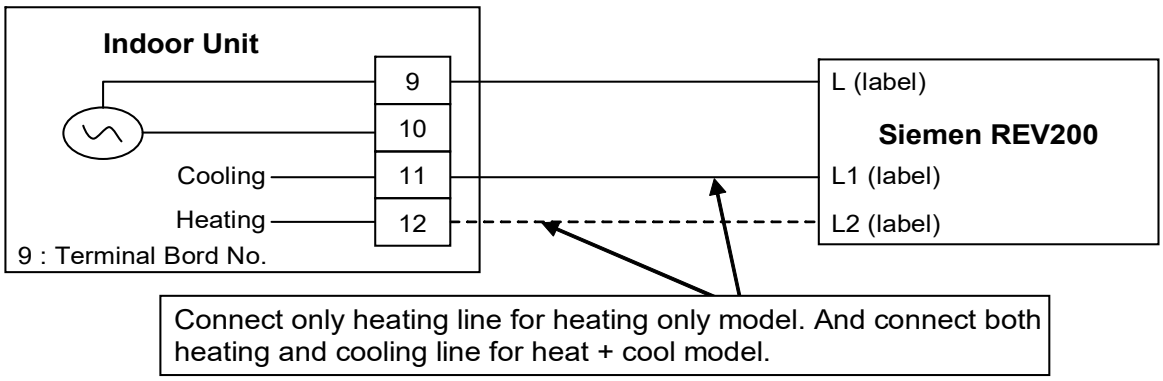
Purpose:

1 Better room temperature control to fulfill different temperature request by external room thermostat.
Recommended external room thermostat:

| Maker | Characteristic |
|---|---|
| Siemen (REV200) | Touch panel |
| Siemen (RAA20) | Analog |

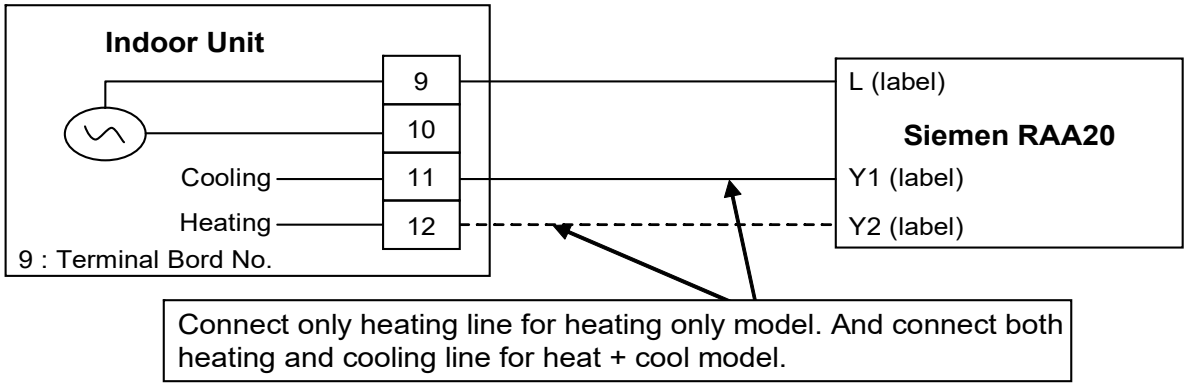
Connection of external room thermostat:
Wire Connection and thermo characteristic of Siemen REV200:

| Setting | L/L1 (H) | Heat Thermo | L/L2 (C) | Cool Thermo |
|---|---|---|---|---|
| Set Temp < Actual Temp | Open Circuit | OFF | Short Circuit | ON |
| Set Temp > Actual Temp | Short Circuit | ON | Open Circuit | OFF |



Wire Connection and thermo characteristic of Siemen RAA20:

| Setting | L/Y1 (H) | Heat Thermo | L/Y2 (C) | Cool Thermo |
|---|---|---|---|---|
| Set Temp < Actual Temp | Open Circuit | OFF | Short Circuit | ON |
| Set Temp > Actual Temp | Short Circuit | ON | Open Circuit | OFF |




Control Content:

- External room thermostat control activate only when remote thermostat connection select YES by Indoor control panel.
- When indoor running heat mode, refer thermo On/Off from heating line feedback. And when indoor running cool mode, refer thermo On/Off from cooling line feedback.
- Heat pump Off immediately when receive thermo off feedback.

## *14.23 Three Ways Valve Control*

Purpose:

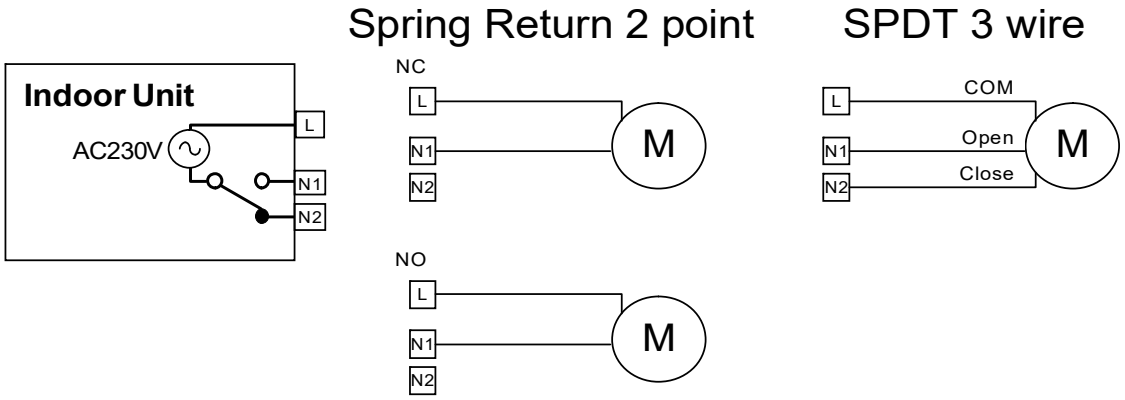
- 3 ways valve is used to change flow direction of hot water from heat pump between heating side and tank side.

Control contents:

1 3 ways valve switch Off:
   - During 3 ways valve switch Off time, the hot water will provide heat capacity to heating side.

2 3 ways valve switch On:
   - During 3 ways valve switch On time, the hot water will provide heat capacity to tank side.

3 Stop condition:
   - During stop mode, 3 ways valve will be in switch off position.




* During pump down and force mode, fix 3 ways valve in close condition.

* Recommended Parts : SFA 21/18 (Siemens)

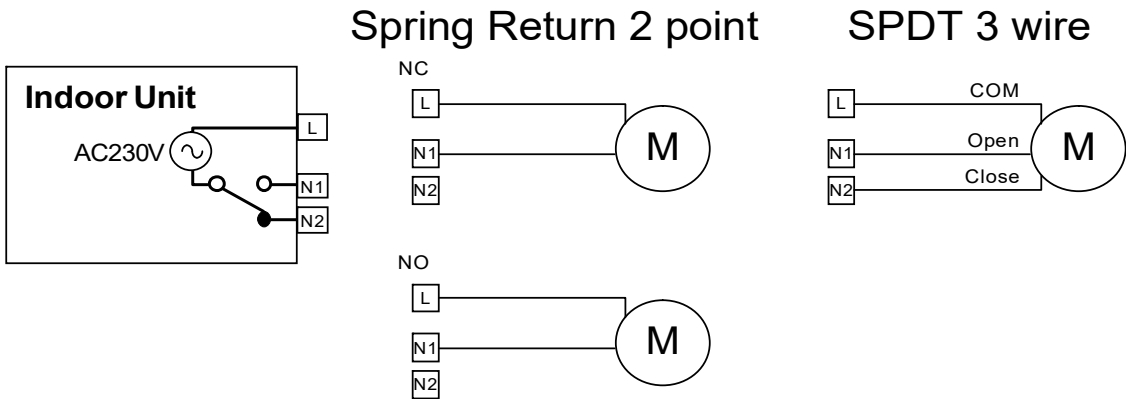
## *14.24 Two Ways Valve Control*

Functionality of 2 ways valve:

- Use to allow hot water to floor heating panel or block cold water to floor heating panel.

Control contents:

1 When indoor running in heat mode, OPEN the 2 ways valve.
2 When indoor running in cool mode, CLOSE the 2 ways valve.
3 Stop condition:
   a. During stop mode, fix 2 ways valve in close condition.

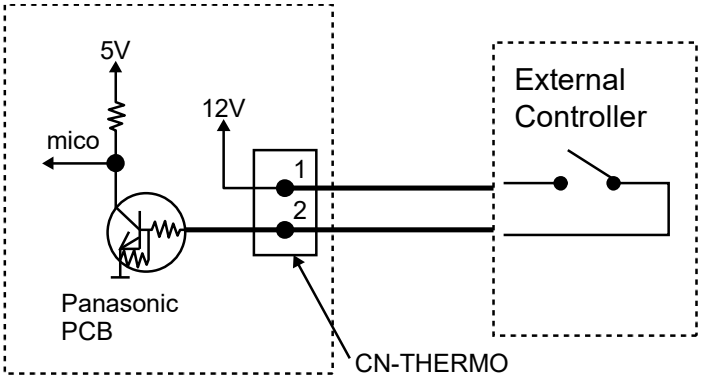



* During pump down mode, fix 2 ways valve in close condition.
* During force mode, open 2 ways valve.

* Recommended Parts : SFA 21/18 (Siemens)

## *14.25 External OFF/ON Control*

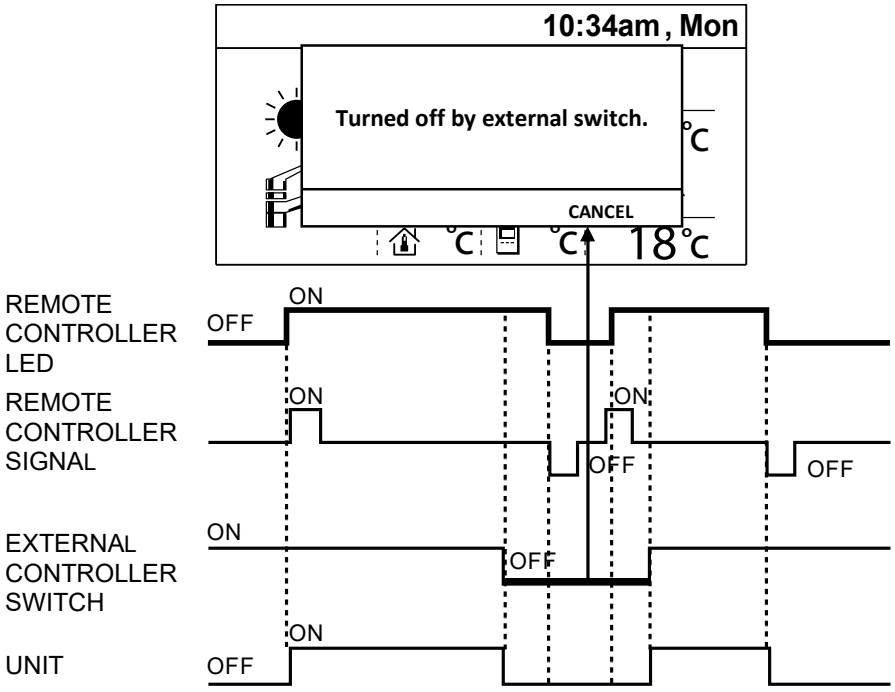
- Communication circuit between indoor unit and external controller is as per below.




- Maximum length of communication cable is 50 meter.
- Control content:

| External Control Switch | Control Panel OFF/ON | Control Panel Power LED | System Status |
|---|---|---|---|
| ON | ON | ON | ON |
| ON | OFF | OFF | OFF |
| OFF | ON | ON | OFF |
| OFF | OFF | OFF | OFF |

Remocon Screen Display and Control Detail:




When External SW connection select "YES" from remocon installer menu:

- Heating or Cooling system will operate normally if the External Switch signal is ON.
- Once the External Switch turn OFF, System Turn OFF ( Heat pump, water pump, heater etc…)
- Remocon LED remain ON or OFF according to the current operation request.
- Pop up menu at remocon main screen as above screen to inform customer system stop by External Switch.
- It is possible to press cancel and return to main screen to do change of operation setting while waiting the External Switch turn ON back.
- Remocon LED will always follow the latest changes from remocon.
- If no action on remocon for continuous 5 minutes, the pop up screen will show again on the screen.
- But once the External Switch Turn ON back, pop up screen will disappear and system can operate normally according to the latest operation setting and request.

## *14.26 External Compressor Switch (Optional PCB)*

External compressor switch port can have two purpose of control as below:

- Heat source ON/OFF function (Dip switch Pin 3 on PCB "OFF")
- Heater ON/OFF function (Dip switch Pin 3 on PCB "ON")

- Heat source ON/OFF function

Purpose:

- Heat pump ON/OFF function is use to turn OFF the high power consumption device (Heat pump, & Heater) when there is energy or electric current limitation. Other optional function still can be operate under heat pump and heater OFF condition.

Control Detail:

- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "OFF" This heat pump ON/OFF function will activate
- The ON/OFF signal of this External Compressor Switch is same as External Switch.
- When the External Compressor Switch is ON:
  - Heat pump system operate normally
- When the External Compressor Switch is OFF:
  - Heat pump, Indoor water pump & Heater ( Booster heater & Backup Heater ) need to turn OFF
  - Solar, Boiler and zone control can be operate follow normal control condition.
    * pump delay OFF also included in this control

    (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

- Heater ON/OFF function

Purpose:
- Heater ON/OFF function is use to turn OFF the heater (backup heater & booster heater) when there is energy or electric current limitation. Heat pump and other optional function still can operate.

Control detail:
- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "ON" This heater ON/OFF function will activate
- When the External Compressor Switch is ON:
  - Heat pump and heater operate normally
- When the External Compressor Switch is OFF:
  - Backup heater and booster heater cannot operate even heater request is ON.
  - Heat pump and option function (Solar, Boiler and zone control) can be operate follow normal control condition.

  (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

## *14.27 Heat/Cool Switch (Optional PCB)*

Purpose:
- User can switch the running mode from heat to cool or cool to heat through external installed Heat/Cool switch. This kind of heat / cool switch may built in inside the field supply room remocon as well.

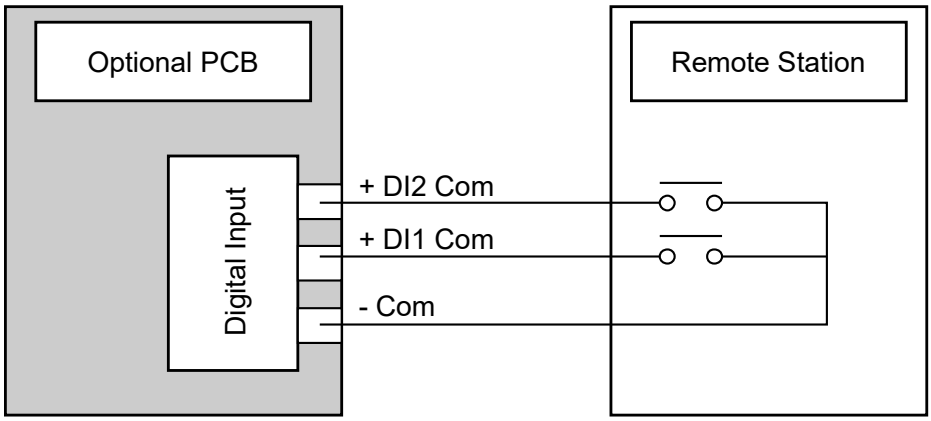
Control contents:
- Heat/Cool Switch can only be set when Cool Function is "enable" at custom menu setting, & Extension PCB select "YES" & Zone 1 not set "Pool" condition.
- This heat/cool switch control will be activate only when installer set the Heat/Cool Switch "USE" through remocon.
- Once the Heat/Cool Switch Set "USE", remocon will check indoor send Signal to judge the option of mode select.
  - When Heat/Cool Switch Contact Open : Remocon only can select Heat Mode, or Heat + Tank Mode, or Tank Mode
  - When Heat/Cool Switch Contact Close : Remocon only can select Cool Mode, or Cool + Tank Mode, or Tank Mode
- Operation ON/OFF will depend on remocon request.
- When Heat Mode is running with Contact Open, user change this setting to contact close, indoor will this signal to remocon judge and change mode to cool and send back to indoor. And it is same as from cool mode change to heat mode.
  * This switch have higher priority, remocon follow indoor send signal when control activated.
  * There is no effect to the operation when the mode running is only Tank Mode.

  (Weekly Timer are ignored and cannot be set during Heat / Cool Switch is "Enable" Condition.)

## *14.28 SG Ready Control (Optional PCB)*

Purpose:

- To set ON/OFF of heat pump and target temperature by digital input of third party device if necessary in field.




Remote control setting

For this function, following items need to be set on R/C (installer menu) –

- SG control = YES or NO
- Capacity up setting 1
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]
- Capacity up setting 2
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]

Control contents:

If SG control on remote control = "Yes", then following control only activate by digital input.

- **While Digital input is " 00 " (Normal operation)**
  - Normal operation. Once detect '00' system will operate back to normal condition.
    (All the target set temperature for heating side and DHW side will return back to previous set temperature when digital signal change from "10' or"11" back to "00".)
- **While digital input is detected " 01 " (HP stop)**
  - Heat pump & room heater & tank heater cannot operate.
    **(Solar control and Boiler back up and 2 Zone control can activate.)**
- **While digital input is detected " 10 " (Capacity 1)**
  - Target water temperature of cooling is changed according to the adjustment value set by Remote control setting.
- **While digital input is detected " 11 " (Capacity 2)**
  - Target water temperature of cooling is changed according to the adjustment value set by Remote control setting.
- **While digital input is detected " 10 " (Capacity 1)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

**If Buffer selection is "YES"**
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 1) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) % * (Max regulation depend on the tank max setting limit)

**If Buffer selection is "NO"**
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 1) %
(Zone 1 and Zone 2 will change according to its own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) %
* (Max regulation depends on the tank max setting limit)
Setting temperature for cooling is changed.
New Target Water Temperature = Target Water Temperature + Remote Control setting for cool (*Capacity 1)
* (Min/Max regulation of cooling water set apply)

- **While digital input is detected " 11 " (Capacity 2)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

**If Buffer selection is "YES"**
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 2) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) %
* (Max regulation depends on the tank max setting limit)

**If Buffer selection is "NO"**
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 2) %
(Zone 1 and Zone 2 will change according to it's own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) %
* (Max regulation depends on the tank max setting limit)
** This function is not applicable for Cooling mode.
Setting temperature for cooling is changed.
New Target Water Temperature = Target Water Temperature + Remote Control setting for cool (*Capacity 2)
* (Min/Max regulation of cooling water set apply

## *14.29 Demand Control (Optional PCB)*

Remote control setting:
- When Optional PCB connection select 'YES", Demand Control function can select "YES" or "NO".

Purpose:
- After the demand control select YES, below control will activated.
  - 0-10V Demand control

0-10V Demand control
- Demand control is use to reduce the current usage of heat pump unit by third party device.

Control start condition:
- Select "YES" at Demand control at installer menu.
- 0-10V input for this electrical current control is detected.

Control content:
- If start condition is fulfilled, indoor will receive the voltage signal from optional PCB. Indoor will send the rate value to outdoor unit.
- Outdoor will change the current limit according to the percentage receive from indoor unit.

## 14.30 Holiday Mode

- Purpose:
  Promotes energy saving by allowing the user to stop the system during holiday and enables the system to resume at the preset temperature after holiday.

- Control details:
  - Indoor operate the unit according running mode request. Target temperature will follow holiday setting temperature.
    - If heat mode request is receive, Target Water Out Temperature will change according to holiday shift temperature set.
      [If heat is set OFF at holiday, unit, water pump and zone control will OFF]
    - If tank mode request is receive, Target Tank Set Temperature will change according to the holiday tank shift temperature set.
      [If tank is set OFF at holiday, heat pump and tank heater will OFF]
  - After days of holiday have been set, heat pump will stop and only resume operation at the end of holiday countdown.

- Start condition:
  - Holiday timer set and the holiday timer start
    * The day holiday mode was set is counted as day 1.

- Stop condition:
  - OFF/ON button is pressed.
  - Holiday timer is reached.

## 14.31 Dry Concrete

- Purpose:
  Provide heat to floor heating panel and dry the wet concrete during installation.

- Setting condition:
  - Dry concrete parameter can be set through remote control under system setup.
  - Parameters are possible to set up to 99 days with different target set temperature

- Control details:
  - Dry concrete mode will be activates when select ON from service setup.
  - Once start dry concrete function, remote control will send step 1 setting temperature to indoor unit.
    * This temperature is set at zone temperature. If system is 2 zones, both zone target temperature is set as same temperature.
  - Heat pump will start heat mode operation to room side with received target water outlet temperature.
    * Heat pump will operate according to Heat pump Target Water Temperature.
  - After complete day 1 setup operation, day 2 data will be send to indoor at 12.00am on the second day.
  - Each preset data will be send every day until dry concrete mode is complete, unit will turns OFF and exit dry concrete function.
  - 3 ways valve and booster heater will turn OFF and 2 ways valve will turns ON.

- Cancel condition:
  - Dry concrete mode is complete and OFF signal is received.
  - OFF signal is received by pressing OFF/ON button.

## 14.32 Flow Sensor

- The water flow sensor serves as an overload protector that shuts down the unit when the water level is detected to below.
- Abnormal flow detection:

| Condition | Abnormal flow | Normal flow |
|---|---|---|
| Normal Case | Flow rate < 5 l/min or >= 69 l/min | >= 5 l/min |
| During Deice Operation OR COOL Mode | Flow rate < 11 l/min or >= 69 l/min | >= 11 l/min |
| During status 2~6 on Anti-freeze deice | Flow rate >= 5 l/min | < 5 l/min |

# 15. Protection Control

## *15.1 Protection Control for All Operations*

### 15.1.1 Time Delay Safety Control

1 The compressor will not start for three minutes after stop of operation.

### 15.1.2 30 Seconds Forced Operation

1 Once the compressor starts operation, it will not stop its operation for 30 seconds.
2 However, it can be stopped using control panel at indoor unit.

### 15.1.3 Total Running Current Control

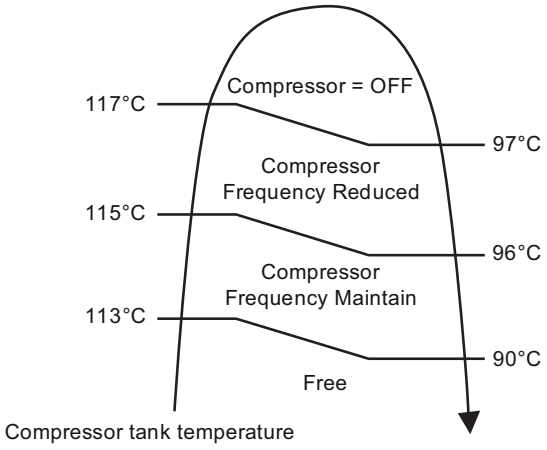
1 When the outdoor running current exceeds X value, the compressor frequency will decrease.
2 If the outdoor running current does not exceed X value, the compressor frequency will return to normal operating frequency.
3 If the outdoor running current continue to increase till exceed Y value, compressor will stop, and if this occurs 3 times within 20 minutes, system will stop operation and OFF/ON control panel LED will blink (F16 error occurs).

| | WH-MXC09J3E5 | | WH-MXC12J6E5 | |
|---|---|---|---|---|
| Operation Mode | X (A) | Y (A) | X (A) | Y (A) |
| Heating | 28.0 | 30.0 | 28.0 | 30.0 |
| Cooling | 20.0 | 30.0 | 20.0 | 30.0 |

A. DC Peak Current Control
1 When the current to IPM exceeds set value, compressor will stop. Compressor will restart after three minutes.
2 If the set value exceeds again for more than 30 seconds after the compressor restarts, operation will restart after two minutes.
3 If the set value exceeds again for within 30 seconds after the compressor restarts, operation will restart after one minute. If this condition repeats continuously for seven times, system will stop operation and OFF/ON control panel LED will blink (F23 error occurs).

### 15.1.4 Compressor Overheating Prevention Control

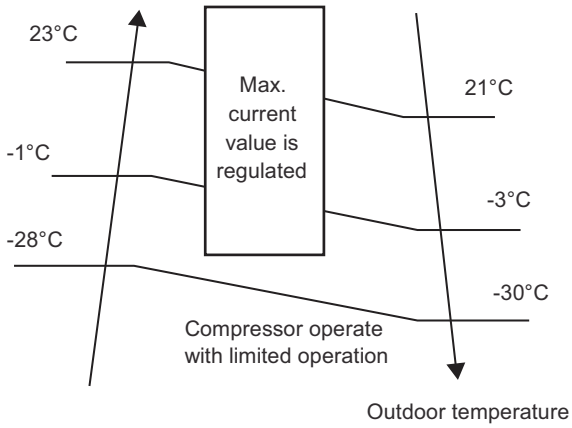
- The compressor operating frequency is regulated in accordance to compressor tank temperature as shown in below figures. When the compressor tank temperature exceeds 107°C, compressor will stop, and if this occurs 4 times within 30 minutes, system will stop operation and OFF/ON control panel LED will blink (F20 error occurs).



### 15.1.5 High Pressure Sensor Control

- Purpose:
  - To protect the system operation.
- Detection period:
  - After compressor on for 1 minute.
- Detection conditions:
  - When abnormal high voltage detection, 5 V or when open circuit detection 0V for 5 seconds continuously.
- After detection:
  - When abnormality is detected continue 5 seconds,, unit stop operation.
  - OFF/ON control panel LED will blink (H64 error occurs).

### 15.1.6 Outside Temperature Current Control




### 15.1.7 Crank Case Heater Control

- Purpose:
  - For compressor protection during low outdoor ambient operation (during heating low temperature operation).
- Control content:
  a. Trigger heater START condition
     - When the outdoor air temperature is below than 4°C, and discharge temperature is 9°C or below.
  b. Resetting heater STOP condition
     1. When the outdoor air temperature exceeds entry condition (6°C)
     2. When the discharge temperature exceeds entry condition (13°C)

## *15.2 Protection Control for Heating Operation*

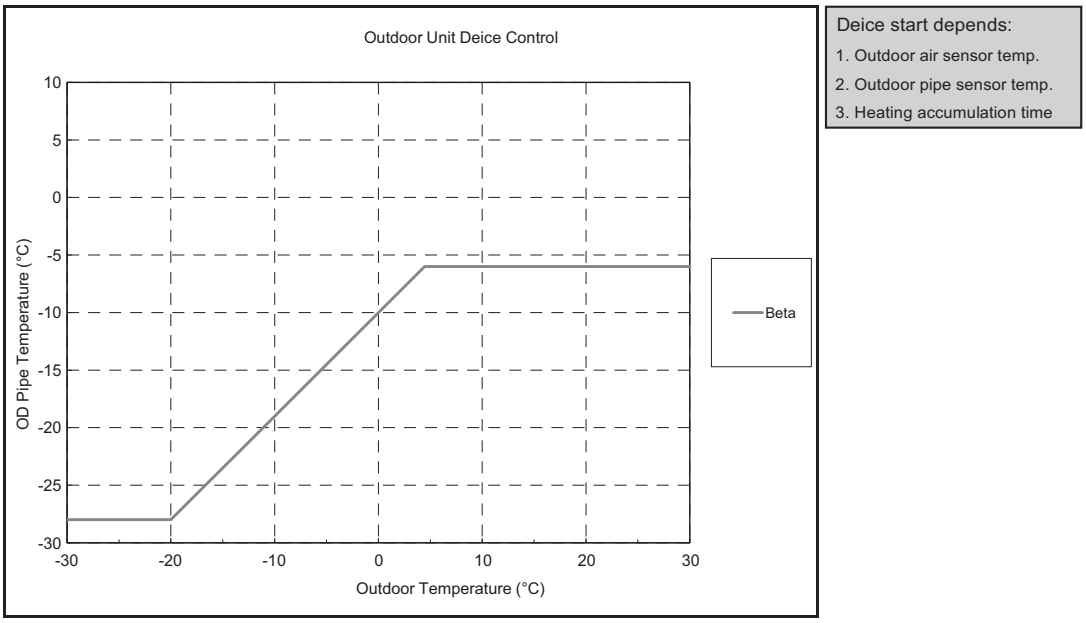
### 15.2.1 Outdoor Air Temperature Control

The maximum current value is regulated when the outdoor air temperature rises above 14°C in order to avoid compressor overloading.

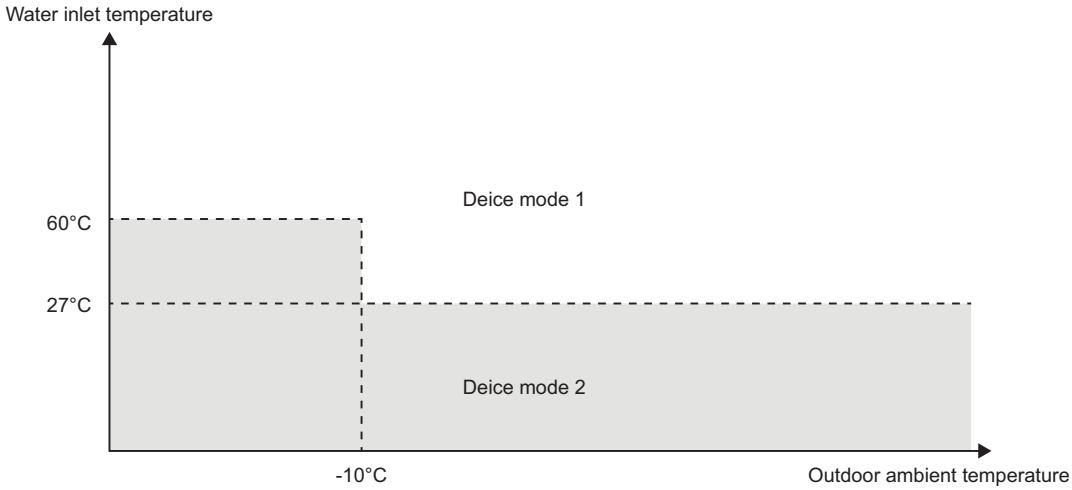
### 15.2.2 Deice Operation

When outdoor pipe temperature and outdoor air temperature is low, deice operation start where outdoor fan motor stop.
Normally, deice start if pipe sensor temperature fulfil deice condition. If remote controller set to AUTO force defrost setting, unit will start force deice after heat pump operate for 3 hours without deice at below outdoor temperature 5°C.

- Deice judging condition




- Deice mode selection condition
  - There are 2 deice modes, according to water inlet temperature and outdoor ambient temperature the deice mode is decided.

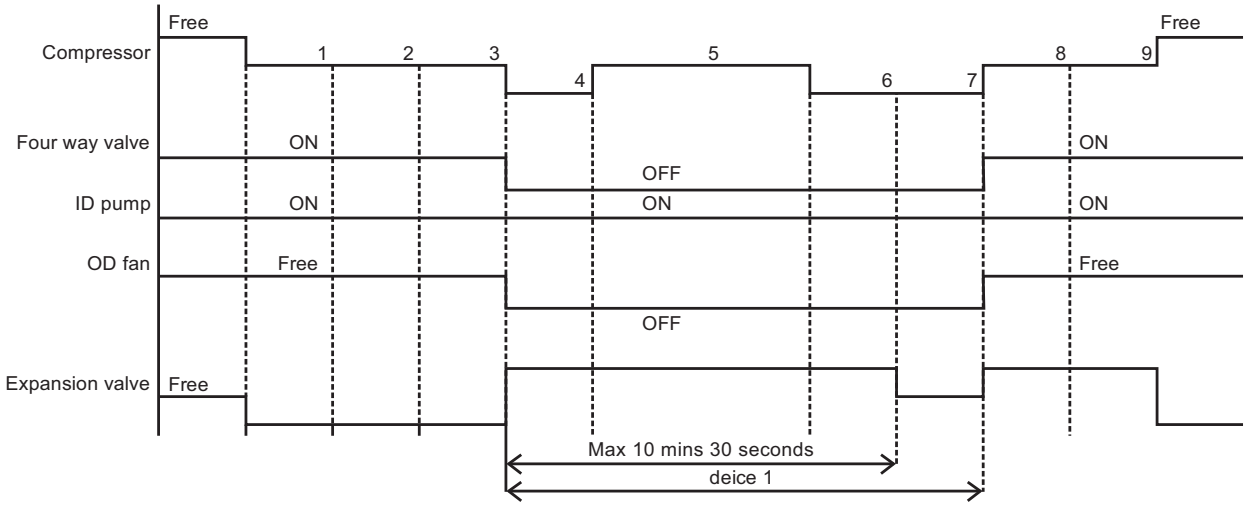



Judgement details:

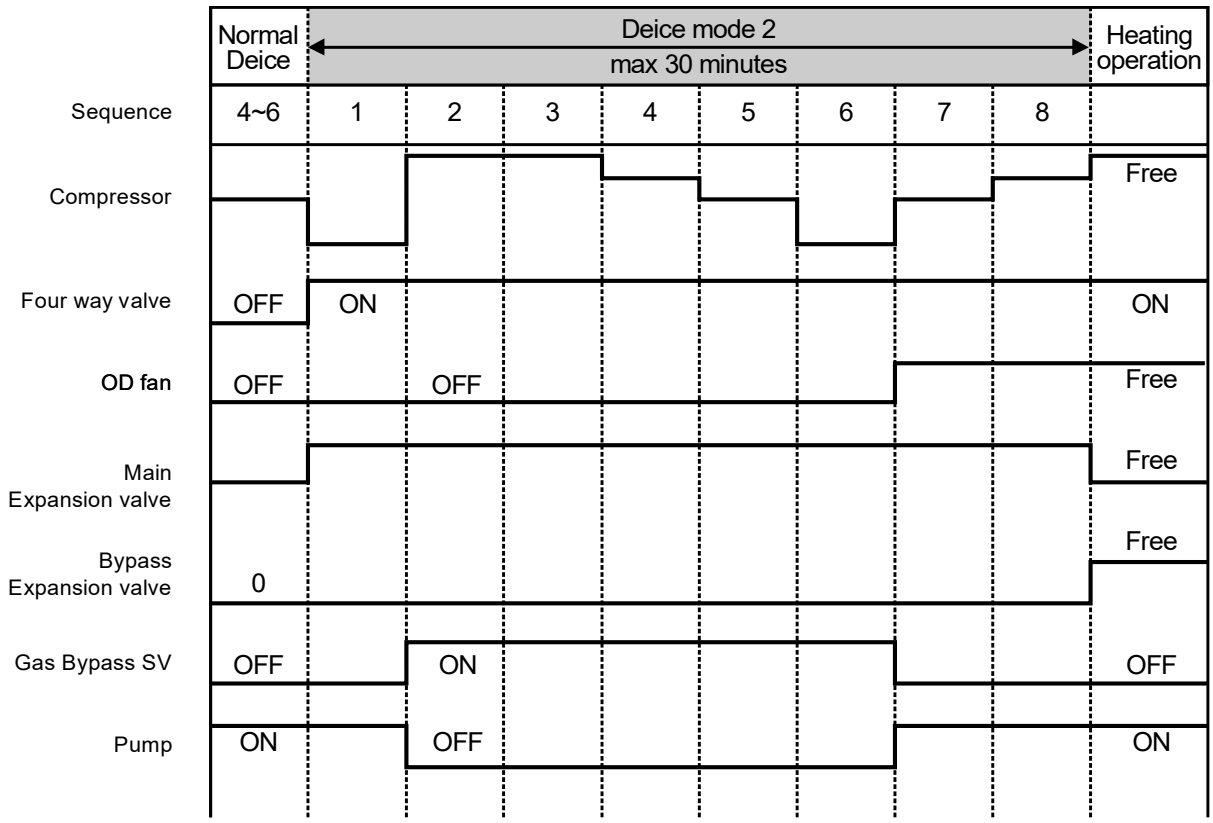
1 When water inlet temperature is more than 60°C, unit will operate deice mode 1.
2 When water inlet temperature is less than 27°C, unit will operate deice mode 2.
3 When water inlet temperature is less than 60°C and outdoor ambient temperature is less than -10°C, unit will operate deice mode 2.
4 When water outlet temperature sensor 2 detect temperature is less than 22°C, unit will operate deice mode 2.

Deice operation time diagram

a. Deice mode 1 control:




b. Deice mode 2 control:




## 15.2.3 Force Defrost Operation

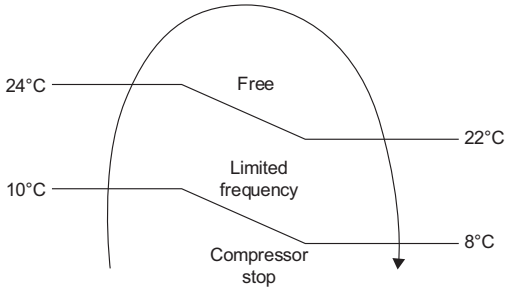
Force defrost can be set through remote control with two selection (Manual OR Auto).

- If Manual defrost set, heat pump only run force defrost at heat mode when force defrost request from quick menu remote control.
- If Auto defrost set, heat pump automatically run force defrost operation after 3 hours heating accumulation time without defrost when ambient below 5°C.

## *15.3 Protection Control for Cooling Operation*

### 15.3.1 Outdoor Air Temperature Control

- The Compressor operating frequency is regulated in accordance to the outdoor air temperature as shown in the diagram below.
- This control will begin 1 minute after the compressor starts.
- Compressor frequency will adjust base on outdoor air temperature.




### 15.3.2 Freeze Prevention Control 1

1 When indoor heat exchanger temperature is lower than 0°C continuously for 10 seconds, compressor will stop operating.
2 Compressor will resume its operation three minutes after the indoor heat exchanger is higher than 1°C.
3 Indoor heat exchanger freeze prevention (H99) will memory in error history.

# 16. Servicing Mode

## *16.1 Expansion Vessel Pre Pressure Checking*

[Upper limit water volume of the system]
The Mono bloc unit has a build-in Expansion Vessel with 10 L air capacity and initial pressure of 1 bar.

- Without antifreeze agent condition

Total amount of water in the system should be below 200 L.
If the total amount of water is more than 200 L, please add expansion vessel (field supply).

- With antifreeze agent condition

In the case of using antifreeze agent, expansion rate ε is different depending on its maker.
Please refer to the antifreeze agent maker for the expansion rate ε before calculate the upper limit water volume of the system.
The expansion vessel capacity required for the system can be calculated from the formula below.

$$V = \frac{\varepsilon \times Vo}{1 - \frac{98 + P_1}{98 + P_2}}$$

V : Required gas volume <expansion vessel volume L>
Vo : System total water volume <L>
ε : Expansion rate 5→60˚C = (depends on antifreeze agent used)
$P_1$ : Expansion tank filling pressure = (100) kPa
$P_2$ : System maximum pressure = 300 kPa

- ( ) Please confirm at actual place
- The gas volume of the sealed type expansion vessel is presented by <V:

❍ It's advised to add 10% margin for required gas volume of calculation.

[Adjustment of the initial pressure of the expansion vessel when there is a difference in installation height]
If the height difference between the Mono bloc unit and the highest point of the system water circuit (H) is more than 7m, please adjust the initial pressure of the expansion vessel (Pg) according to the following formula.

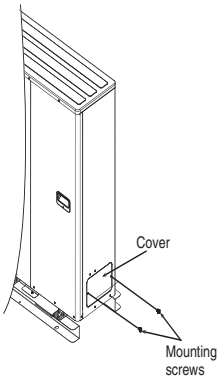
Pg= (H*10+30) kPa

## *16.2 Maintenance*

- In order to ensure optimal performance of the unit, seasonal inspections on the unit, functional check of RCCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer.

### 16.2.1 Maintenance for Water Filter Set

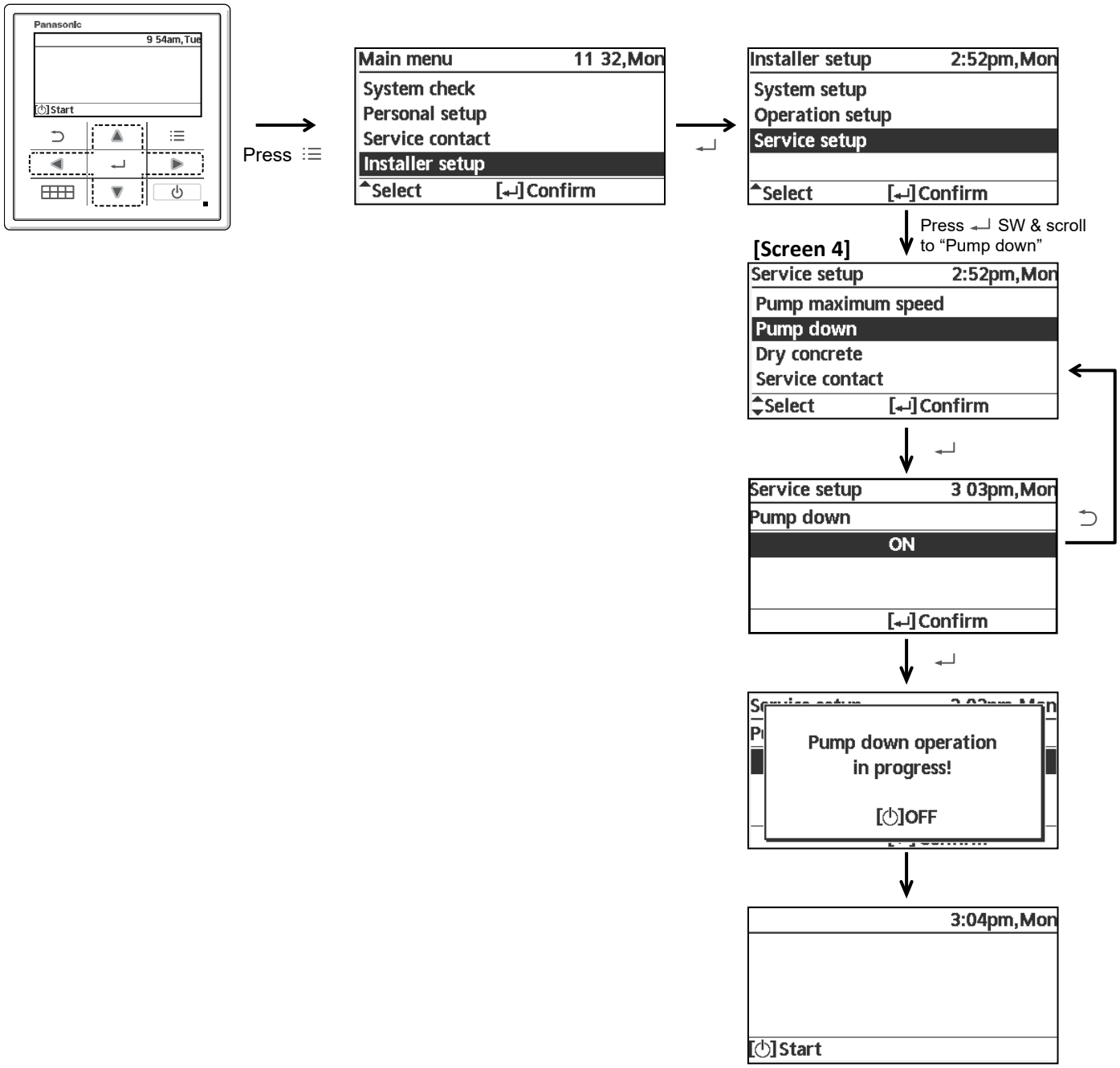
1. Remove the Cover by loosening the mounting screws to access to the Magnetic Water Filter Set.
2. Turn OFF power supply.
3. Set the two valves for the Magnetic Water Filter Set to "CLOSE".
4. Take off the clip, then gently pull out the mesh. Beware of small amount water drain out from it.
5. Clean the mesh with warm water to remove all the stain. Use soft brush if necessary.
6. Reinstall the mesh to the Magnetic Water Filter Set and set back the clip on it.
7. Set the two valves for the Magnetic Water Filter Set to "OPEN".
8. Turn ON power supply.
9. After cleaning, reinstall the Cover by tightening the mounting screws properly.



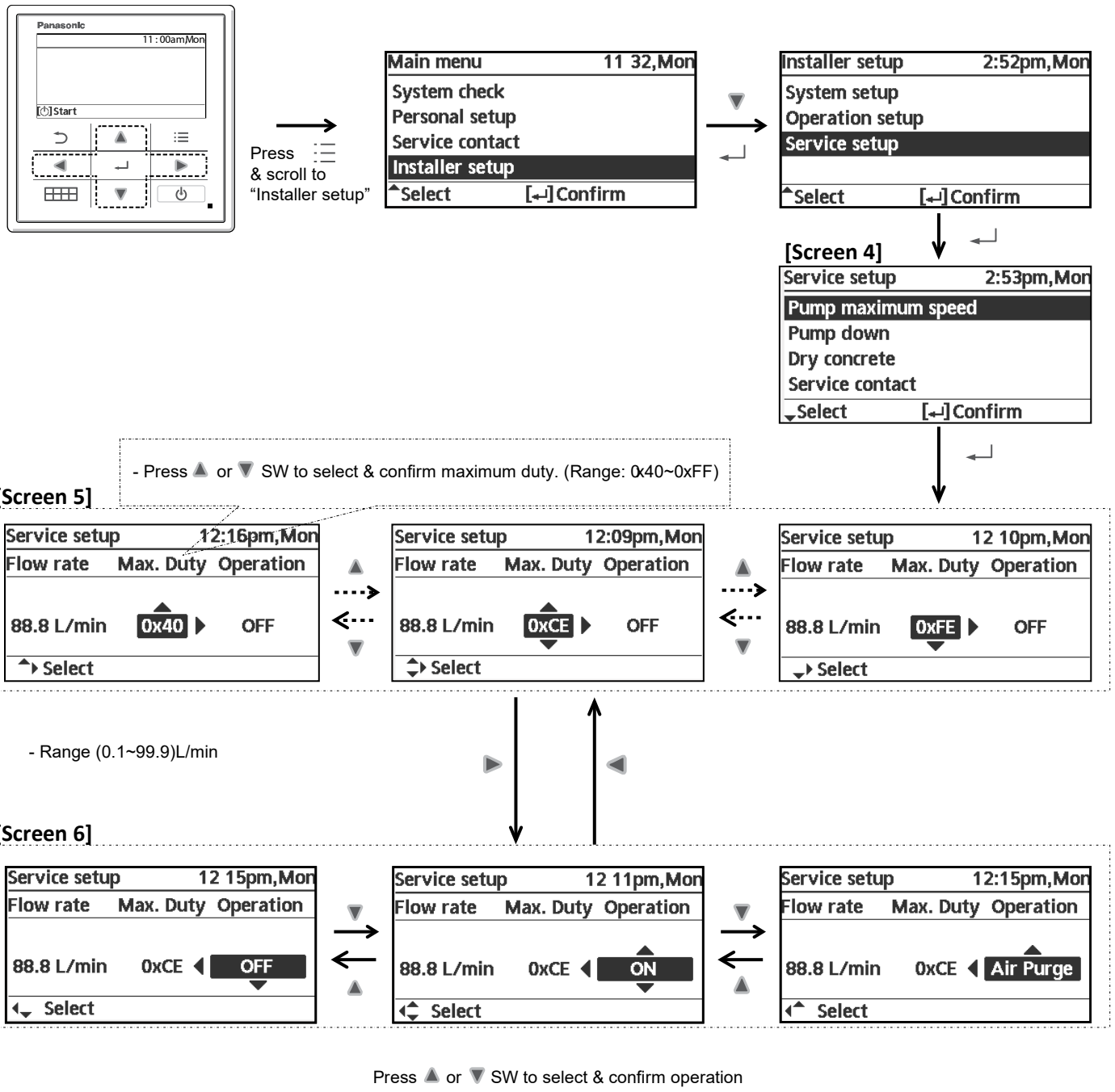

| ⚠ WARNING |
| --- |
| Do not add or replace other than R32 type. It may cause product damage, burst, injury and etc. Use compatible R32 tools for refrigerant piping work and refrigerant charging during installation or servicing. |

## *16.3 Pump Down Procedures*

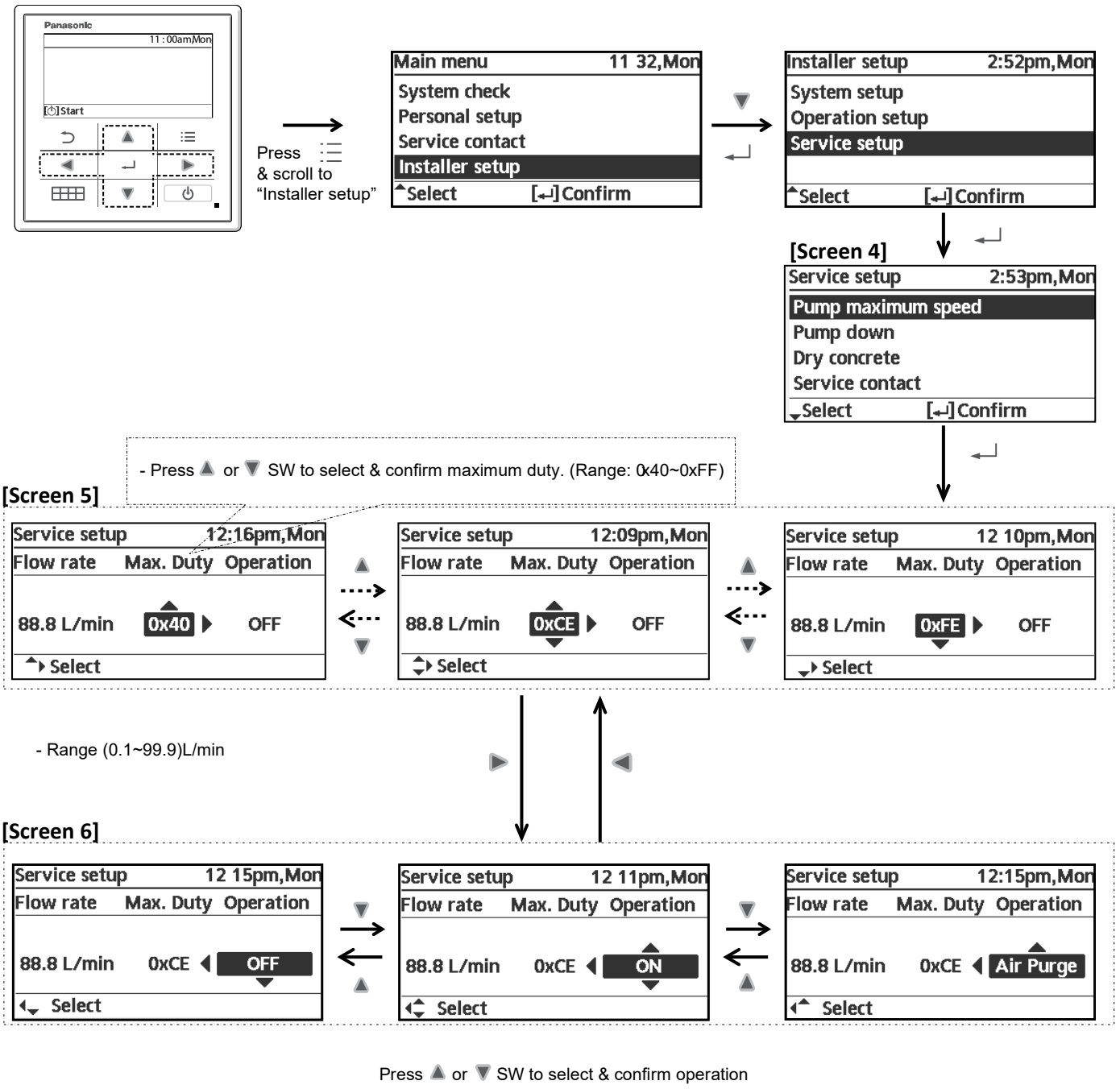
Refer below steps for proper pump down procedure.



## 16.4 How To Adjust Pump Speed



## 16.5 How To Adjust Pump Speed

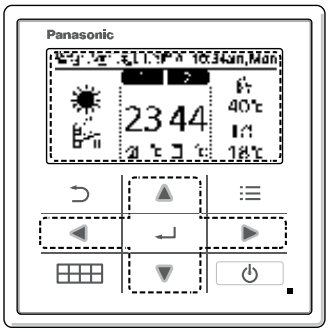



**NOTE:**

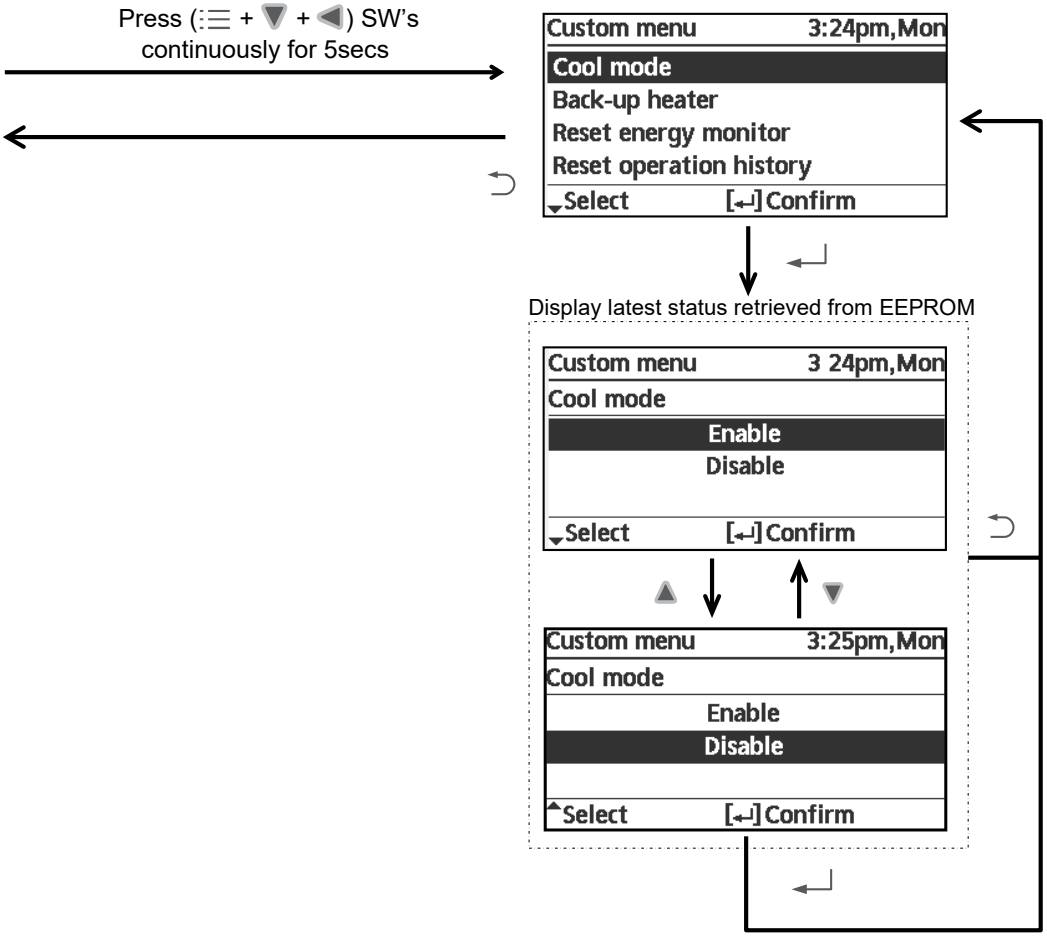
1. Whenever at [Screen 5], if press ⏻ SW to OFF, pump operation should be turned OFF.
2. Whenever at [Screen 6], if press ⏻ SW to OFF, pump operation should be turned OFF.

## 16.6 How To Unlock Cool Mode

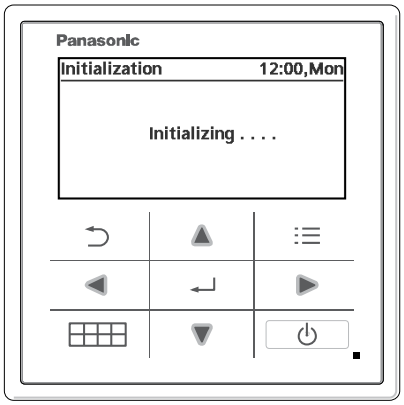
Operation must be OFF




Press (☰ + ▼ + ◀) SW's continuously for 5secs

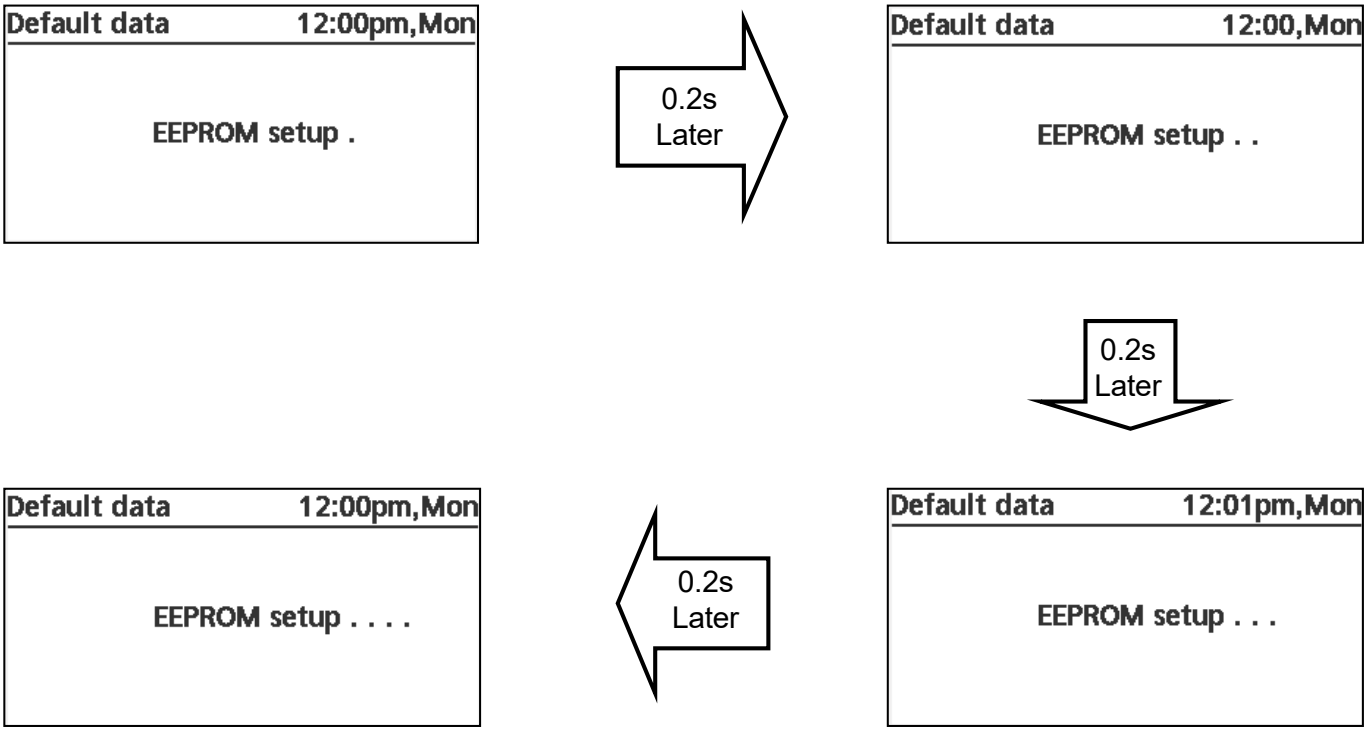


## 16.7 EEPROM Factory Default Data Setup Procedure

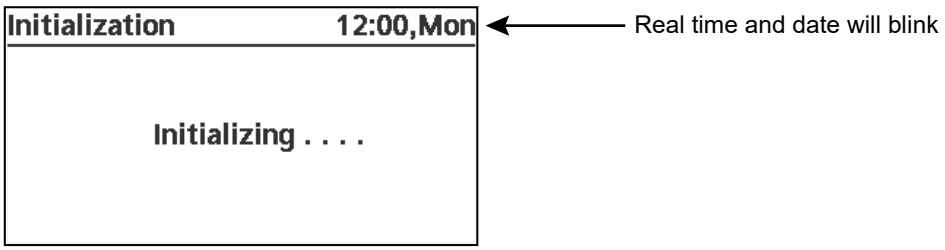



- EEPROM default data setup is only possible during initialization process.
- Press ( ▲, ▼, ◀, ▶ )simultaneously for 5secs continuously, initialization process will stop & EEPROM default data setup process will start.

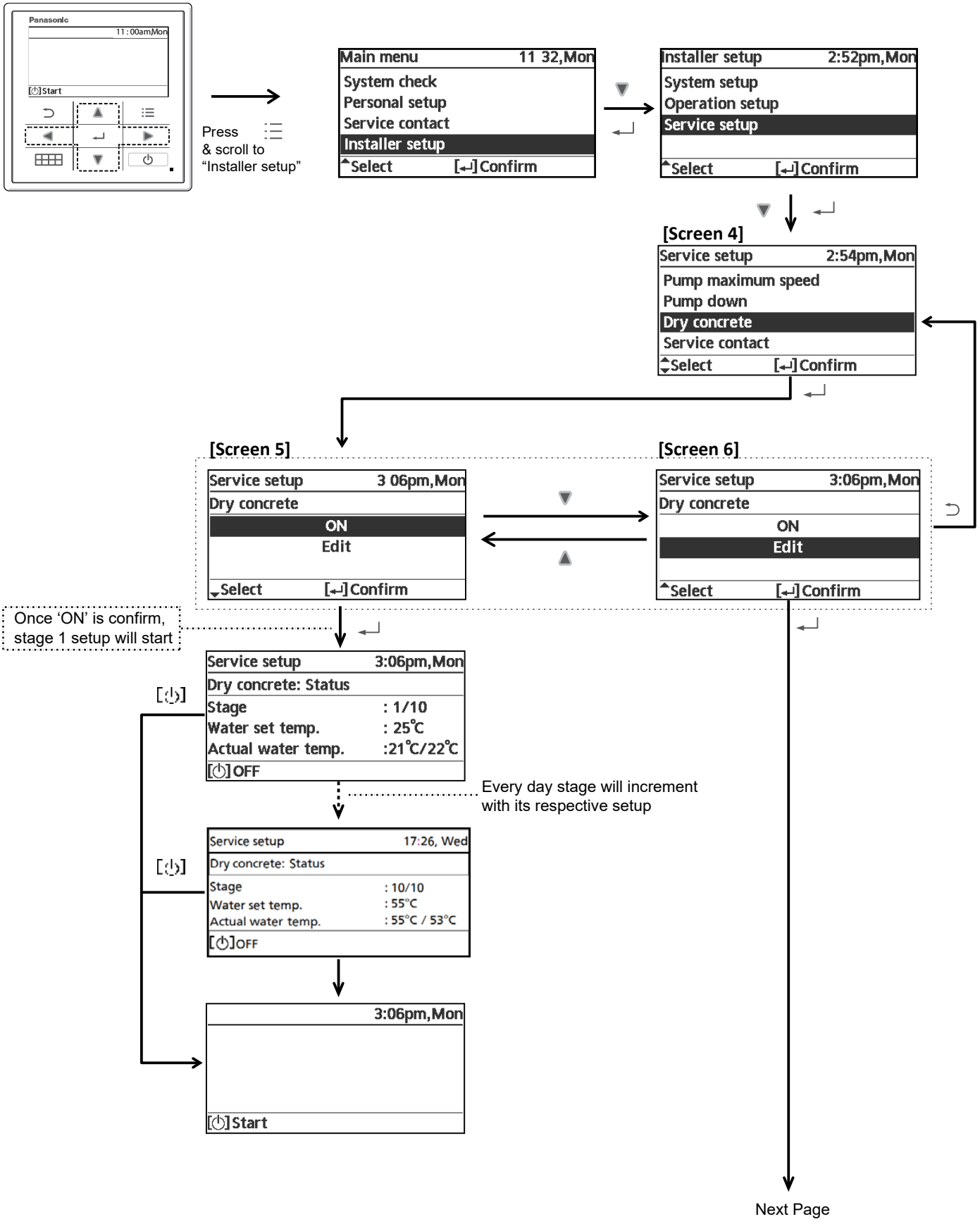
During EEPROM default data setup process, display should be as shown below.

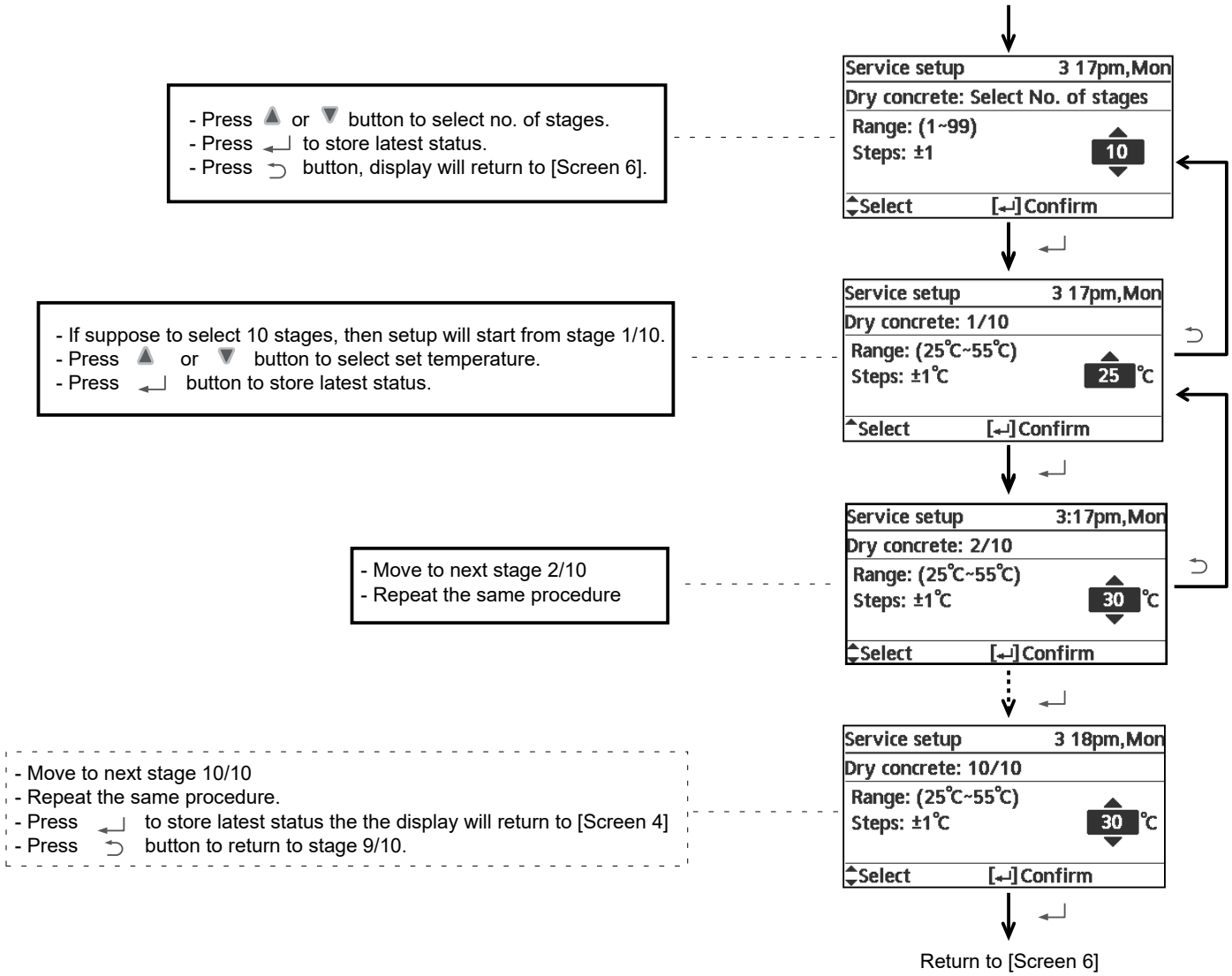



- Once EEPROM default data setup process is complete, initialization process will re-start from beginning.



## 16.8 Dry Concrete Setup

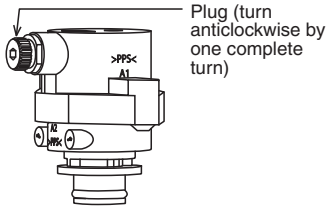



- Press ▲ or ▼ button to select no. of stages.
- Press ↵ to store latest status.
- Press ⮌ button, display will return to [Screen 6].
Service setup 3 17pm,Mon
Dry concrete: Select No. of stages
Range: (1~99)
Steps: ±1
10
Select [↵]Confirm
- If suppose to select 10 stages, then setup will start from stage 1/10.
- Press ▲ or ▼ button to select set temperature.
- Press ↵ button to store latest status.
Service setup 3 17pm,Mon
Dry concrete: 1/10
Range: (25℃~55℃)
Steps: ±1℃
25 ℃
Select [↵]Confirm
- Move to next stage 2/10
- Repeat the same procedure
Service setup 3:17pm,Mon
Dry concrete: 2/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
- Move to next stage 10/10
- Repeat the same procedure.
- Press ↵ to store latest status the the display will return to [Screen 4]
- Press ⮌ button to return to stage 9/10.
Service setup 3 18pm,Mon
Dry concrete: 10/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
Return to [Screen 6]

# 17. Maintenance Guide

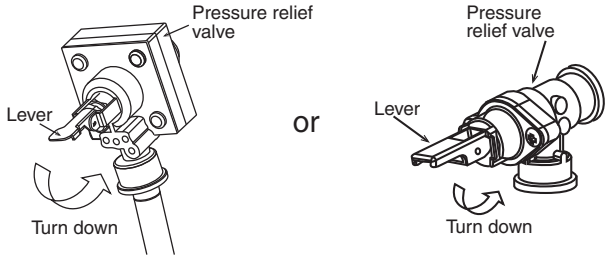
In order to ensure safety and optimal performance of the unit, seasonal inspections on the unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

1 Air purge valve
Make sure all the piping installations are properly done before carry out below steps.
- Open Cabinet front plate to access to the Pressure Relief Valve and Air Purge Valve.
- Turn the plug on the Air Purge Valve outlet anticlockwise by one complete turn from fully closed position.



Air purge valve

- Set the Pressure Relief Valve level “DOWN”.



Pressure relief valve
Pressure relief valve

- Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Mono bloc unit via water inlet. Stop filling water if the free water flow through Pressure Relief Valve drain hose.
- Turn ON the power supply and make sure Water Pump is running.
- Check and make sure no water leaking at the tube connecting points.
- Reinstall the Cabinet front plate by tightening the 2 mounting screws.

2 Water pressure *(0.1 MPa = 1 bar)
Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure Gauge). If necessary add tap water into the water circuit.

3 Pressure relief valve
- Check for correct operation of Pressure Relief Valve by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keeps drained out from the unit, switch off the system, and then contact your local authorized dealer.

4 Expansion vessel

[Upper limit water volume of the system]

The Mono bloc unit has a build-in Expansion Vessel with 10 L air capacity and initial pressure of 1 bar.

- Without antifreeze agent condition
  Total amount of water in the system should be below 200 L.
  If the total amount of water is more than 200 L, please add expansion vessel (field supply).
- With antifreeze agent condition
  In the case of using antifreeze agent, expansion rate ε is different depending on its maker.
  Please refer to the antifreeze agent maker for the expansion rate ε before calculate the upper limit water volume of the system.
  The expansion vessel capacity required for the system can be calculated from the formula below.

$$V = \frac{\varepsilon \times Vo}{1 - \frac{98 + P_1}{98 + P_2}}$$

V : Required gas volume <expansion vessel volume L>
Vo : System total water volume <L>
ε : Expansion rate 5 → 60˚C = (depends on antifreeze agent used)
$P_1$ : Expansion tank filling pressure = (100) kPa
$P_2$ : System maximum pressure = 300 kPa

- ( ) Please confirm at actual place
- The gas volume of the sealed type expansion vessel is presented by <V>.

❍ It's advised to add 10% margin for required gas volume of calculation.

[Adjustment of the initial pressure of the expansion vessel when there is a difference in installation height]

If the height difference between the Mono bloc unit and the highest point of the system water circuit (H) is more than 7m, please adjust the initial pressure of the expansion vessel (Pg) according to the following formula.

Pg= (H*10+30) kPa

5 RCCB

Ensure the RCCB set to "ON" condition before check RCCB.
Turn on the power supply to the Mono bloc unit.
This testing could only be done when power is supplied to the Mono bloc unit.

| ⚠ WARNING |
| --- |
| Be careful not to touch parts other than RCCB test button when the power is supplied to Mono bloc unit. Else, electrical shock may happen. |

- Push the "TEST" button on the RCCB. The lever would turn down and indicate "0", if it functions normal.
- Contact authorized dealer if the RCCB malfunction.
- Turn off the power supply to the Mono bloc Unit.
- If RCCB functions normal, set the lever to "ON" again after testing finish.

6 Test run

Before test run, make sure below items have been checked:-

a. Pipework are properly done.
b. Electric cable connecting work are properly done.
c. Mono bloc unit is filled up with water and trapped air is released.

Antifreeze agent must be added into water circuit to prevent freezing of water when outdoor ambient temperatures is low. Recommended antifreeze:
Propylene glycol: 40% (equivalent to -20°C)

- Set ON to the Mono bloc unit and RCCB 2. Then, for remote control operation please refers to (Mono bloc) Air-to-Water Heatpump's operation instruction.

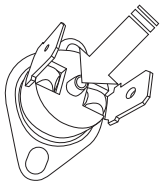
Note:
• During winter, turn on the power supply and standby the unit for at least 15 minutes before test run. Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

For normal operation, Water Pressure Gauge reading should be in between 0.05 MPa and 0.3 MPa.
After test run, please clean the Magnetic Water Filter Set. Reinstall it after finish cleaning.

7 Water flow of water circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger "H75" during defrost process.]

8 Reset overload protector
Overload Protector a serves the safety purpose to prevent the water over heating. When the Overload Protector trip at high water temperature, take below steps to reset it.
a. Take out the cover.
b. Use a test pen to push the centre button gently in order to reset the Overload Protector.
c. Fix the cover to the original fixing condition.



Use test pen to push this button for reset Overload Protector.

# *17.1 Maintenance Menu*

## 17.1.1 Service and maintenance

**If forget Password and cannot operate remote controller**

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

## 17.1.2 Maintenance menu

**Setting method of Maintenance menu**

| Maintenance menu | 12:00am,Mon |
|---|---|
| Actuator check | |
| Test mode | |
| Sensor setup | |
| Reset password | |
| ▼ Select | [↵] Confirm |

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated.
It affects temperature control.

④ Reset password (Reset password)

## 17.1.3 Custom menu

**Setting method of Custom menu**

| Custom menu | 12:00am,Mon |
|---|---|
| Cool mode | |
| Back-up heater | |
| Reset energy monitor | |
| Reset operation history | |
| ▼ Select | [↵] Confirm |

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer.
When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history).
Please use when moving house and handover the unit.

### 17.1.4 Specifications

#### *17.1.4.1 Specifications of fresh water was heat transfer medium in brazed heat exchanger*

| Parameter | Quality Limits for Tap Water on the Secondary Side |
|---|---|
| Temperature | Below 60°C |
| pH | 7 to 9 |
| Alkalinity | 60mg/l <$HCO_3$ <300mg/l |
| Conductivity | < 500µS/cm |
| Hardness | [$Ca^+$, $Mg^+$] / [$HCO_3^-$] > 5 |
| Chloride | < 200mg/l at 60°C |
| Sulphate | [$SO_4^{2-}$] > 100mg/l and [$HCO_3^-$] / [$SO_4^{2-}$] > 1 |
| Nitrate | $NO_3$ < 100mg/l |
| Chlorine | < 0.5mg/l |

#### *17.1.4.2 External filter*

Solids in the water must be filtered.
Minimum filter mesh size required for the field supply external filter in the water inlet is 20 mesh.